

GSDML

Technical Specification for PROFINET IO

Draft in PI Review

Version 2.3 – Date June 2011
Order No.: 2.352

File name : GSDML_Spec_2352_d23_Jun11

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Prepared by the Project Group 1 "GSD" in the Committee 4 "Integration".

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may: indicates flexibility of choice with no implied preference.

should: indicates flexibility of choice with a strongly preferred implementation.

shall: indicates a mandatory requirement. Designers shall implement such mandatory requirements to ensure interoperability and to claim conformance with this specification.

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Revision Log

Version	Originator	Date	Change Note / History / Reason
2.3	C4/PG1 "GSD"	2011-06-09	Version for PI Review.

1 Management Summary - Scope of this Document

The GSDML (General Station Description Markup Language) is an XML based description language for PROFINET IO Devices.

1.1 Goals of GSDML

- GSDML must be able to describe the features of the PROFINET device model.
- GSDML editing can be accomplished with standard XML editors.
- Compliance with ISO 15745: The IEC 61158/61784 standard refers to the ISO 15745 standard as base for device descriptions.
- Additional information within a GSDML should allow a better integration of devices into an engineering tool.

It is not an objective of the GSDML format to describe technological functions or the graphical user interface of a device. For this purpose already established concepts (for example TCI, FDT, EDD) shall be used.

1.2 Structure of a GSDML document

The ISO 15745 standard indicates the basic structure of a GSDML document.

- The GSDML implements a structured approach. In Chapter 5 this model is described in UML notation.
- The device access point (DAP) has a special role within a PROFINET IO Device. So for this module an own section within a GSDML is provided.
- A GSDML file can contain more than one DAP descriptions. This allows building one GSD file for a family of devices sharing the same modules.

2 List of affected patents

There is no affected patent known by the members of the GSD Working Group. The list is empty. No patent search, neither external nor internal, has been done by the members of the GSD Working Group up to now. PROFIBUS & PROFINET International does not guarantee the completeness of this list.

3 Related documents and references

3.1 References

The following referenced documents are indispensable for the application of this document. For dated references, only the edition cited applies. For undated references, the latest edition of the referenced document (including any amendments) applies.

ANSI/IEEE Std 754-1985, *IEEE Standard for Binary Floating-Point Arithmetic*

ISO 639-1:2002, *Codes for the representation of names of languages – Part 1: Alpha-2 code*

ISO/IEC 646:1991, *Information technology – ISO 7-bit coded character set for information interchange*

ISO 15745-1:2003/Amd 1:2007, *Industrial automation systems and integration – Open systems application integration framework – Part 1: Generic reference description*

ISO 15745-3:2003, *Industrial automation systems and integration – Open systems application integration framework – Part 3: Reference description for IEC 61158-based control systems*

WD-SVG11-20110512, *Scalable Vector Graphics (SVG) 1.1 (Second Edition) – W3C Working Draft 12 May 2011*, available at <http://www.w3.org/TR/SVG/>

REC-xml-20081126, *Extensible Markup Language (XML) 1.0 (Fifth Edition) – W3C Recommendation 26 November 2008*, available at <http://www.w3.org/TR/xml/>

REC-xmlschema-1-20041028, *XML Schema Part 1: Structures Second Edition – W3C Recommendation 28 October 2004*, available at <http://www.w3.org/TR/xmlschema-1/>

REC-xmlschema-2-20041028, *XML Schema Part 2: Datatypes Second Edition – W3C Recommendation 28 October 2004*, available at <http://www.w3.org/TR/xmlschema-2/>

REC-xpath20-20101214, *XML Path Language (XPath) 2.0 (Second Edition) – W3C Recommendation 14 December 2010*, available at <http://www.w3.org/TR/xpath20/>

REC-xmlsig-core-20080610, *XML Signature Syntax and Processing (Second Edition) – W3C Recommendation 10 June 2008*, available at <http://www.w3.org/TR/xmlsig-core/>

UML V2.3, *OMG - Unified Modeling Language Specification (Version 2.3, May 2010)*, available at http://www.omg.org/technology/documents/modeling_spec_catalog.htm

The Unicode Standard, V6.0.0, available at <http://www.unicode.org/>

PNO-2.702 *Overview and guidance for PROFINET specifications, Technical Specification for PROFINET, Version 2.3, October 2010, PROFIBUS & PROFINET International, Order Number 2.702*, corresponds to IEC 61158-1

PNO-2.712 *Application Layer services for decentralized periphery and distributed automation, Technical Specification for PROFINET, Version 2.3, October 2010, PROFIBUS & PROFINET International, Order Number 2.712*, corresponds to IEC 61158-5-10

PNO-2.722 *Application Layer protocol for decentralized periphery and distributed automation, Technical Specification for PROFINET, Version 2.3, October 2010, PROFIBUS & PROFINET International, Order Number 2.722*, corresponds to IEC 61158-6-10

PNO-2.732 *Profiles for distributed automation, Technical Specification for PROFINET, Version 2.2, October 2006, PROFIBUS & PROFINET International, Order Number 2.732*, corresponds to IEC 61784-1

64 PNO-2.742 *Profiles for decentralized periphery, Technical Specification for PROFINET, Version 2.3,*
65 *October 2010, PROFIBUS & PROFINET International, Order Number 2.742, corresponds to IEC 61784-2*

66 Related documents

67 ANSI INCITS 4-1986 (R2007), *Information Systems – Coded Character Sets – 7-Bit American National*
68 *Standard Code for Information Interchange (7-Bit ASCII)* (predecessor of ISO/IEC 646)

69 IEEE 802.1AB-2009, *IEEE Standard for local and metropolitan networks – Station and Media Access*
70 *Control Connectivity Discovery*, available at [http://standards.ieee.org/getieee802/download/802.1AB-](http://standards.ieee.org/getieee802/download/802.1AB-2009.pdf)
71 [2009.pdf](http://standards.ieee.org/getieee802/download/802.1AB-2009.pdf)

72 IEEE 802.3-2008, *IEEE Standard for Information technology – Telecommunications and information*
73 *exchange between systems – Local and metropolitan area networks – Specific requirements – Part 3:*
74 *Carrier sense multiple access with Collision Detection (CSMA/CD) Access Method and Physical Layer*
75 *Specifications*, available at http://standards.ieee.org/getieee802/download/802.3-2008_section2.pdf

76 ISO 8601:2004, *Data elements and interchange formats – Information interchange – Representation of*
77 *dates and times*

78 ISO/IEC 10646:2011, *Information technology – Universal Multiple-Octet Coded Character Set (UCS)*

79 ISO/IEC 11578:1996, *Information technology – Open Systems Interconnection – Remote Procedure Call*
80 *(RPC)*

81 IEC 60559:1989, *Binary floating-point arithmetic for microprocessor systems*

82 IEC 61158-5-10:2010, *Industrial communication networks – Fieldbus specifications – Part 5-10:*
83 *Application layer service definition – Type 10 elements*

84 IEC 61158-6-10:2010, *Industrial communication networks – Fieldbus specifications – Part 6-10:*
85 *Application layer protocol specification – Type 10 elements*

86 IEC 61784-2:2010, *Industrial communication networks – Profiles – Part 2: Additional profiles for ISO/IEC*
87 *8802-3 based communication networks in real-time applications*

88 IEC 61804-3:2010, *Function blocks (FB) for process control – Part 3: Electronic Device Description*
89 *Language (EDDL)*

90 IETF RFC 768, *User Datagram Protocol*, available at <http://tools.ietf.org/html/rfc768>

91 IETF RFC 791, *Internet Protocol*, available at <http://tools.ietf.org/html/rfc791>

92 IETF RFC 894, *A Standard for the Transmission of IP Datagrams over Ethernet Networks*, available at
93 <http://tools.ietf.org/html/rfc894>

94 IETF RFC 1034, *Domain names – concepts and facilities*, available at <http://tools.ietf.org/html/rfc1034>

95 IETF RFC 1157, *A Simple Network Management Protocol (SNMP)*, available at
96 <http://tools.ietf.org/html/rfc1157>

97 IETF RFC 2131, *Dynamic Host Configuration Protocol*, available at <http://tools.ietf.org/html/rfc2131>

98 IETF RFC 3629, *UTF-8, a transformation format of ISO 10646*, available at
99 <http://tools.ietf.org/html/rfc3629>

100 REC-html401-19991224, *HTML 4.01 Specification – W3C Recommendation 24 December 1999,*
101 *available at <http://www.w3.org/html/>*

- 102 REC-xslt-19991116, *XSL Transformations (XSLT), Version 1.0, W3C Recommendation 16 November*
103 *1999*, available at <http://www.w3.org/TR/xslt>
- 104 PNO-2.312 *Profile for HART on PROFIBUS – Application Guideline, Version 1.0, May 2002*
- 105 PNO-2.602 *Tool Calling Interface Specification for PROFIBUS and PROFINET, Version 1.1, October*
106 *2008, Order Number 2.602*
- 107 PNO-3.102 *Profile for HART on PROFIBUS, Version 1.0, July 2001*
- 108 PNO-3.192b-PROFIsafe Profile V2.5, *PROFIsafe – Profile for Safety Technology on PROFIBUS DP and*
109 *PROFINET IO, Draft Version 2.5c, February 2010*, PROFIBUS & PROFINET International, Order Number
110 3.192b
- 111 PNO-3.502-Profile Guidelines Part 1 – *Identification & Maintenance Functions, Version 1.2, October 2009,*
112 *PROFIBUS & PROFINET International, Order Number 3.502*
- 113 PNO-7.012 *Fieldbus integration in PROFINET IO, Guideline, Draft Version 2.0, November 2010*
- 114 PNO-7.112 *PROFINET IO Configure in Run for Distributed Control Systems, Draft Version 0.95, October*
115 *2010*, PROFIBUS & PROFINET International, Order Number 7.112
- 116 PNO-7.122 *PROFINET IO System Redundancy for Distributed Control Systems, Guideline for*
117 *PROFINET, Draft Version 0.97, November 2010*, PROFIBUS & PROFINET International, Order Number
118 7.122

4 Definitions and Abbreviations

4.1 Definitions

NOTE The UML terminology and notation used in this document is described in Annex A of ISO 15745-1:2003/Amd 1:2007.

For the purposes of this document, the terms and definitions given in ISO 15745-1:2003/Amd 1:2007 apply.

GSD	General Station Description The device description (file) of a specific device (either PROFIBUS or PROFINET).
PB-GSD	PROFIBUS GSD The device description of a PROFIBUS Masters or Slave. Use this term only to distinguish between PB-GSD and PN-GSD, otherwise use the term GSD.
PN-GSD	PROFINET GSD The device description of a PROFINET IO Device. Use this term only to distinguish between PB-GSD and PN-GSD, otherwise use the term GSD.
GSDL	GSD Language The language in which PB-GSDs are written (ASCII, line-oriented, keywords carry information). Out of historical reasons, the GSDL specification is still called "GSD Specification".
GSDML	GSD Markup Language The language in which PN-GSDs are written (XML, elements and attributes carry information).

4.2 Abbreviated terms

ANSI	American National Standards Institute (http://www.ansi.org/)
APDU	Application PDU
API	Application Process Identifier
AR	Application Relationship
ASCII	American Standard Code for Information Interchange (see ANSI INCITS 4-1986 (R2007) and the US variant of ISO/IEC 646:1991)
ASE	Application Service Element
ASIC	Application Specific Integrated Circuit
CiR	Configuration in Run
CR	Communication Relationship
CRC	Cyclic Redundancy Check
DAP	Device Access Point
DCP	Discovery and basic Configuration Protocol
DFP	Dynamic Frame Packing
DHCP	Dynamic Host Configuration Protocol (see RFC 2131)
DNS	Domain Name System (see RFC 1034)
EDD	Electronic Device Description (see IEC 61804-3:2010)
FAP	Fieldbus Access Point
FCS	Frame Check Sequence
FDT	Field Device Tool (see http://www.fdt-jig.org/)

161	FFW	Fast Forwarding
162	FSO	Frame Sending Offset
163	GSD	General Station Description
164	GSDML	GSD Markup Language
165	HART	Highway Adressable Remote Transducer
166	HMI	Human Machine Interface
167	HTML	Hypertext Markup Language (see REC-html401-19991224)
168	IAS	Industrial Automation Systems
169	ID	Identification
170	IEC	International Electrotechnical Commision (http://www.iec.ch/)
171	IEEE	Institute of Electrical and Electronics Engineers (http://www.ieee.org/)
172	IETF	Internet Engineering Task Force (http://www.ietf.org/)
173	IFG	Interframe Gap
174	IM or I&M	Identification & Maintenance (see Profile Guidelines <i>Part 1 – Identification & Maintenance Functions</i>)
175		
176	IO or I/O	Input / Output
177	IOCS	Input Output Object Consumer Status
178	IOPS	Input Output Object Producer Status
179	IOXS	IOCS and IOPS
180	IP	Internet Protocol (see RFC 791 and RFC 894)
181	IR	Isochronous Relay
182	IRT	Isochronous RT
183	ISO	International Organisation for Standardization (http://www.iso.org/)
184	LLDP	Link Layer Discovery Protocol (see IEEE 802.1AB-2009)
185	MAU	Medium Attachment Unit
186	MIB	Management Information Base
187	MSXML	Microsoft XML Core Services (see http://msdn.microsoft.com/en-us/library/ms763742%28VS.85%29.aspx)
188		
189	MRPD	Media Redundancy with Planned Duplication of frames
190	MRT	Media Redundancy Timer for cyclic realtime data MRPD extension
191	MSOT	Max Switchover Time (Device specific time to perform a switchover)
192	MTOT	Max Takeover Time (Monitoring of the recovery time by the IO Controller)
193	NAP	Network Access Point (PNIO interface with its port submodules, see PNO-7.122 <i>PROFINET IO System Redundancy</i>)
194		
195	NC	Numerical Control
196	PA	Process Automation
197	PDU	Protocol Data Unit
198	PI	PROFIBUS & PROFINET International (see PNO)
199	PNO	PROFIBUS Nutzerorganisation e.V. (http://www.profibus.com/)
200	PSVI	Post-Schema-Validation Infoset (see REC-xmlschema-1-20041028)
201	PTCP	Precision Transparent Clock Protocol
202	RC	Robot Control

203	RFC	Request for Comments
204	RPC	Remote Procedure Call (see ISO/IEC 11578:1996)
205	RT	Real Time Protocol
206	SAP	Service Access Point
207	SR	System Redundancy (see PNO-7.122 <i>PROFINET IO System Redundancy</i>)
208	SIL	Safety Integrity Level (see PROFIsafe Profile V2.5)
209	SNMP	Simple Network Management Protocol (see RFC 1157)
210	SVG	Scalable Vector Graphics (see WD-SVG11-20110512)
211	TCI	Tool Calling Interface (see <i>Tool Calling Interface Specification</i>)
212	UCS	Universal Multiple-Octet Coded Character Set (see <i>The Unicode Standard</i> or ISO/IEC
213		10646:)
214	UDP	User Datagram Protocol (see RFC 768)
215	UML	Unified Modeling Language (see UML V2.3)
216	UTF	UCS Transformation Format (see <i>The Unicode Standard</i> or ISO/IEC 10646:)
217	UUID	Universally Unique Identifier (see ISO/IEC 11578:1996)
218	W3C	World Wide Web Consortium (http://www.w3.org/)
219	XML	Extensible Markup Language (see REC-xml-20081126)
220	XPath	XML Path Language (see REC-xpath20-2010121)
221	XSD	XML Schema Definition (see REC-xmlschema-1-20041028 and REC-xmlschema-2-
222		200410289)
223	XSL	Extensible Stylesheet Language
224	XSLT	XSL Transform (see REC-xslt-19991116)

5 UML notation of the GSDML object model

The following chapter describes the structure of a PN-GSD in UML notation. The semantics of the elements and attributes are described in the Annex A of this document.

NOTE In addition to the UML terminology and notation in ISO 15745-1:2003/Amd 1:2007, Annex A, the following diagrams make use of the multiplicity notation (see UML V2.3): The multiplicity of an attribute is displayed in squared brackets.

5.1 Top level structure

Figure 1 shows the top level structure of GSDML. The root element shall be “ISO15745Profile”.

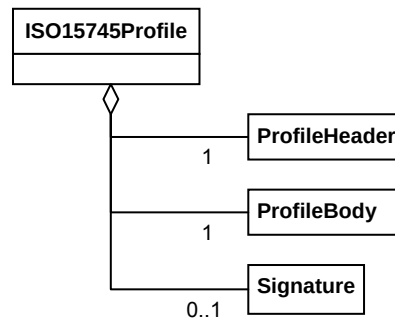


Figure 1 — Class diagram of top level structure

The structure below “ISO15745Profile” is according to the definitions of ISO 15745-1:2003/Amd 1:2007. Please refer this document for the description of the “ProfileHeader” class.

The XML schema representing the GSDML device profile template is defined in A.5. The profile header of the GSDML device profile shall have the following content:

```

<ProfileHeader>
  <ProfileIdentification>PROFINET Device Profile</ProfileIdentification>
  <ProfileRevision>1.00</ProfileRevision>
  <ProfileName>Device Profile for PROFINET Devices</ProfileName>
  <ProfileSource>PROFIBUS Nutzerorganisation e. V. (PNO)</ProfileSource>
  <ProfileClassID>Device</ProfileClassID>
  <ISO15745Reference>
    <ISO15745Part>4</ISO15745Part>
    <ISO15745Edition>1</ISO15745Edition>
    <ProfileTechnology>GSDML</ProfileTechnology>
  </ISO15745Reference>
</ProfileHeader>
  
```

The optional element “Signature” (supported since GSDML version 2.0) allows an integrity check of a PN-GSD. The structure of this element is defined in REC-xmlsig-core-20080610.

5.2 Device profile

5.2.1 General

Figure 2 shows the ProfileBody class structure of a GSDML device profile. UML classes without an attribute field are detailed in a separate diagram.

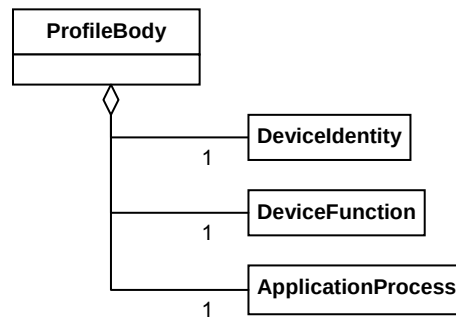


Figure 2 — GSDML device profile class diagram

5.2.2 Device identity

Figure 3 shows the structure of the DeviceIdentity class.

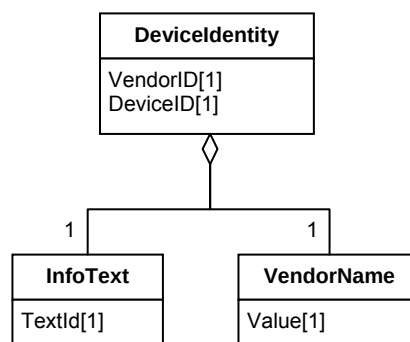


Figure 3 — DeviceIdentity class diagram

Attributes and semantics of the classes are defined in A.4.2.

5.2.3 Device function

Figure 4 shows the structure of the DeviceFunction class.

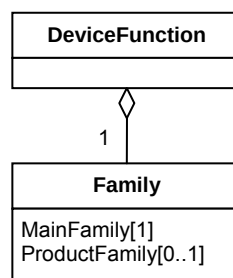


Figure 4 — DeviceFunction class diagram

Attributes and semantics of the classes are defined in A.4.3.

5.2.4 Application process

5.2.4.1 General

Figure 5 describes the structure of the ApplicationProcess element. UML classes without an attribute field are detailed in a separate diagram. Attributes and semantics of the classes are defined in A.4.4.

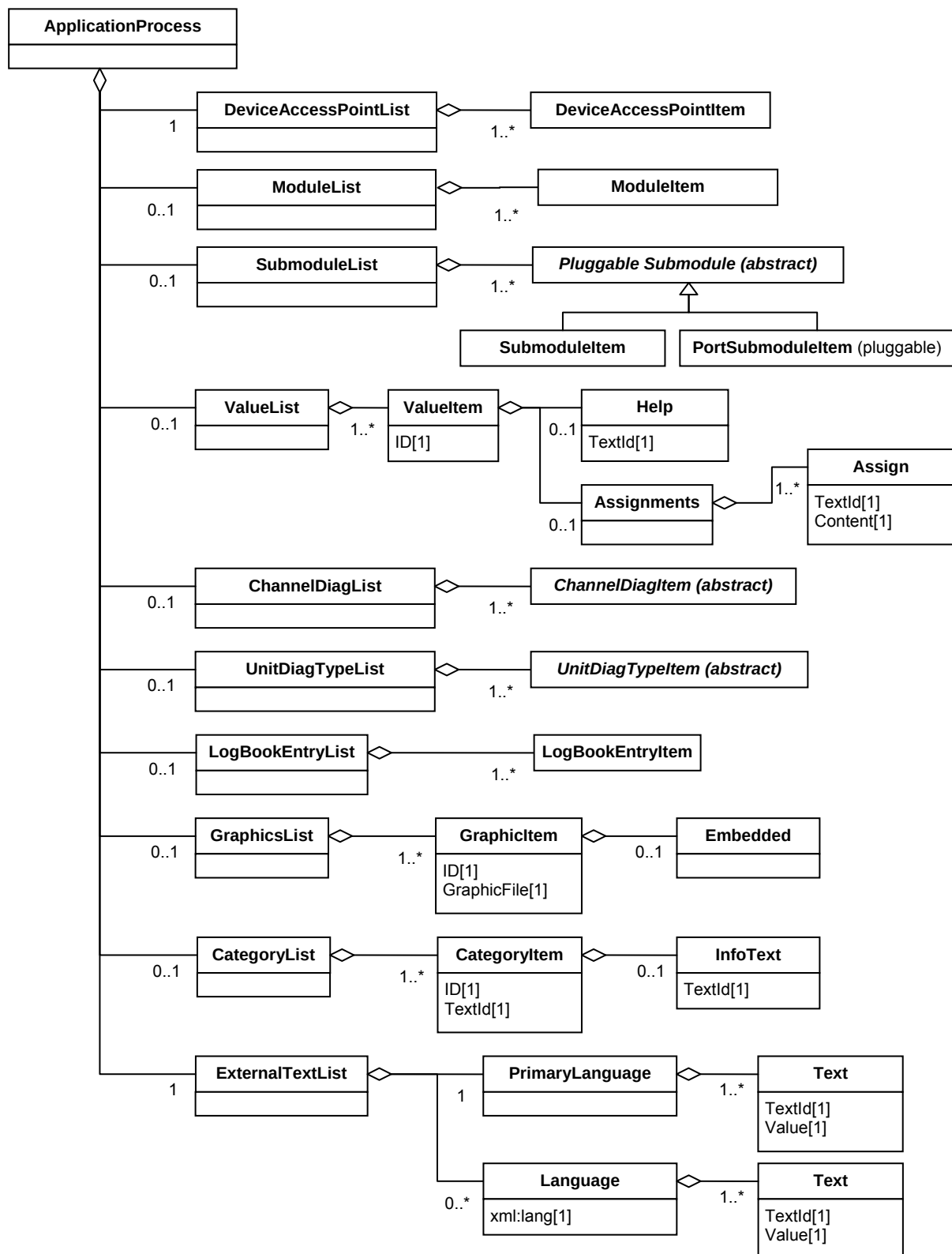


Figure 5 — ApplicationProcess class diagram

5.2.4.2 DeviceAccessPointItem

Figure 6 describes the structure of the DeviceAccessPointItem element. UML classes without an attribute field are explained in a separate diagram in detail. Attributes and semantics of the classes are defined in A.4.5.

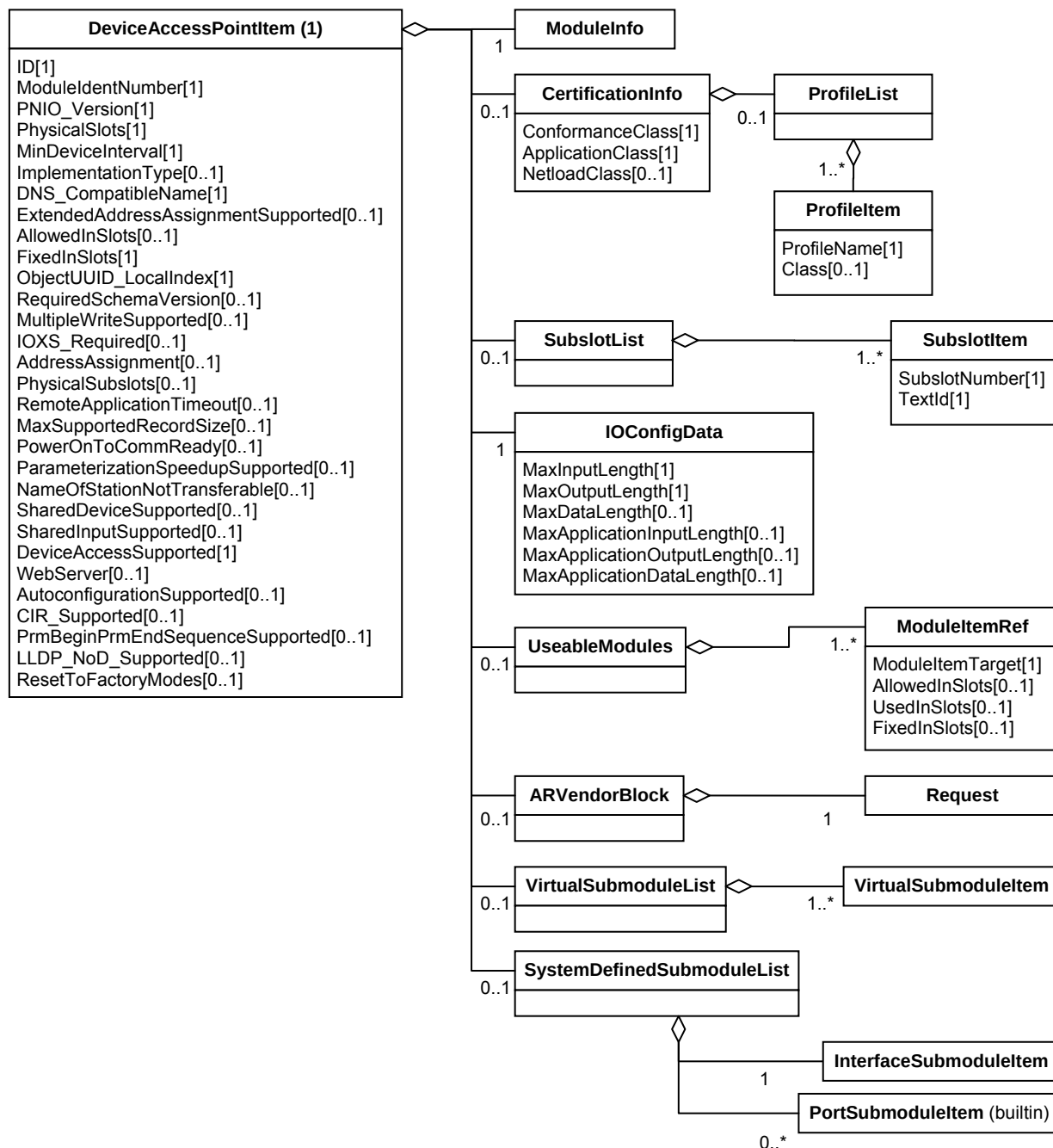


Figure 6 — DeviceAccessPointItem class diagram (part 1)

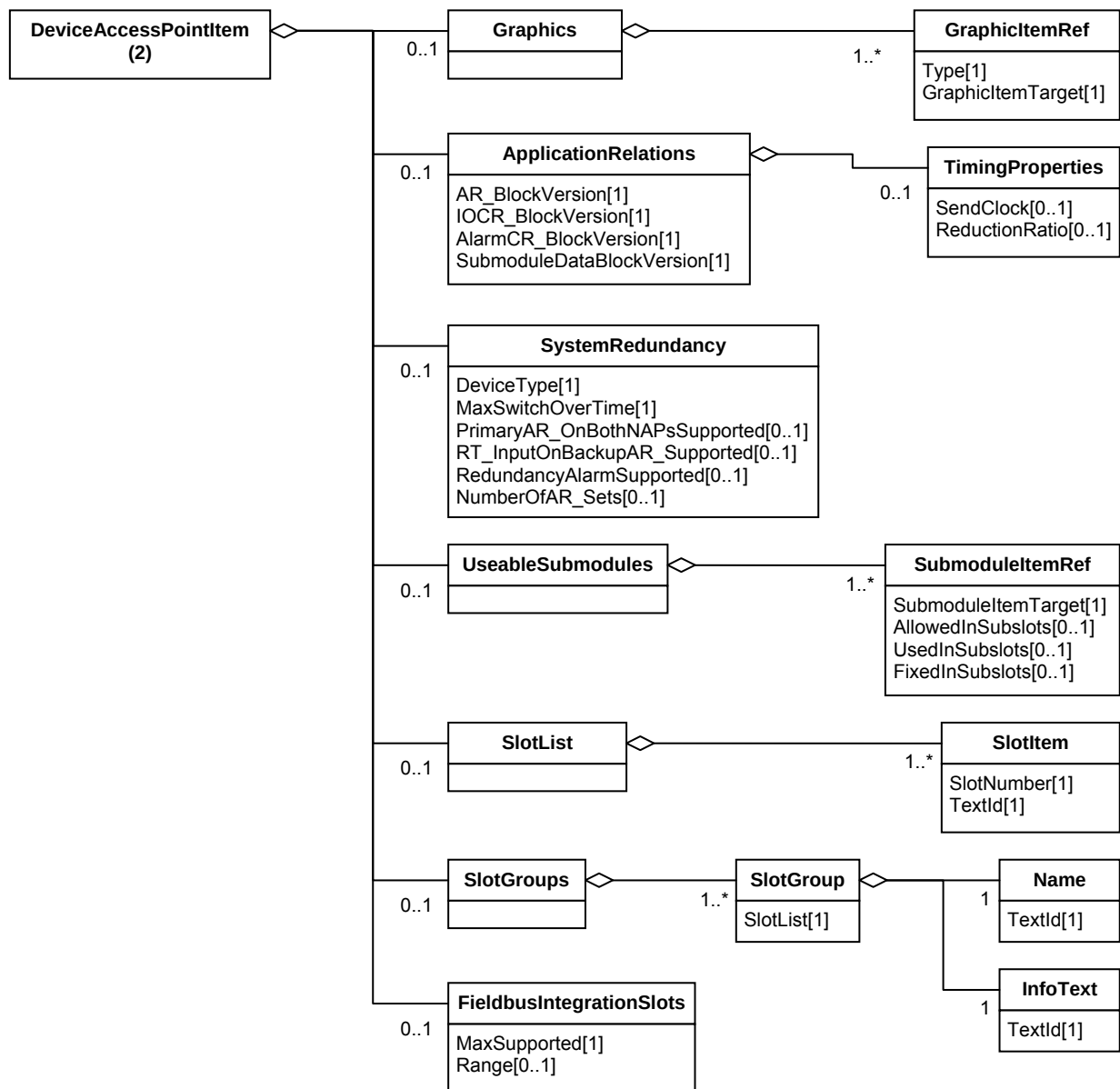


Figure 7 — DeviceAccessPointItem class diagram (part 2)

5.2.4.3 ModuleItem

Figure 8 describes the structure of the ModuleItem element. UML classes without an attribute field are explained in a separate diagram in detail. Attributes and semantics of the classes are defined in A.4.6.

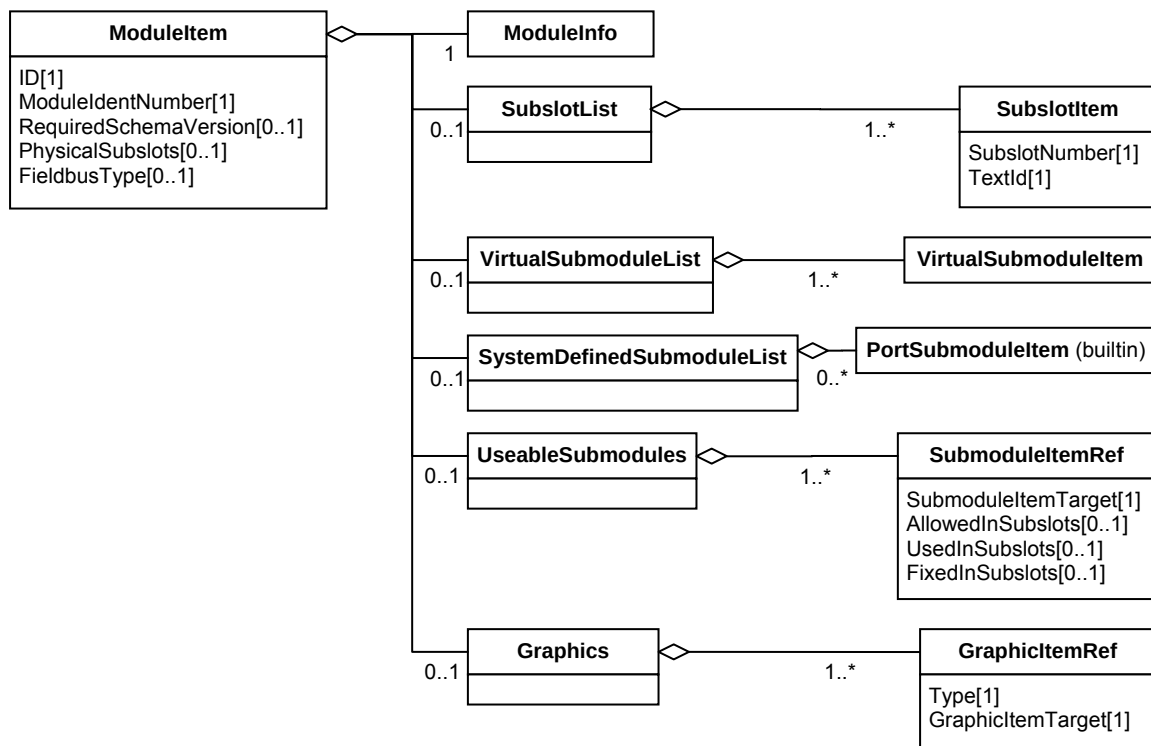


Figure 8 — ModuleItem class diagram

5.2.4.4 SubmoduleItem

Figure 9 describes the structure of the SubmoduleItem element. UML classes without an attribute field are explained in a separate diagram in detail. Attributes and semantics of the classes are defined in A.4.7.

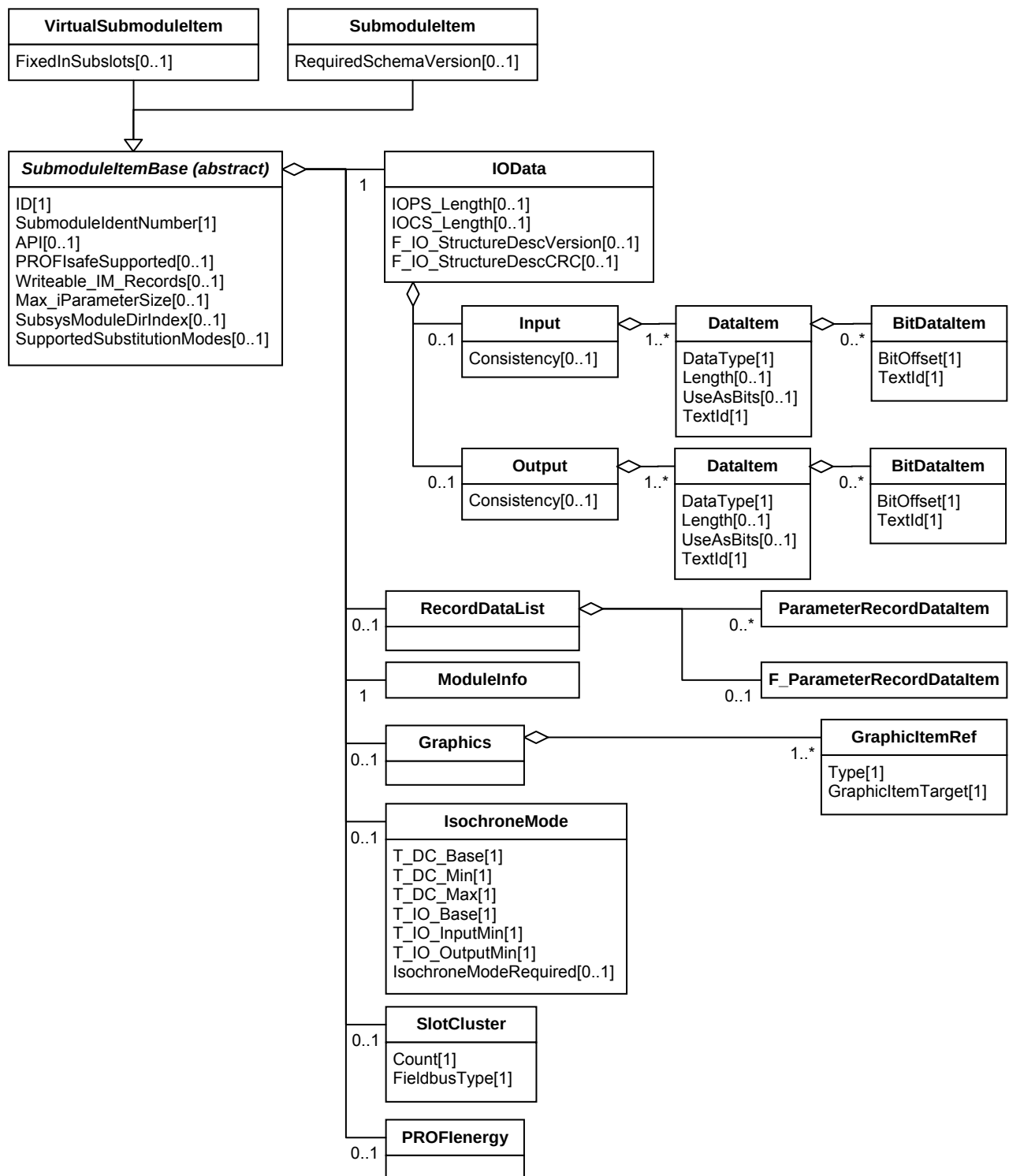


Figure 9 — SubmoduleItem class diagram

5.2.4.5 InterfaceSubmoduleItem

Figure 10 shows the structure of the InterfaceSubmoduleItem element. Attributes and semantics of the classes are defined in A.4.8.

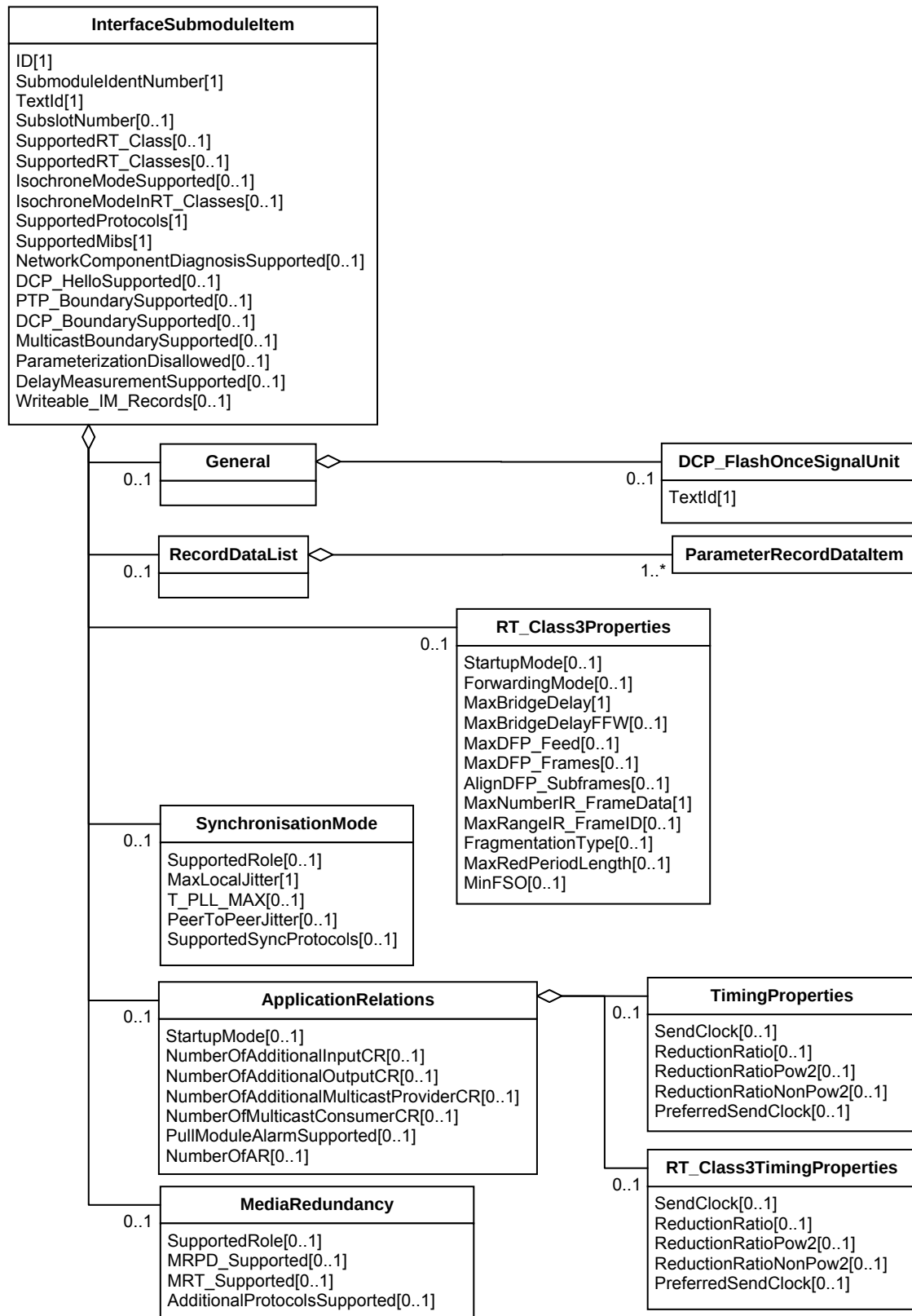


Figure 10 — InterfaceSubmodule class diagram

5.2.4.6 PortSubmoduleItem

Figure 11 shows the structure of the PortSubmoduleItem element. Attributes and semantics of the classes are defined in A.4.9.

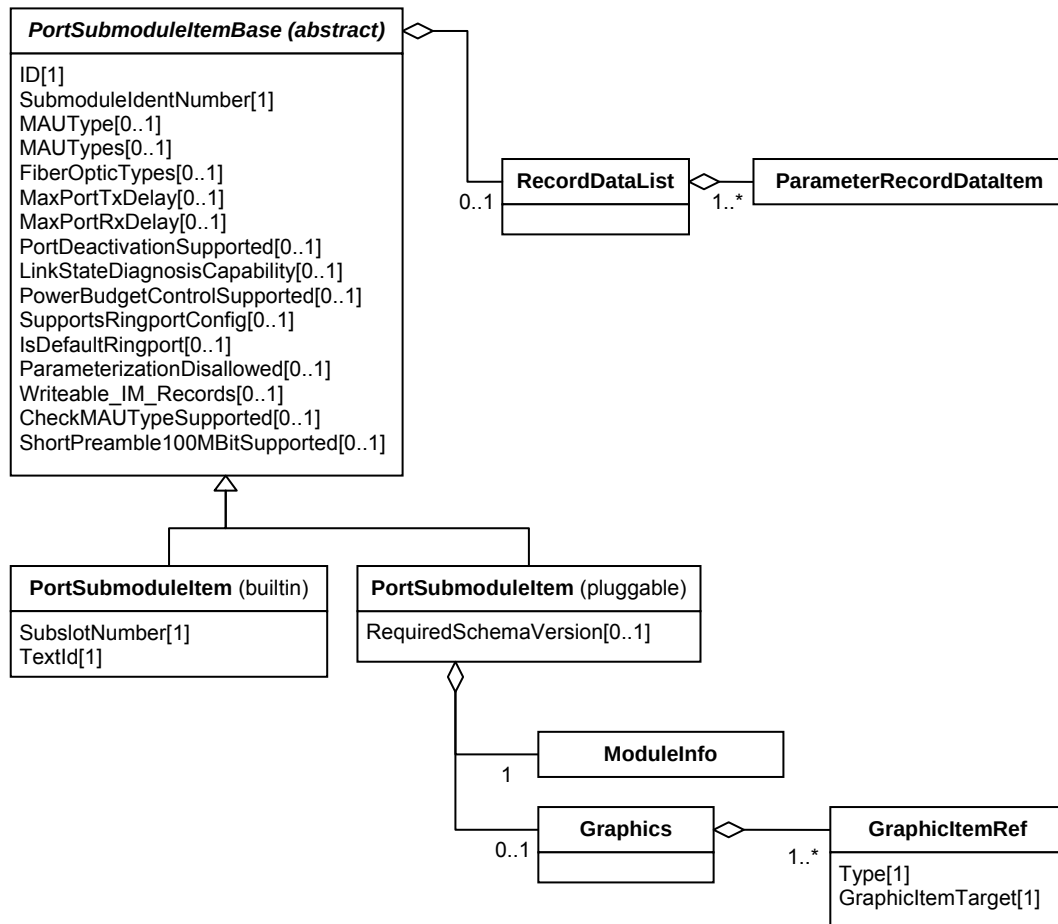


Figure 11 — PortSubmodule class diagram

5.2.4.7 ARVendorBlock/Request and ParameterRecordDataItem

Figure 12 and Figure 13 describe the structure of the ARVendorBlock/Request and the ParameterRecordDataItem element. Attributes and semantics of the classes are defined in A.4.10.

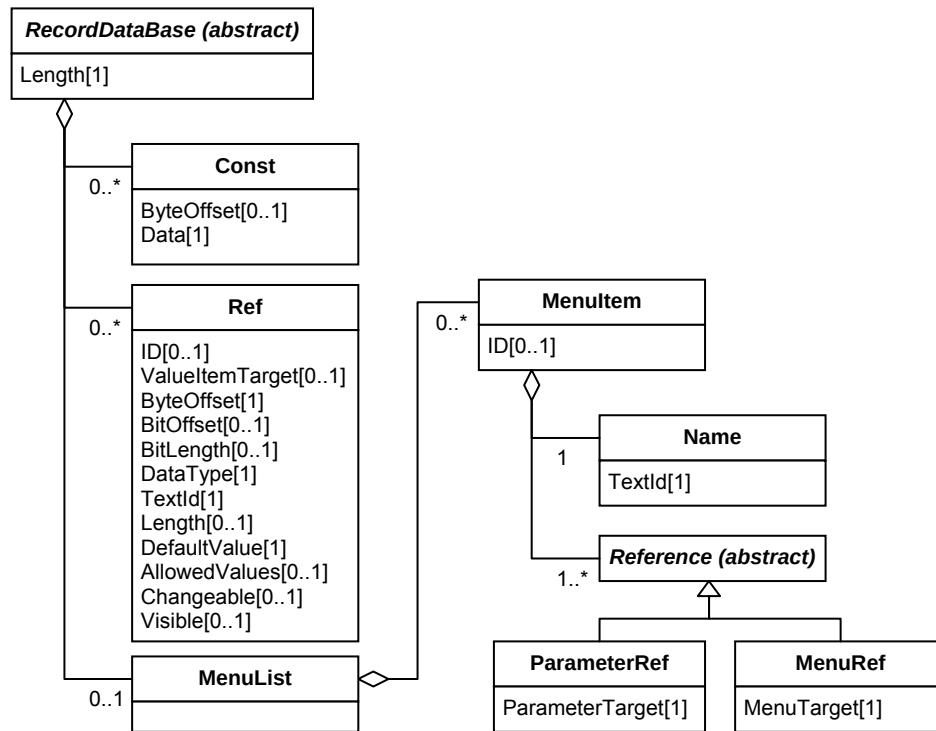


Figure 12 — RecordDataBase class diagram

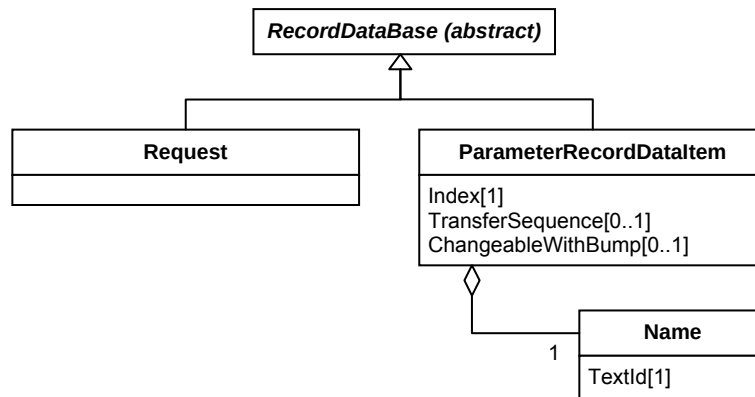


Figure 13 — ARVendorBlock/Request and ParameterRecordDataItem class diagram

5.2.4.8 F_ParameterRecordDataItem

Figure 14 describes the structure of the ParameterRecordDataItem element. Attributes and semantics of the classes are defined in A.4.10.4.

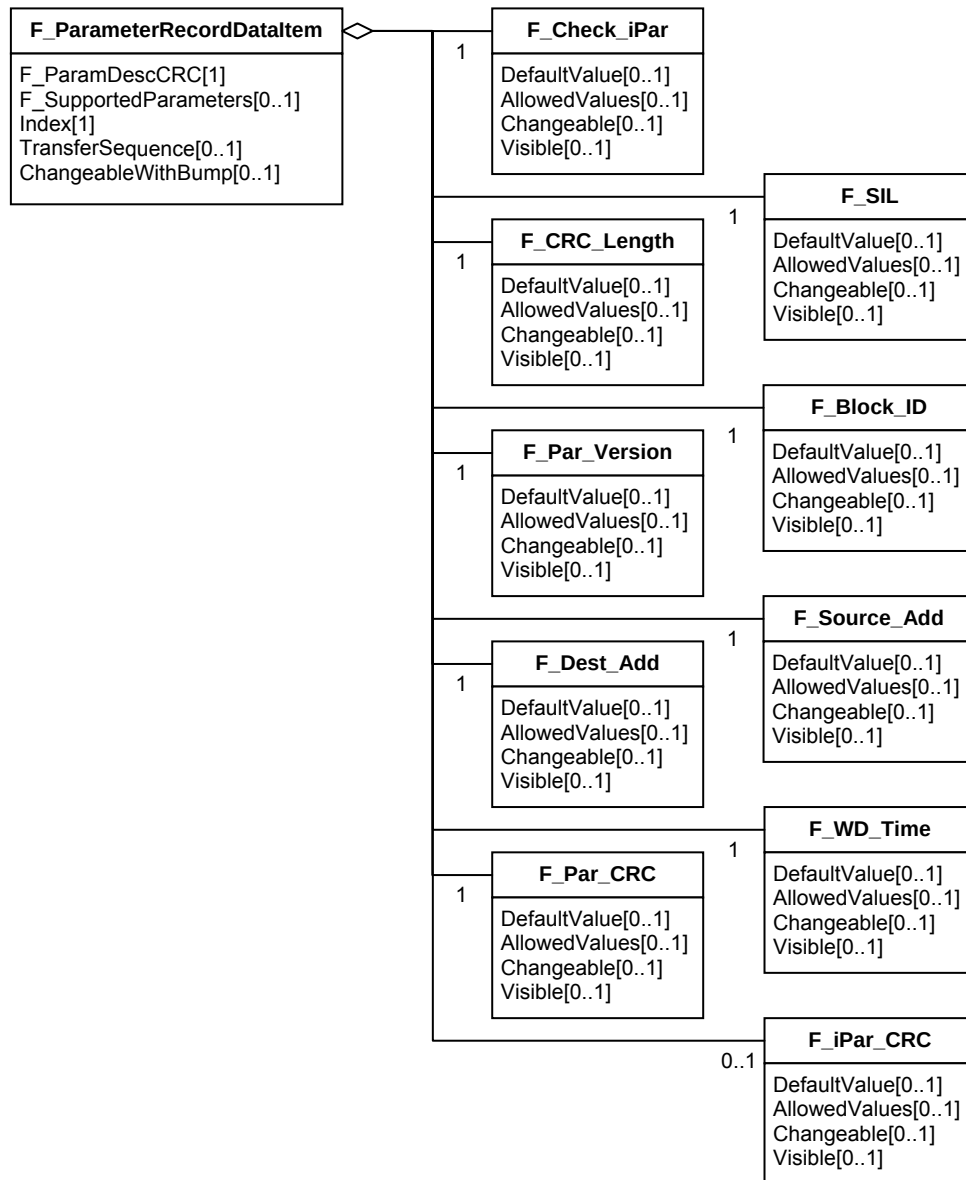
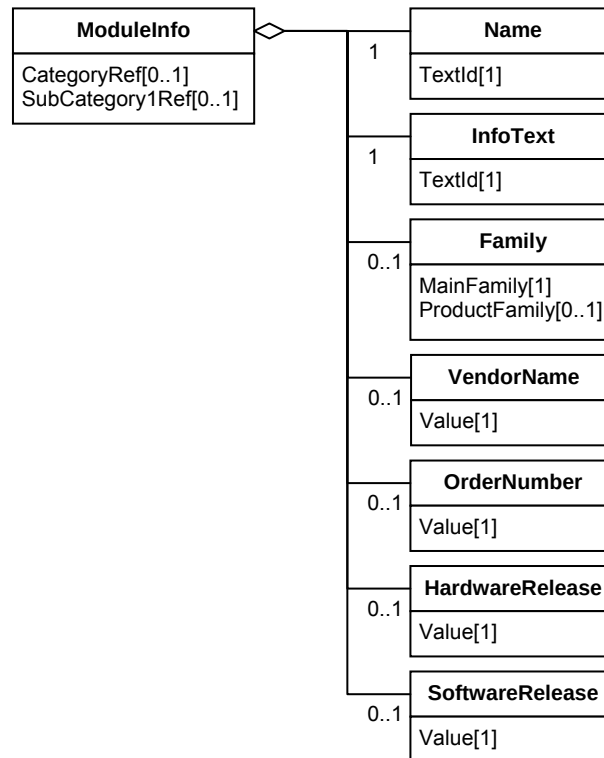


Figure 14 — F_ParameterRecordDataItem class diagram

315 **5.2.4.9 ModuleInfo**

316 Figure 15 describes the structure of the ModuleInfo element. Attributes and semantics of the classes are
317 defined in A.4.12.



318
319 **Figure 15 — ModuleInfo class diagram**

5.2.4.10 ChannelDiagItem

Figure 16 describes the structure of the ChannelDiagItem element. Attributes and semantics of the classes are defined in A.4.13.

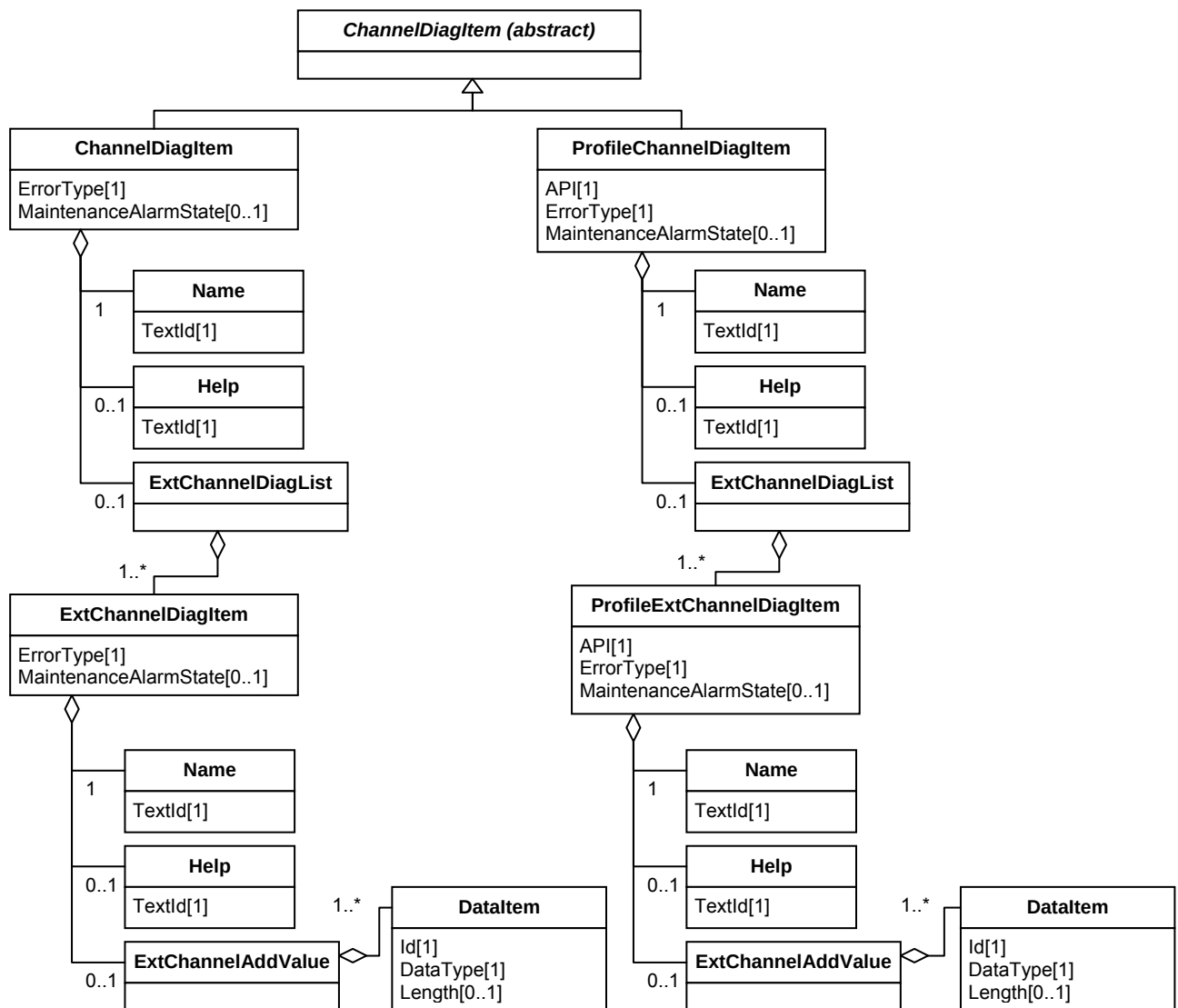


Figure 16 — ChannelDiagItem class diagram

5.2.4.11 UnitDiagTypeItem

Figure 17 describes the structure of the UnitDiagTypeItem element. Attributes and semantics of the classes are defined in A.4.14.

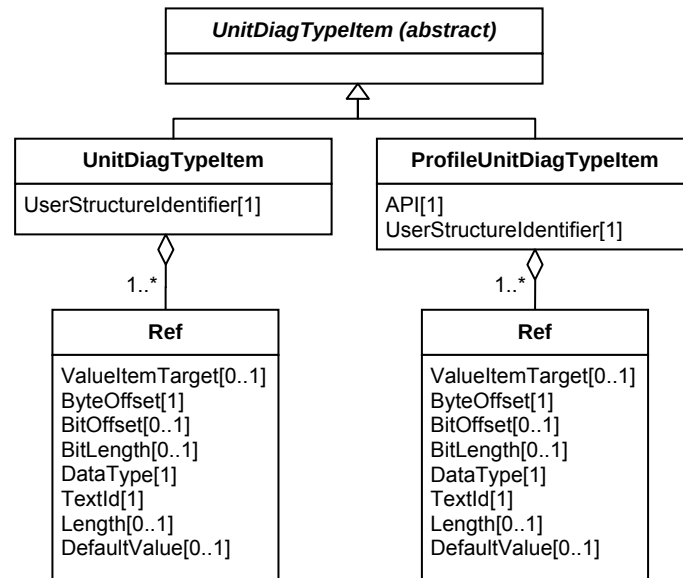


Figure 17 — UnitDiagTypeItem class diagram

5.2.4.12 LogBookEntryItem

Figure 18 describes the structure of the LogBookEntryItem element. Attributes and semantics of the classes are defined in A.4.15.

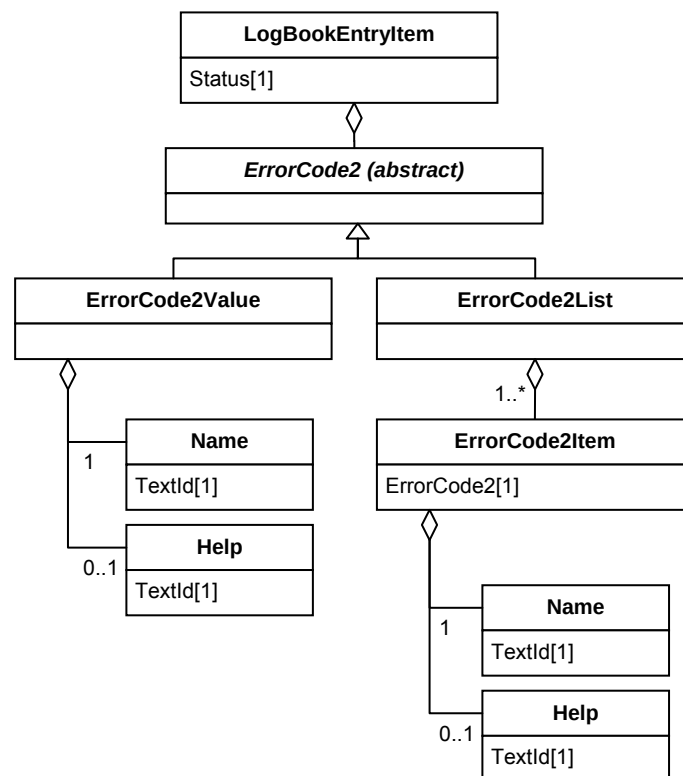


Figure 18 — LogBookEntryItem class diagram

5.3 Communication network profile

Figure 19 shows the class structure of a GSDML communication network profile.

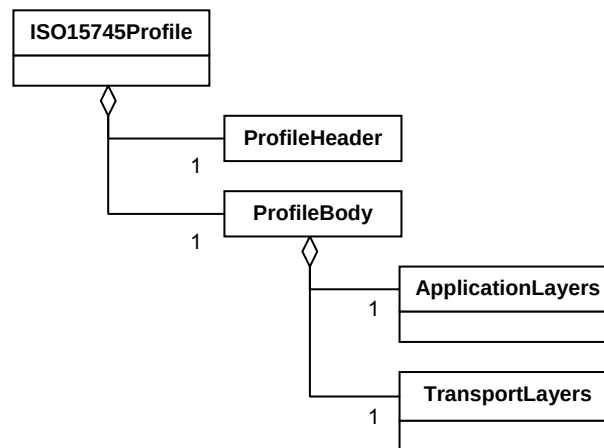


Figure 19 — communication network profile class diagram

NOTE In GSDML the classes of the communication network profile are empty. The reason is that no options of the communication properties of a PROFINET device are provided.

The XML schema representing the GSDML communication network profile template (GSDML-CommNetwork-v2.3.xsd) is provided together with this specification as a separate file.

6 Requirement for certification tests

The device description file shall be checked according to the functionality of the device and the actual specification of the file. This check is precondition for doing the projecting of the PROFINET controller and thus the interoperability testing.

A certification test has to ensure that all GSDs of Version 2.3 comply with the XML schemas defined in A.4.13. The GSD of the device shall be tested with that schema. This test can be processed automatically with the help of an XML parser.

The validation of the GSD with GSDML schema V2.3 shall be done with the following parsers:

- MSXML 4.0 Service Pack 3 (Microsoft XML Core Services), Date Published 9/29/2009, <http://www.microsoft.com/downloads/details.aspx?familyid=7F6C0CB4-7A5E-4790-A7CF-9E139E6819C0&displaylang=en>
- XmlReader class of Microsoft .NET Framework Version 2.0 with schema validation switched on, i.e. Settings.ValidationType set to ValidationType.Schema.
 - o Microsoft .NET Framework Version 2.0 Redistributable Package – Date Published: 1/22/2006, <http://www.microsoft.com/downloads/details.aspx?familyid=0856EACB-4362-4B0D-8EDD-AAB15C5E04F5&displaylang=en>
 - o Microsoft .NET Framework 2.0 Service Pack 2 – Date Published: 1/16/2009, <http://www.microsoft.com/downloads/details.aspx?FamilyID=5b2c0358-915b-4eb5-9b1d-10e506da9d0f&displaylang=en>
- Xerces-C++ Version 3.1.1, <http://xerces.apache.org/xerces-c/>

Additionally the GSD has to follow all “shall” rules that are specified in this document. This test can be processed with a dedicated checker tool (e.g. PROFINET XML Viewer) or by analyzing the XML content of a GSD file.

Annex A (normative)

PROFINET profile templates

A.1 Overview

PROFINET is an Ethernet based network compliant to PNO-2.742 *Profiles for decentralized periphery*, CP 3/3.

PROFINET uses the profile description based on ISO 15745-1:2003/Amd 1:2007. The profile technology name is GSDML (General Station Description Markup Language).

It is not the purpose of the GSDML format to describe technological functions or the graphical user interface of a device. For this purpose already established concepts (for example Electronic Device Description Language (EDDL), see IEC 61804-3:2010) are recommended.

By using the GSDML a GSD (General Station Description) file is created. In order to distinguish from the PROFIBUS GSD format described in ISO 15745-3:2003, Annex B, the term PN-GSD is used.

A PN-GSD can contain more than one Device Access Points (DAP). A DAP is a special module which connects the device to the network. This allows building one file for a family of devices sharing the same modules (see A.4.4.1 and A.4.5).

When building a device according to a PROFINET profile (e.g. PROFIsafe, PROFIenergy, PROFIdrive, ..) please keep in mind that in addition to rules for the design of the device these profiles usually also specify rules for the GSD of the device. These rules have to be followed in addition to the rules set in this specification.

A.2 Data types and Data representation

A.2.1 Data types of attributes

Table 1 describes the data types used for attributes in the GSDML.

Regular expressions are used as defined in REC-xml-20081126.

The GSDML specific types defined in GSDML-Primitives-v2.3.xsd have a “T” for “type” at the end of their name which is omitted in this specification for better readability.

Table 1 — Data types

Name	Definition Schema	Description
Boolean	http://www.w3.org/2001/XMLSchema	See REC-xmlschema-2-20041028
Integer	http://www.w3.org/2001/XMLSchema	See REC-xmlschema-2-20041028
String	http://www.w3.org/2001/XMLSchema	See REC-xmlschema-2-20041028
NormalizedString	http://www.w3.org/2001/XMLSchema	See REC-xmlschema-2-20041028
Token	http://www.w3.org/2001/XMLSchema	See REC-xmlschema-2-20041028
Id	GSDML-Primitives-v2.3.xsd	Used for object identification, is xsd:token.
RefId	GSDML-Primitives-v2.3.xsd	Used for object reference, is xsd:token.
Unsigned8	GSDML-Primitives-v2.3.xsd	Unsigned integer in range 0..255
Unsigned16	GSDML-Primitives-v2.3.xsd	Unsigned integer in range 0..65535
Unsigned32	GSDML-Primitives-v2.3.xsd	Unsigned integer in range 0..4294967295
Unsigned16hex	GSDML-Primitives-v2.3.xsd	Unsigned integer in range 0x0..0xFFFF
Unsigned32hex	GSDML-Primitives-v2.3.xsd	Unsigned integer in range 0x0..0xFFFFFFFF
Unsigned32hex0	GSDML-Primitives-v2.3.xsd	Unsigned integer in range 0x1..0xFFFFFFFF
ValueList	GSDML-Primitives-v2.3.xsd	Base type for a list including ranges of unsigned values. The value list shall comply with the following regular expression: $((\backslash d+\backslash \backslash \backslash d+) (\backslash d+))((\backslash d+\backslash \backslash \backslash d+) (\backslash d+))^*$ If a range is defined, the value left from “..” shall be less than the value right from “..” (see Example 1).
SignedOrFloatValueList	GSDML-Primitives-v2.3.xsd	Base type for a list including ranges of signed values or floating-point values. The value list shall comply with the following regular expression: $((-\backslash d+(\backslash \backslash \backslash d+)?([eE]-\backslash d+)?\backslash \backslash \backslash d+(\backslash \backslash \backslash d+)?([eE]-\backslash d+)?)(-\backslash d+(\backslash \backslash \backslash d+)?([eE]-\backslash d+)?)(-\backslash d+(\backslash \backslash \backslash d+)?([eE]-\backslash d+)?\backslash \backslash \backslash d+(\backslash \backslash \backslash d+)?([eE]-\backslash d+)?)(-\backslash d+(\backslash \backslash \backslash d+)?([eE]-\backslash d+)?)(-\backslash d+(\backslash \backslash \backslash d+)?([eE]-\backslash d+)?\backslash \backslash \backslash d+(\backslash \backslash \backslash d+)?([eE]-\backslash d+)?))^*$ Each value shall appear only once in this list, i.e. - single values shall not appear twice - there shall be no range that includes a given single value - ranges shall not overlap. If a range is defined, the value left from “..” shall be less than the value right from “..” (see Example 2).
TokenList	GSDML-Primitives-v2.3.xsd	Base type for a list of token values. The token list shall comply with the following regular expression:

Name	Definition Schema	Description
		$([0-9a-zA-Z_+;]*([0-9a-zA-Z_+])?)$ Each token shall be separated by a semicolon character and shall be interpreted case sensitive. The list may be empty.
TokenList1	GSDML-Primitives-v2.3.xsd	Base type for a list of token values. The token list shall comply with the following regular expression: $([0-9a-zA-Z_+;]*[0-9a-zA-Z_+])$ Each token shall be separated by a semicolon character and shall be interpreted case sensitive. The list shall contain at least one token.
VersionString	GSDML-Primitives-v2.3.xsd	Base type for a version information. Shall comply with the regular expression: $V\d+\.\d+$
Enumeration	GSDML-Primitives-v2.3.xsd	String with defined content
EXAMPLE 1 Valid ValueList contents: <pre> 7 1 2 3 4 5 12211 0..6 5..12 0..34 36 38 </pre> EXAMPLE 2 Valid FloatOrSignedValueList contents: <pre> 7 1 2 3 4 -12 5 12211 0..6 -5..12 0..34 36 38 0.34 -12.5..122.11 0.0..3e-4 3.6E2..3.8 </pre>		

394

395 A.2.2 Representation of values in the GSDML

396 Sometimes there is the need to specify a value whose type depends on a separate attribute (e.g.
397 "DataType"). In this case, the attribute is of type String to be able to host all relevant representations. This
398 list describes the representation of values for each supported data type.

399 The lexical representation of floating point values follows REC-xmlschema-2-20041028 except that "INF",
400 "-INF" and "NaN" and the '+' sign are not supported. For the allowed value ranges see ANSI/IEEE Std
401 754-1985.

402 The date / time values follow REC-xmlschema-2-20041028 which uses ISO 8601:2004 as base.

403 **Table 2 — Representation of numerical values**

Data type	SV	Regular Expression	Description
Bit	1.0	0 1	
BitArea	1.0	\d+	Represented as decimal unsigned integer in the range 0 to $(2^{\text{BitLength}})-1$. The BitOffset (position of the BitArea in an octet stream) does not influence the representation.
Integer8	1.0	-?\d+	Decimal integer in range -128..127
Integer16	1.0	-?\d+	Decimal integer in range -32768..32767
Integer32	1.0	-?\d+	Decimal integer in range -2147483648..2147483647
Integer64	1.0	-?\d+	Decimal integer in range -9223372036854775808..9223372036854775807
Unsigned8	1.0	\d+	Decimal integer in range 0..255

Data type	SV	Regular Expression	Description
Unsigned16	1.0	\d+	Decimal integer in range 0..65535
Unsigned32	1.0	\d+	Decimal integer in range 0..4294967295
Unsigned64	1.0	\d+	Decimal integer in range 0..18446744073709551615
Float32	2.2	-?\d+(\.\d+)?([eE]-?\d+)?	Single-precision floating point number
Float64	2.2	-?\d+(\.\d+)?([eE]-?\d+)?	Double-precision floating point number

404

405

Table 3 — Representation of non-numerical values

Data type	SV	Regular Expression	Description
Date	2.2	\d{4}-\d{2}-\d{2}(T\d{2}:\d{2}:\d{2}(\.\d{1,3})?)?	yyyy-mm-dd[Thh:mm:ss[.fff]] fff = fraction of a second (up to millisecond)
TimeOfDay with date indication	2.2	\d{4}-\d{2}-\d{2}(T\d{2}:\d{2}:\d{2}(\.\d{1,3})?)?	yyyy-mm-dd[Thh:mm:ss[.fff]] fff = fraction of a second (up to millisecond)
TimeOfDay without date indication	2.2	\d{2}:\d{2}:\d{2}(\.\d{1,3})?	hh:mm:ss[.fff] fff = fraction of a second (up to millisecond)
TimeDifference with date indication	2.2	P(\d+D)?(T(\d+H)?(\d+M)?(\d+(\.\d{1,3})?S)?)?	Range: up to 65536 days minus 1 ms Resolution: 1 ms EXAMPLE: "P445DT12H30M17.100S"
TimeDifference without date indication	2.2	P(\d+H)?(\d+M)?(\d+(\.\d{1,3})?S)?	Range: up to one day minus 1 ms Resolution: 1 ms EXAMPLE: "P12H30M17.100S"
NetworkTime	2.2	\d{4}-\d{2}-\d{2}(T\d{2}:\d{2}:\d{2}(\.\d{1,9})?)?	yyyy-mm-dd[Thh:mm:ss[.fffffffff]] fffffffff = fraction of a second (up to nanosecond)
NetworkTimeDifference	2.2	-?P(\d+D)?(T(\d+H)?(\d+M)?(\d+(\.\d{1,9})?S)?)?	Range: from about -24855 to 24855 days Resolution: 1 ns EXAMPLE: "P445DT12H30M17.100S"
VisibleString	2.2	.*	No restriction, just the string. Note that the XML syntax requires coding of some characters: & → & ' → ' > → > < → < " → " LF →
 LF = linefeed Only linefeed is allowed for formatting text.
OctetString	2.2	(0x[0-9a-fA-F][0-9a-fA-F])*0x[0-9a-fA-F][0-9a-fA-F]	Same format as ../RecordDataList/ParameterRecordDataItem/Const/Data

406

A.3 General rules

A.3.1 Version control

If a GSD is already released, it is important that the identification of objects remains unchanged. Therefore the content of the attributes corresponding to the following XPath expressions (see REC-xpath20-2010121) shall not be changed in a new version of a GSD:

//DeviceAccessPointItem/@ID

//ModuleItem/@ID

//SubmoduleItem/@ID

//VirtualSubmoduleItem/@ID

//InterfaceSubmoduleItem/@ID

//PortSubmoduleItem/@ID

//ValueItem/@ID

//Ref/@ID

//MenuItem/@ID

//GraphicItem/@ID

//CategoryItem/@ID

Engineering tools shall be able to manage different GSD files for the same device. In the device catalog of the tool, only the objects (DAPs, modules and submodules) contained in the newest GSD file shall be presented. It is the tool vendors decision what “newest” means exactly. The tool may decide strictly on the date / time given in the GSD file name regardless of the version of the GSDML that the GSD is based on, or it may favor newer GSDML versions.

Device instances in projects shall refer to the GSD file and version on which they are based. When installing a newer GSD version for a device, the device instances in existing projects shall not be affected; i.e. there is no automatic update. A change of the GSD file / version on which a device instance is based may be provided as an explicit user action.

A.3.2 Compatibility with older GSDML versions

To ensure compatibility with GSDs based on older GSDML versions, an engineering tool shall always use the schema version defined in the GSD filename for validation (both XML schema and business logic). For example if a tool supports GSDML schema V2.1 it is recommended to validate a GSDML V1.0 file with schema version and business logic V1.0 and not with version V2.1.

Previous GSDML schema versions are: V1.0, V2.0, V2.1, V2.2 and V2.25.

A.3.3 Compatibility with newer GSDML versions

Engineering tools shall also accept GSDs based on a GSDML version which is newer than what they support (note that the version comparison must be done textually, so that e.g. V2.25 is older than V2.3). In this case there shall be no XML schema validation. All objects (DAPs, modules and submodules) whose attribute “RequiredSchemaVersion” specifies a higher version than what the tool supports shall be ignored. Their description shall not be checked and they shall not be put into the device catalog. If the “RequiredSchemaVersion” is lower or equal to the highest GSDML version that the tool supports, the

object's description shall be checked with business logic of the highest GSDML version that the tool supports. If all right the object shall be put into the catalog. Note that for attributes of type TokenList, in this compatibility mode unknown Tokens shall be ignored – a warning shall be output, but no error.

It may be necessary to re-install the GSD after an update of the tool which adds support for a newer GSDML version to get all objects into the catalog and take full advantage of the new features.

A.3.4 Rules for the name of a PN-GSD

The name of a GSD shall be composed of the seven fields below in the following order:

1. "GSDML"
2. The version ID in format Vx.y whereby "x" and "y" are unsigned numbers. The version ID refers to the ID of the GSDML Schema used.
3. Vendor name
4. Device family name
5. Release date of the GSD in format yyyyymmdd
6. Release time of the GSD in format hhmmss (Optional). "hh" is "00" up to "24".
7. ".xml" (file extension)

Fields 1-6 shall be separated by the dash character "-" (ASCII 45 decimal). The vendor name and device family name may contain any character allowed by Windows¹, including the space " " and the dash "-". So an engineering tool has to parse the release date and time relative to the end of the filename string. The length of the filename is only limited by Windows.

The vendor name may contain the VendorID to ensure a unique filename.

EXAMPLE "GSDML-V2.3-Vendor-Device-20110315.xml"

Already released files shall not be changed without changing the filename. When building a new version of a GSD, the release date shall be changed.

If more than one version of a GSD is installed, the engineering tool can use the release date to determine the newest version. If the release time is not present in the filename, the time "00:00:00" shall be used instead.

To avoid conflicting filenames with tool generated GSD files the field 3 shall start with the "#" character (ASCII 35) when the GSD is used to describe a configuration instead of a device. It shall not be used as first character when the GSD is used to describe one ore more PROFINET devices.

A.3.5 Schema location in a PN-GSD

As prefix for the schema instance namespace, <http://www.w3.org/2001/XMLSchema-instance>, "xsi:" shall be used in the GSDML.

An XML schema validator needs information about the location of the assigned schema file. Therefore the attribute xsi:schemaLocation of the ISO15745Profile root element is specified.

To use the same location for all PN-GSDs, the relative path "..\xsd" shall be used for the schema files.

¹ The operating system Microsoft Windows NT or one of its successors (2000, XP, Vista, 7, ..)

Engineering tools shall compare the version contained in the schema filename in the schema location with the version in the GSDML file name. If both version informations do not match, they shall issue a warning and they should prefer the version in the schema filename.

A.3.6 Identification of objects

XML is a strictly hierarchical description language. It implicitly establishes a father-child relation between the parts of the description. In principle, it is possible to describe a PROFINET IO Device with only this relation, but then e.g. a module would have to be described again and again at each DAP it is pluggable with. That would be unnecessarily big and hard to maintain.

So the father-child-relation is only used within an object to group the information into a partial XML tree. The top element has an ID attribute that identifies the object within its category. This ID may be referenced by other objects to establish supplemental relations between the objects which can't be expressed directly in XML.

Objects of the same type are grouped into a list. The DeviceAccessPointList is the most important, as it is the starting point when configuring a device.

The Identification of objects shall be kept unique over all elements of the same category as described in the following table. (For example, all identification of objects of type "ModuleItem" shall be kept unique.) As an exception to this rule, the IDs of all submodules shall be kept unique over the categories SubmoduleItem, VirtualSubmoduleItem, InterfaceSubmoduleItem and PortSubmoduleItem. This is to allow tools to manage all submodules in a uniform way.

Document wide unique IDs are not necessary.

Table 4 shows the addressable elements. The right column shows all those objects where the items of the left column are being referenced in. These references use the IDs in question as a means of addressing.

Table 4 — Object identification

Type	Referring Elements
DeviceAccessPointItem	Not used yet
ModuleItem	Attribute "ModuleItemTarget" of element "ModuleItemRef"
SubmoduleItem	Attribute "SubmoduleItemTarget" of element "SubmoduleItemRef"
VirtualSubmoduleItem	Not used yet
InterfaceSubmoduleItem	Not used yet
PortSubmoduleItem	Attribute "SubmoduleItemTarget" of element "SubmoduleItemRef" for PortSubmoduleItems placed in the SubmoduleList Not used yet for PortSubmoduleItems placed in the SystemDefinedSubmoduleList
ValueItem	Attribute "ValueItemTarget" of all "Ref" elements.
Ref	Attribute "ParameterTarget" of the "ParameterRef" element.
MenuItem	Attribute "MenuTarget" of the "MenuRef" element.
GraphicItem	Attribute "GraphicItemTarget" of the "GraphicItemRef" element.
CategoryItem	Attributes "CategoryRef" and "SubCategory1Ref" of the "ModuleInfo" element.

A.3.7 Language support

The language support is based on the concepts of XML. Language dependent strings can be kept within the GSD or separated in another file. Both strategies can be combined.

508 The different strings are placed in the form of dictionaries inside the GSD. Every language dependent text
 509 shall have a "TextId" attribute referencing an entry of a dictionary.

510 EXAMPLE 1

```

511 <ChannelDiagItem ErrorType="256">
512   <Name TextId="ID_COMM_ERROR"/>
513 </ChannelDiagItem>
514 <ExternalTextList>
515   <PrimaryLanguage>
516     <Text TextId="ID_COMM_ERROR" Value="Communication error"/>
517   </PrimaryLanguage>
518   <Language xml:lang="de">
519     <Text TextId="ID_COMM_ERROR" Value="Kommunikationsfehler"/>
520   </Language>
521   <Language xml:lang="fr">
522     <Text TextId="ID_COMM_ERROR" Value="Erreur de communication"/>
523   </Language>
524 </ExternalTextList>

```

525 The ExternalTextList shall have an element PrimaryLanguage. The text strings defined within the
 526 PrimaryLanguage element are used if a text string is missing in the selected language. For a GSD the
 527 primary language shall be English.

528 The Language elements of the ExternalTextList shall have an attribute "xml:lang" for identification of the
 529 selected language. The codes for the representation of names of languages shall comply with ISO 639-
 530 1:2002 (two-letter-codes). Since the language of the "PrimaryLanguage" block is English, a separate
 531 "Language" element with English language shall not be used.

532 In addition strings can reside in an external file – no changes are needed in the GSD itself to support a
 533 new language. The name of the external file shall be built by the name of the corresponding GSD name
 534 appending the string "-Text-" and the ISO 639-1:2002 compliant two letter code.

535 EXAMPLE 2 "GSDML-V2.3-Vendor-Device-2011031-Text-de.xml"

536 External files shall be located relative to the GSD in a subdirectory. The name of the subdirectory shall be
 537 built of the two letter language code of the language of the external file. Because the external text files are
 538 located one level deeper in the directory hierarchy, the schema location shall use the relative path
 539 "..\..\xsd".

540 Example for an external text file:

```

541 <?xml version="1.0" encoding="UTF-8"?>
542 <ExternalTextDocument xml:lang="de"
543   xmlns="http://www.profibus.com/GSDML/2003/11/Primitives"
544   xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
545   xsi:schemaLocation="http://www.profibus.com/GSDML/2003/11/Primitives ..\..\xsd\GSDML-
546   Primitives-v2.3.xsd">
547   <Text TextId="IDT_FamilyDescription" Value="Dieser Text beschreibt die
548   Eigenschaften einer Gerätefamilie"/>
549   <Text TextId="IDT_MODULE_NAME_DAP1" Value="PNO Beispiel DAP - EA Daten"/>
550   <Text TextId="IDT_INFO_DAP1" Value="Hier können weitere Informationen zum DAP in
551   lesbarer Form stehen"/>
552   <Text TextId="IDT_MODULE_NAME_Module1" Value="Ein/Ausgangsmodul"/>
553   <Text TextId="IDT_INFO_Module1" Value="Beschreibung des Ein/Ausgabemoduls"/>
554   <Text TextId="IDT_DATAITEM_Status" Value="Status"/>
555   <Text TextId="IDT_DATAITEM_Value" Value="Messwert"/>
556   <Text TextId="IDT_DATAITEM_Digital Outputs" Value="Binäre Ausgänge"/>
557 </ExternalTextDocument>

```

558 The coding of the XML file (e.g. UTF-8) is not regulated by the GSDML. Any coding compliant with the
 559 rules of the XML specification REC-xml-20081126 is allowed.

Attributes may contain all characters that are allowed by the chosen XML coding unless restricted by the GSDML schema or specification. Note however, that the characters <, >, &, ' and " shall be escaped according to the XML rules as <; >; &; '; and ";, respectively.

563 A.3.8 Notation of elements and attributes for schema extensions

564 In GSDML schema extensions the names of elements and attributes shall be composed as follows:

- 565 1. First character shall be either a decimal number ("0" to "9") or a capital letter in the range "A" to
566 "Z".
- 567 2. The following characters shall be in the range "0" to "9" or "a" to "z". Capital letters in the Range
568 "A" to "Z" can be used in order to improve the readability.
- 569 3. Abbreviations shall be used in capital form followed by an "_" character when one or more
570 characters are following.
- 571 4. The first character after the "_" character shall be either a decimal number ("0" to "9") or a capital
572 letter in the range "A" to "Z"

573 NOTE In GSDML the "_" character is often used to separate parts of keywords in order to make the keywords
574 more readable. Sometimes upper and lower case letters were used for the same purpose.

575 A.3.9 Handling of defaults

576 It is highly recommended that engineering tools check the GSD for compliance with this specification
577 before using it. Such a check is done in two steps:

- 578 1. Validation with the GSDML schema
- 579 2. Checks which can't be expressed in XML Schema and thus must be programmed (business
580 logic).

581 Both steps may be very time-consuming. So it is wise to do these checks only once, e.g. in an "install" or
582 "import" action in the engineering tool and use the installed GSDs without any checks later.

583 Reading the GSD file with or without schema validation influences the handling of default values, so care
584 must be taken:

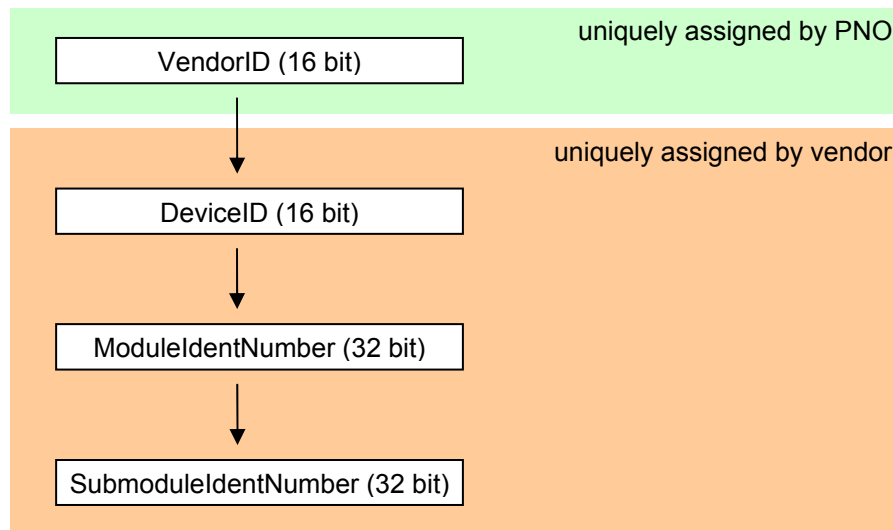
- 585 • With schema validation, optional attributes that have a default value will be automatically added
586 with their default value. In previous versions of the GSDML schema there were situations where
587 the behaviour of the engineering tool depended on the presence of an optional attribute, but there
588 was a default value in the schema for this attribute. To find out whether the attribute actually was
589 in the GSD or was added by the default in the schema, the post-schema-validation info set (PSVI)
590 must be consulted.
- 591 • Without schema validation, optional attributes that have a default value will not be added. All
592 defaults must be programmed in the engineering tool.
- 593 • Some attributes have defaults which can't be expressed in XML schema. These have to be
594 programmed in the engineering tool regardless of schema validation.

595 A.3.10 PROFINET IO Device Identification

596 A.3.10.1 PROFINET IO specification requirements

597 Identification numbers

598 The PROFINET IO specification establishes the following hierarchy of identification numbers:



600 **Figure 20 — Hierarchy of identification numbers**

601 Note: Instance (16 bit), SlotNumber (16 bit) and SubslotNumber (16 bit) are only used for addressing, but
 602 not for identification purposes.

603 Using these identification numbers it shall be possible to uniquely identify each hardware item from the
 604 engineering tool's perspective. In other words, an engineering tool shall be able to find the appropriate
 605 device description for the hardware item. Different hardware items may obtain identical identification
 606 numbers when they only differ in properties which are not relevant for engineering, e.g. certificates,
 607 environmental conditions, mechanical connections. In this case, the field ORDER_ID of the I&M0 data
 608 should be different to identify the exact hardware item.

609 That means:

- 610 • The DeviceID shall uniquely designate the hardware of a specific vendor.
- 611 • The ModuleIdentNumber shall uniquely designate all modules which may appear in a specific
 612 device.
- 613 • The SubmoduleIdentNumber shall uniquely designate all submodules which may appear in a
 614 specific module.

615 For exceptions to this rule, see below.

616 Please note:

- 617 • If a module can be placed in different devices, the ModuleIdentNumber shall be assigned
 618 uniquely across all these devices.
- 619 • If a submodule can be placed in different modules, the SubmoduleIdentNumber shall be assigned
 620 uniquely across all these modules.

621 For the SubmoduleIdentNumber there is an additional rule:

622 This number does not only identify the submodule hardware, but also the representation of the cyclic data
 623 (i.e. length and structure). If the same hardware submodule may have different representations of cyclic
 624 data, the submodule shall have different SubmoduleIdentNumbers for each of the possible
 625 representations. Note that each of these SubmoduleIdentNumbers shall still be able to identify the
 626 specific submodule, so each of these ident numbers shall be different from the ident numbers of other
 627 submodule hardware on the same module.

628 So, while the VendorID, DeviceID and ModuleIdentNumber shall be one-to-one, i.e. there can be no more
629 than one ident number for the same hardware, the submodule may have several unique
630 SubmoduleIdentNumbers.

631 In case a vendor's portfolio includes PROFINET IO devices with a modular hardware configuration it may
632 be useful to extend the vendor-unique assignment of DeviceID numbers also to ModuleIdentNumbers
633 and SubmoduleIdentNumbers to prevent conflicts. The vendor may assign numbers individually for each
634 new hardware or he may assign non-overlapping number ranges in advance to the different departments
635 for decentralized assignment.

636 All these identification numbers are stored in the hardware and can be read out, e.g. via the RPC Object-
637 UUID (Vendor- and DeviceID), via the I&M functions (Vendor- and DeviceID) and via
638 RealIdentificationData (ModuleIdentNumber and SubmoduleIdentNumber).

639 Exceptions

640 There are some cases where a unique identification of hardware is just impossible.

641 a) Existing modular devices become PROFINET IO Devices

642 Modular, non-PROFINET IO fieldbus devices can be made to PROFINET IO devices by just exchanging
643 the head module. In this case, existing modules may only support insufficient identification information or
644 none at all (depending on the requirements of the previously supported fieldbus).

645 Even newly designed modules are exempted from supporting appropriate identification information, if the
646 protocol between the modules is not able to transfer it and extending the protocol may break compatibility.

647 **Advice:** Do as best as you can. The PROFINET IO head module shall follow the rules because it is newly
648 designed and is not hampered by the protocol between the modules. Extend the protocol between the
649 modules if that does not break backward compatibility, so that at least newly designed modules will
650 identify properly. Use whatever information is available to distinguish the different (sub)modules as their
651 ident numbers.

652 b) Gateways (Links)

653 Devices (or modules) which integrate other networks (e.g. other fieldbuses) into PROFINET IO are called
654 Links (or Linking Modules). These Links map other devices into the PROFINET IO address space, either
655 as proxy devices, proxy modules or as proxy submodules.

656 It is clear that, depending on the properties of the other network, identification information may be
657 insufficient, or too big to fit into the PROFINET IO ident numbers.

658 What to do in this case will be detailed in the Guideline "*Fieldbus integration in PROFINET IO*" and its
659 various amendments for specific fieldbuses.

660 A.3.10.2 Effects on the GSD

661 Identification numbers

662 The GSD file for a PROFINET IO device shall contain exactly these identification numbers discussed
663 before that are readable from the hardware.

664 The set of **Vendor-** and **DeviceID** designates a device one-to-one. This set can be used by an
665 engineering tool to find the GSD file which matches the device. Thus for each such set there may be only
666 exactly one current GSD file.

667 Head modules are modelled as special modules, the DeviceAccessPoints (DAPs). The DAPs contain the
668 PROFINET interface and carry the slots, in which "normal" modules can be plugged.

669 Regarding the **ModuleIdentNumber** the DAPs are not different from “normal” modules. That means, in a
 670 GSD file the ModuleIdentNumber shall be kept unique across all DAPs and all pluggable modules.

671 An automated check of the uniqueness of the ModuleIdentNumber could easily be achieved e.g. by
 672 means of XML schema. Unfortunately, there are legal exceptions that break uniqueness. Therefore a
 673 check by means of XML schema or PROFINET XML-Viewer is impossible.

674 The **SubmoduleIdentNumber** appears in different places inside the GSDML because submodules may
 675 be defined in different places inside the GSDML:

- 676 • Interface- and PortSubmodule at the DAP in the SystemDefinedSubmoduleList
- 677 • PortSubmodule at “normal” modules in the SystemDefinedSubmoduleList
- 678 • “normal”, built-in submodules at the DAP or modules in the VirtualSubmoduleList
- 679 • PortSubmodules and “normal” submodules in the (global) SubmoduleList

680 Only the submodules from the SubmoduleList are described globally and associated to the DAPs /
 681 modules via reference. They exist therefore only once per type (class). All other submodules are
 682 described at the respective DAP / module and shall therefore be described separately for each DAP /
 683 each module and each occurrence (per instance).

684 For each DAP und each module, the SubmoduleIdentNumbers of all submodules from their
 685 SystemDefinedSubmoduleList, their VirtualSubmoduleList, plus all submodules from the (global)
 686 SubmoduleList pluggable according to the plugging rules (“UseableSubmodules”), shall be kept unique.

687 Recommendation: Different submodules inside the GSD file should have different 688 SubmoduleIdentNumbers regardless on which DAP or module they are defined and regardless whether a 689 conflict will be prevented by the plugging rules.

690 If the submodule allows to change the representation of cyclic data via parameterization, then there shall
 691 be a separate submodule description in the GSD for each representation. This description shall contain
 692 the SubmoduleIdentNumber which matches the representation, and the ParameterRecord shall contain
 693 the setting of the representation in a hard-coded way. The name of the hardware item will then be
 694 complemented by a name of the representation of the cyclic data. The submodule has multiple entries in
 695 the device catalog. The selection of the representation of the cyclic data is done by selecting the
 696 appropriate entry of the catalog.

697 An automated check for uniqueness of the SubmoduleIdentNumber is very difficult because of the
 698 heterogenous possibilities of describing a submodule and also because of the formation of subsets by the
 699 slot rules. In addition, there are exceptions which defeat uniqueness. Therefore, an automated check by
 700 e.g. XML schema or PROFINET XML-Viewer is impossible.

701 Exceptional case: Some hardware items have ParameterRecords where changing the first part (the
 702 “selector”) completely changes the structure of the rest of the ParameterRecord. Such a dependency
 703 can’t be described with GSD means. The work-around for this situation is to have a separate (sub)module
 704 description for each setting of the selector, i.e. for each different structure of the ParameterRecord, and
 705 including the selector in a hard-coded way. The name of the hardware item will then be complemented by
 706 a name of the setting of the selector. The situation is very similar to the one described above
 707 (representation of cyclic data changeable), but here the SubmoduleIdentNumber will not change.

708 Recommendation: Hardware / firmware should be built in such a way that the structure of the 709 ParameterRecords is fixed.
--

710 Other IDs

711 Apart from the ModuleIdentNumber and SubmoduleIdentNumber the following holds for a GSD file:

712 The XML technology initially only allows a strictly hierarchical structure. If a more complex structure
 713 needs to be represented, additional relations between the objects need to be implemented as references.
 714 This is what the “**ID**” attributes are for in the GSD file. These IDs give the objects a name which can be
 715 used for referencing from other objects. The IDs do not exist on the hardware item and are not displayed
 716 in the engineering tool. These IDs shall be kept unique for each object type; this is also checked by the
 717 XML schema.

718 The language-dependent text referenced by the “**ModuleInfo/Name/@TextId**” attribute serves for the
 719 catalog entry. Naturally, the name shall be kept unique for all supported languages (at least within the
 720 category or subcategory), so that the user can choose the correct hardware item.

721 **A.3.11 PROFINET IO Diagnosis**

722 PROFINET IO uses AlarmItems to signal diagnosis information (see PNO-2.722 *Application Layer*
 723 *protocol*).

724 The alarms can be roughly grouped into

- 725 • manufacturer specific alarms
- 726 • profile specific alarms
- 727 • system defined alarms.

728 An engineering tool needs to know the structure of the AlarmItems and additional texts for displaying the
 729 diagnostic information in a human readable manner.

730 Manufacturer specific alarms shall be described in the GSDML file of a device.

731 Profile specific alarms can't be implemented in the engineering tool because otherwise a new profile
 732 would require an update of the tool. A profile specific description (separate file) would be nice, but is
 733 currently not available. So all profile specific alarms produced by a device shall be described in its
 734 GSDML file, too.

735 System defined alarms are considered “well-known” and stable; their structure and additional texts should
 736 be implemented in the engineering tool. Changes in system defined alarms require an update of the tool.

737 This following table shows which of the AlarmItem types are described in the GSDML file by which
 738 elements:

739 **Table 5 — AlarmItem types**

UserStructureIdentifier	ChannelErrorType	ExtChannelErrorType	GSDML Element
0x0000-0x7FFF ManufacturerSpecific	–	–	UnitDiagTypeList/UnitDiagTypeItem, enumerated by UserStructureIdentifier
0x8000 ChannelDiagnosis	0x0000-0x000E Reserved or system defined	–	–
	0x000F-0x001F Manufacturer specific, but with recommendation	–	ChannelDiagList/ChannelDiagItem (no ExtChannelDiagList), enumerated by (Channel)ErrorType
	0x0020-0x00FF Reserved for common profiles	–	ChannelDiagList/ChannelDiagItem (no ExtChannelDiagList), enumerated by (Channel)ErrorType
	0x0100-0x7FFF Manufacturer specific	–	ChannelDiagList/ChannelDiagItem (no ExtChannelDiagList), enumerated by (Channel)ErrorType
	0x8000-0x8FFF Reserved or system defined	–	–
	0x9000-0x9FFF Reserved for profiles	–	ChannelDiagList/ProfileChannelDiagItem (no ExtChannelDiagList), enumerated by API and

UserStructureIdentifier	ChannelErrorType	ExtChannelErrorType	GSDML Element
			(Channel)ErrorType
	0xA000-0xFFFF Reserved	–	–
0x8001 Multiple	Will be split into single channel diagnosis alarms and decoded accordingly.		
0x8002 ExtChannelDiagnosis	0x0000-0x000E Reserved or system defined	–	–
	0x000F-0x001F Manufacturer specific, but with recommendation	0x0001-0x7FFF Manufacturer specific	ChannelDiagList/ChannelDiagItem, enumerated by (Channel)ErrorType, with ExtChannelDiagList/ExtChannelDiagItem, enumerated by (ExtChannel)ErrorType
	0x0020-0x00FF Reserved for common profiles	0x0001-0x7FFF Manufacturer specific	ChannelDiagList/ChannelDiagItem, enumerated by (Channel)ErrorType, with ExtChannelDiagList/ExtChannelDiagItem, enumerated by (ExtChannel)ErrorType
	0x0100-0x7FFF Manufacturer specific	0x0001-0x7FFF Manufacturer specific	ChannelDiagList/ChannelDiagItem, enumerated by (Channel)ErrorType, with ExtChannelDiagList/ExtChannelDiagItem, enumerated by (ExtChannel)ErrorType
	0x8000-0x8FFF Reserved or system defined	–	–
	0x9000-0x9FFF Reserved for profiles	0x9000-0x9FFF Reserved for profiles	ChannelDiagList/ProfileChannelDiagItem, enumerated by (Channel)ErrorType, with ExtChannelDiagList/ProfileExtChannelDiagItem, enumerated by API and (ExtChannel)ErrorType
	0xA000-0xFFFF Reserved	–	–
0x8003 QualifiedChannelDiagnosis	The structure of the QualifiedChannelDiagnosis is that of the ExtChannelDiagnosis except for an additional system defined QualifiedChannelQualifier. So the QualifiedChannelDiagnosis is described with the same GSDML elements as the ExtChannelDiagnosis.		
0x8004-0x8FFF Reserved or system defined	–	–	–
0x9000-0x9FFF Reserved for profiles	–	–	UnitDiagTypeList/ProfileUnitDiagTypeItem, enumerated by API and UserStructureIdentifier
0xA000-0xFFFF Reserved	–	–	–

740

741 Recommendations for Diagnosis

742 Diagnosis shall be as specific as possible. Whenever possible, standard (system-defined) diagnosis shall
 743 be used. These principles lead to the following list, showing the diagnosis options in decreasing
 744 recommendation:

- 745 • Channel diagnosis (well defined)
- 746 • Channel diagnosis (manufacturer specific)
- 747 • Extended channel diagnosis (well defined)
- 748 • Extended channel diagnosis (manufacturer specific)
- 749 • As a last resort: Manufacturer specific diagnosis (unit diagnosis)

750 Diagnosis is not appropriate for:

- 751 • Configuration errors – these must be reported via the ModDiffBlock
- 752 • Data errors – these must be reported via the Qualifier

A.3.12 Slot Cluster concept

The upcoming version 2.0 of the specification *Fieldbus integration in PROFINET IO* describes how devices of other fieldbuses can be integrated into a PROFINET environment via linking devices:

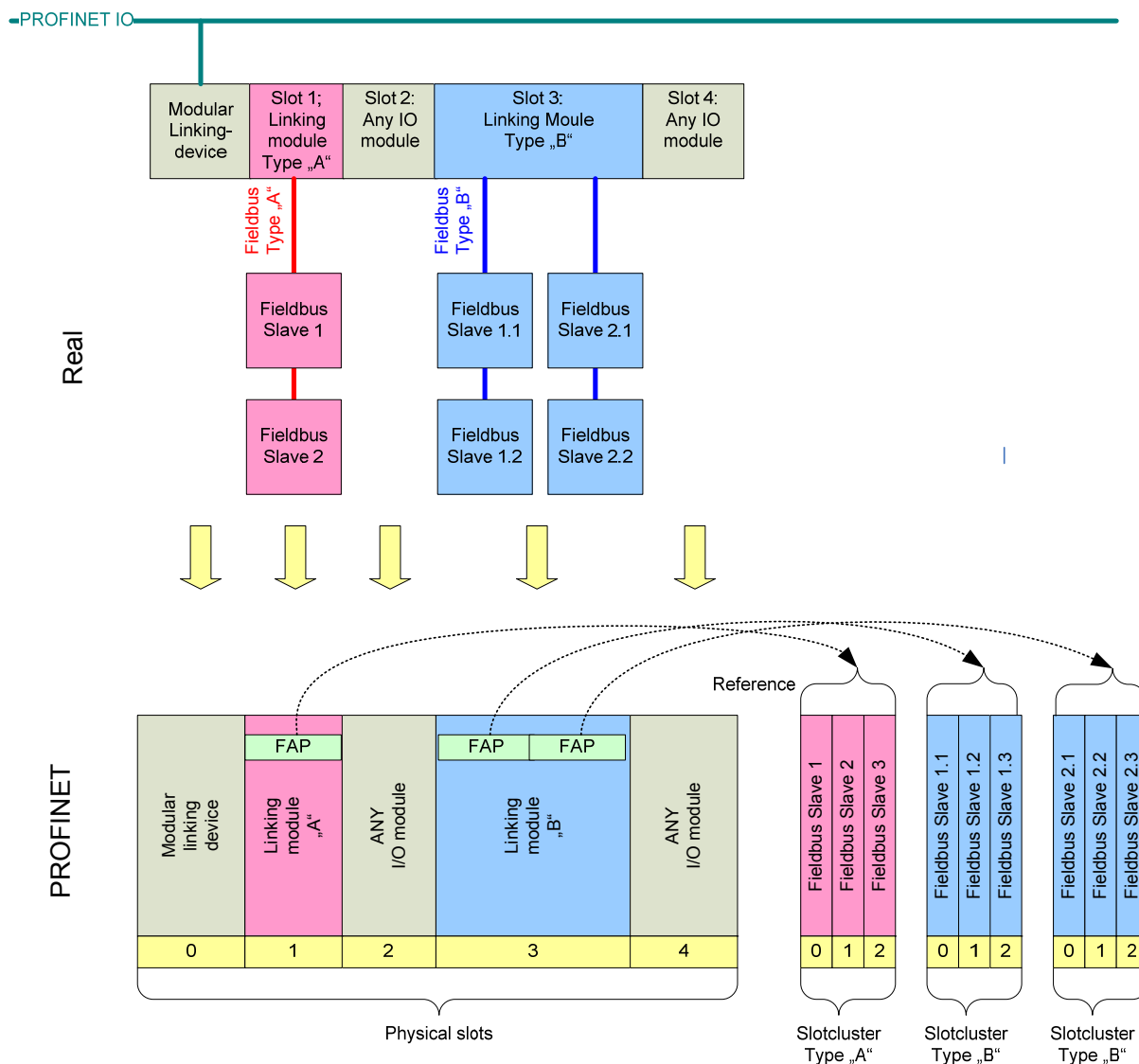


Figure 21 — Mapping of subordinated fieldbus devices to PROFINET

The integration uses modular mapping, i.e. each device of a subordinated fieldbus is represented as a module in PROFINET. If the device has any substructures, these are mapped to submodules inside the fieldbus integration module.

It must be prevented that fieldbus integration modules are configured into physical slots and vice versa. To achieve this, there is a new type of slot – fieldbus integration slots. These slots are in a different, continuous, device specific slot number range above the physical slots. The position and size of this range is specified in the element `DeviceAccessPointItem/FieldbusIntegrationSlots`. “Normal” modules may only be configured into physical slots, according to the rules in element `DeviceAccessPointItem/UseableModules`.

The following figure shows the details of a linking module:

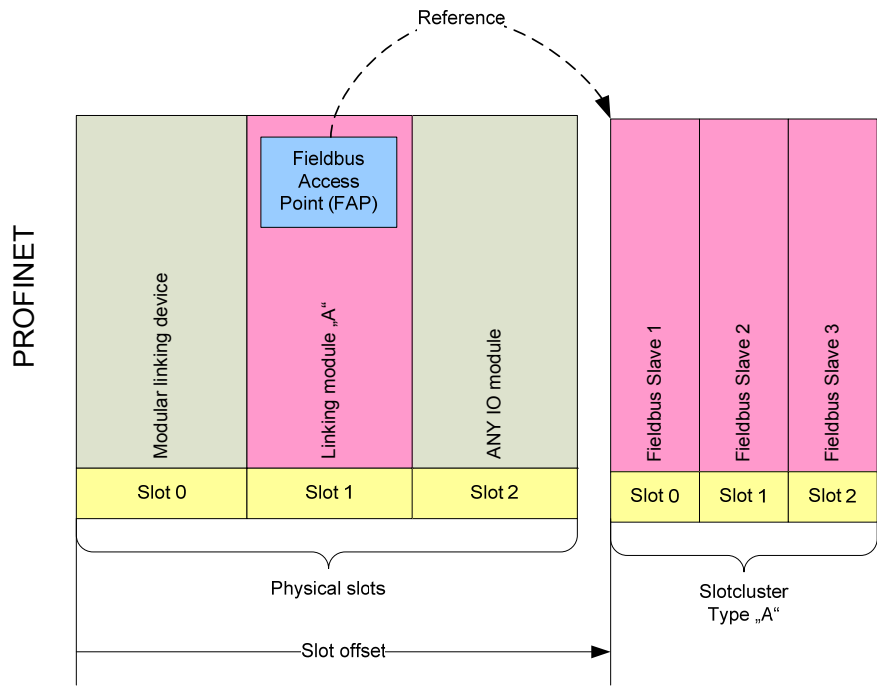


Figure 22 — Relation between FAP and Slot Cluster

For each fieldbus line that goes out of the linking module, there must be a dedicated submodule called FAP (Fieldbus Access Point). In the GSD, a submodule becomes a FAP by specifying the element “SlotCluster”. This element describes the type of the fieldbus line (“FieldbusType”) and the size of a contiguous range of fieldbus integration slots – the slot cluster – that must be provided for this line. The fieldbus type (e.g. PROFIBUS, INTERBUS, ..) is defined as the profile specific type of the fieldbus integration profile. A link to a list of these types will be added to the table http://www.profibus.com/IM/Profile_ID_Table.xml in line “Fieldbus Integration”, row “Profile Specific Type Specification” after the release of the profile.

When configuring the linking module, for each FAP the engineering tool shall find an unused slot number range in the range of fieldbus integration slots and assign this range to the FAP. A reference to the assigned slot cluster is downloaded to each FAP of the linking device and may be read out by a profile-specific service at runtime. Slot numbers are often referenced in the application program, so they must be kept unchanged when possible. Thus, when a linking module is unconfigured, the slot cluster of each FAP will be freed, but other slot clusters of the linking device shall not be moved to close the gap in the slot numbers. The engineering tool shall provide a user action for moving / compacting slot clusters instead.

Engineering tools should display the slot clusters as separate tables. In this case, the slots in a slot cluster shall be numbered relative to the slot offset starting with 0.

A module becomes a fieldbus integration module by specifying the attribute “FieldbusType”. This module may be configured into any fieldbus integration slot which is part of a slot cluster of the same fieldbus type.

A.4 Element description

A.4.1 General

Regular Expressions are used as defined in REC-xml-20081126. The notation of the elements location is according REC-xpath20-2010121.

The XML schema language REC-xmlschema-1-20041028 allows “true”, “false” but also “1”, “0” as valid content for boolean datatypes. In this document the terms “true” and “false” also stand for “1” and “0”.

The column “SV” describes in which schema version an attribute was introduced.

A.4.2 DeviceIdentity related elements

A.4.2.1 DeviceIdentity

This element contains general information about a device.

Location:

///ISO15745Profile/ProfileBody/DeviceIdentity

Use: Required

Each element shall contain attributes as shown in Table 6.

Table 6 — Attributes of element DeviceIdentity

Name	Type	Use	SV	Description
VendorID	Unsigned16hex	Required	1.0	Contains the vendor specific part of the Device Ident Number as defined in PNO-2.712 <i>Application Layer services</i> . The VendorID is assigned by vendor association. Manufacturers of devices have to apply for the VendorID at the appropriate vendor association. For PROFINET the responsible association to assign a unique VendorID is the tradename owner.
DeviceID	Unsigned16hex	Required	1.0	Contains the device specific part of the DeviceIdentNumber. The DeviceID is assigned by the vendor. It uniquely identifies a device amongst all devices of this vendor.

A.4.2.2 InfoText

This element contains human readable additional text information about a device.

Location:

///ISO15745Profile/ProfileBody/DeviceIdentity/InfoText

Use: Required.

Each element shall contain the attribute as shown in Table 7.

Table 7 — Attributes of element InfoText

Name	Type	Use	SV	Description
TextId	RefId	Required	1.0	Contains the ID of a text as a reference into the “ExternalTextList”.

814 A.4.2.3 VendorName

815 This element contains the name of the device vendor.

816 Location:

817 //ISO15745Profile/ProfileBody/DeviceIdentity/VendorName

818 Use: Required

819 Each element shall contain the attribute as shown in Table 8.

820 **Table 8 — Attributes of element VendorName**

Name	Type	Use	SV	Description
Value	Token	Required	1.0	Contains the name of the device vendor.

821

822 A.4.3 DeviceFunction related elements**823 A.4.3.1 DeviceFunction**

824 The DeviceFunction element shall contain the element "Family".

825 Location:

826 //ISO15745Profile/ProfileBody/DeviceFunction

827 Use: Required

828 Attributes: None

829 A.4.3.2 Family

830 The device shall be assigned to a function class. In addition to the main family, a device can be assigned
831 to a vendor specific product family.

832 Location:

833 //ISO15745Profile/ProfileBody/DeviceFunction/Family

834 Use: Required

835 Each element shall contain at least one attribute out of Table 9.

836 **Table 9 — Attributes of element Family**

Name	Type	Use	SV	Description
MainFamily	Enumeration	Required	1.0	Contains the assignment to a function class. On of the following values are allowed: "General" "Drives" "Switching Devices" "I/O" "Valves" "Controllers" "HMI" "Encoders" "NC/RC" "Gateway" "PLCs" "Ident Systems" "PA Profiles"

Name	Type	Use	SV	Description
				"Network Components" "Sensors" The function class that best fits to the device shall be chosen. There is no fixed mapping between these classes and the application profiles.
ProductFamily	NormalizedString	Optional	1.0	Contains the vendor specific assignment of the device to a product family.

837 **A.4.4 ApplicationProcess related elements**

838 **A.4.4.1 DeviceAccessPointList**

839 A GSD shall contain information about one or more different access points of the same family. This
 840 element contains the list of specified DAP.

841 Location:

842 //ISO15745Profile/ProfileBody/ApplicationProcess/DeviceAccessPointList

843 Use: Required

844 Attributes: None

845 **A.4.4.2 DeviceAccessPointItem**

846 Location:

847 //ISO15745Profile/ProfileBody/ApplicationProcess/DeviceAccessPointList/DeviceAccessPointItem

848 Use: One or more

849 Attributes and child elements: See A.4.5.1

850 **A.4.4.3 ModuleList**

851 This list contains all modules that are described within this GSD.

852 Location:

853 //ISO15745Profile/ProfileBody/ApplicationProcess/ModuleList

854 Use: Optional

855 Since up to GSDML V2.0 this element is required, a GSD file without this element may be rejected by an
 856 engineering tool without GSDML V2.1 support.

857 Attributes: None

858 **A.4.4.4 ModuleItem**

859 This element contains subelements to describe the properties of a module.

860 Location:

861 //ISO15745Profile/ProfileBody/ApplicationProcess/ModuleList/ModuleItem

862 Use: One or more

863 Attributes and child elements: See A.4.6.1

864 A.4.4.5 SubmoduleList

865 The SubmoduleList element describes all submodules which can be placed on subslots of a DAP or
866 module by the engineering tool.

867 Location:

868 //ISO15745Profile/ProfileBody/ApplicationProcess/SubmoduleList

869 Use: Optional

870 Supported since GSDML version: 2.1

871 Attributes: None

872 A.4.4.6 Pluggable Submodule (abstract)

873 Describes a submodule which can be placed on subslots of a DAP or module by the engineering tool.

874 Either a SubmoduleItem or a PortSubmoduleItem.

875 Location:

876 //ISO15745Profile/ProfileBody/ApplicationProcess/SubmoduleList/SubmoduleItem or

877 //ISO15745Profile/ProfileBody/ApplicationProcess/SubmoduleList/PortSubmoduleItem

878 Use: One or more

879 Supported since GSDML version: 2.1

880 Attributes and child elements: See A.4.7.1 or A.4.9.1, respectively.

881 A.4.4.7 ValueList

882 The ValueList element contains elements for the assignment of numerical values to text strings.

883 Location:

884 //ISO15745Profile/ProfileBody/ApplicationProcess/ValueList

885 Use: Optional

886 Attributes: None

887 NOTE See A.4.10.3 for an example of the ValueList.

888 A.4.4.8 ValueItem

889 The ValueItem element groups all value objects and can be referenced from the "Ref/ValueItemTarget"
890 element.

891 Location:

892 //ISO15745Profile/ProfileBody/ApplicationProcess/ValueList/ValueItem

893 Use: One or more

894 Each element shall contain attributes as shown in Table 10.

895 **Table 10 — Attributes of element ValueItem**

Name	Type	Use	SV	Description
ID	Id	Required	1.0	Contains the ID of the ValueItem element to be referenced from the attribute "ValueItemTarget" of the "Ref" element.

896

897 **A.4.4.9 Help**

898 The Help element contains additional help information about the ValueItem parameter.

899 Location:

900 //ISO15745Profile/ProfileBody/ApplicationProcess/ValueList/ValueItem/Help

901 Use: Optional

902 Each element shall contain attributes as shown in Table 11.

903 **Table 11 — Attributes of element Help**

Name	Type	Use	SV	Description
TextId	RefId	Required	1.0	Contains the ID of a text as a reference into the "ExternalTextList".

904

905 **A.4.4.10 Assignments**

906 This element contains one or more "Assign" elements.

907 Location:

908 //ISO15745Profile/ProfileBody/ApplicationProcess/ValueList/ValueItem/Assignments

909 Use: Optional

910 Attributes: None

911 **A.4.4.11 Assign**912 The Assign element contains the assignment between the content of a parameter and the textual
913 representation.

914 Location:

915 //ISO15745Profile/ProfileBody/ApplicationProcess/ValueList/ValueItem/Assignments/Assign

916 Use: One or more

917 Each element shall contain attributes as shown in Table 12.

918 **Table 12 — Attributes of element Assign**

Name	Type	Use	SV	Description
Content	String	Required	1.0	Contains the value to assign to a text reference. Within an "Assignments" element all "Assign" child elements shall have different numerical values in their "Content" attribute. The value shall comply with the following regular expression: -?\d+(\.\d+)?([eE]-?\d+)? For the representation of values see Table 2.

Name	Type	Use	SV	Description
				Be careful: Support of floating point values was introduced to this attribute in GSDML version 2.2. Entering floating point values here breaks backward compatibility to engineering tools that don't support GSDML V2.2.
TextId	RefId	Required	1.0	Contains the ID of the language dependent text in the "ExternalTextList". Within an "Assignments" element the "TextId" attribute of all "Assign" child elements shall be unique.

919

920 **A.4.4.12 ChannelDiagList**

921 Specifies a list of channel type specific error texts.

922 Location:

923 //ISO15745Profile/ProfileBody/ApplicationProcess/ChannelDiagList

924 Use: Optional

925 Attributes: None

926 **A.4.4.13 ChannelDiagItem** (abstract)

927 Either a ChannelDiagItem or a ProfileChannelDiagItem.

928 Location:

929 //ISO15745Profile/ProfileBody/ApplicationProcess/ChannelDiagList/ChannelDiagItem or

930 //ISO15745Profile/ProfileBody/ApplicationProcess/ChannelDiagList/ProfileChannelDiagItem

931 Use: One or more

932 Attributes and child elements: See A.4.13.1 or A.4.13.2, respectively

933 **A.4.4.14 UnitDiagTypeList**

934 The UnitDiagTypeList assigns diagnostic values to manufacturer specific status and error messages.

935 Location:

936 //ISO15745Profile/ProfileBody/ApplicationProcess/UnitDiagTypeList

937 Use: Optional

938 Attributes: None

939 **A.4.4.15 UnitDiagTypeItem** (abstract)

940 Either a UnitDiagTypeItem or a ProfileUnitDiagTypeItem.

941 Location:

942 //ISO15745Profile/ProfileBody/ApplicationProcess/UnitDiagTypeList/UnitDiagTypeItem or

943 //ISO15745Profile/ProfileBody/ApplicationProcess/UnitDiagTypeList/ProfileUnitDiagTypeItem

944 Use: One or more

945 Attributes and child elements: See A.4.14.1 or A.4.14.2, respectively

946 A.4.4.16 LogBookEntryList

947 This element contains a list of descriptions for LogBook entries.

948 Location:

949 //ISO15745Profile/ProfileBody/ApplicationProcess/LogBookEntryList

950 Use: Optional

951 Supported since GSDML version: 2.3

952 Attributes: None

953 A.4.4.17 LogBookEntryItem

954 A LogBookEntryItem defines a manufacturer specific PNIOStatus in the LogBook (see PNO-2.722
955 *Application Layer protocol*, "Coding section related to PNIOStatus").

956 Location:

957 //ISO15745Profile/ProfileBody/ApplicationProcess/LogBookEntryList/LogBookEntryItem

958 Use: One or more

959 Supported since GSDML version: 2.3

960 Attributes and child elements: See A.4.15

961 GraphicsList

962 This element contains a list of GraphicItems.

963 Location:

964 //ISO15745Profile/ProfileBody/ApplicationProcess/GraphicsList

965 Use: Optional

966 Attributes: None

967 A.4.4.18 GraphicItem

968 A GraphicItem carries the information for the symbolic representation of a device, module, or submodule.

969 Location:

970 //ISO15745Profile/ProfileBody/ApplicationProcess/GraphicsList/GraphicItem

971 Use: One or more

972 Each element shall contain attributes as shown in Table 13.

973 **Table 13 — Attributes of element GraphicItem**

Name	Type	Use	SV	Description
ID	Id	Required	1.0	Contains the ID of the GraphicItem element to be referenced from the "GraphicItemRef" element.
GraphicFile	String	Required	1.0	Contains the file name without file extension. The format depends on the type of the GraphicItem which is defined by the "Type" attribute of the "GraphicItemRef" element. The name shall be composed of the four fields below in the

Name	Type	Use	SV	Description
				following order: 1) "GSDML" 2) The VendorID as defined in A.4.2.1 without "0x" prefix. 3) The DeviceID as defined in A.4.2.1 without "0x" prefix. 4) A vendor specific extension As a delimiter between the fields the dash character "-" (ASCII 45 decimal) shall be used. Left-hand zeros of VendorID and DeviceID shall be used.

974

975 **A.4.4.19 GraphicsList/GraphicItem/Embedded**

976 This element is used to describe the graphical information inside the GSD in Scalable Vector Graphics
 977 (SVG) format (see WD-SVG11-20110512).

978 NOTE SVG is a language for describing two-dimensional vector and mixed vector/raster graphics in
 979 XML.

980 Location:

981 //ISO15745Profile/ProfileBody/ApplicationProcess/GraphicsList/GraphicItem/Embedded

982 Use: Optional

983 Attributes: None

984 **A.4.4.20 CategoryList**

985 This element contains a list of CategoryItems.

986 NOTE 1 GSDML allows building of module and submodule categories. These categories can be used to group the
 987 modules and submodules within a catalog of an engineering tool. For example all analog input modules can be
 988 placed in one section of the catalog. This makes it easier to find the required module for the user or the engineering
 989 tool.

990 NOTE 2 Assigning module does not impact the runtime characteristics of a module or submodule.

991 Location:

992 //ISO15745Profile/ProfileBody/ApplicationProcess/CategoryList

993 Use: Optional

994 Attributes: None

995 **A.4.4.21 CategoryItem**

996 A CategoryItem defines the information within a single category.

997 Location:

998 //ISO15745Profile/ProfileBody/ApplicationProcess/CategoryList/CategoryItem

999 Use: One or more

1000 Each element shall contain attributes as shown in Table 14.

1001

Table 14 — Attributes of element CategoryItem

Name	Type	Use	SV	Description
ID	Id	Required	1.0	Contains the ID of the CategoryItem element to be referenced from the attribute "CategoryRef" or "SubCategory1Ref" of the "ModuleInfo" element.
TextId	RefId	Required	1.0	Contains the ID of a text as a reference into the "ExternalTextList".

1002

1003 A.4.4.22 InfoText

1004 This element contains human readable additional text information about a category or subcategory 1.

1005 Location:

1006 //ISO15745Profile/ProfileBody/ApplicationProcess/CategoryList/CategoryItem/InfoText

1007 Use: Optional

1008 Supported since GSDML version: 2.2

1009 Each element shall contain attributes as shown in Table 15.

1010

Table 15 — Attributes of element InfoText

Name	Type	Use	SV	Description
TextId	RefId	Required	2.2	Contains the ID of a text as a reference into the "ExternalTextList".

1011

1012 A.4.4.23 ExternalTextList

1013 The ExternalTextList contains language dependent text strings.

1014 Location:

1015 //ISO15745Profile/ProfileBody/ApplicationProcess/ExternalTextList

1016 Use: Required

1017 Attributes: None

1018 A.4.4.24 PrimaryLanguage

1019 The PrimaryLanguage element contains the text definitions of the primary language which shall be used,
1020 if a text in the selected language is not available. For GSDML the primary language is English.

1021 Location:

1022 //ISO15745Profile/ProfileBody/ApplicationProcess/ExternalTextList/PrimaryLanguage

1023 Use: Required

1024 Attributes: None

1025 A.4.4.25 Text

1026 The Text element contains the language dependent text and can be referenced by the TextId attribute of
1027 other elements.

- 1028 Location:
 1029 //ISO15745Profile/ProfileBody/ApplicationProcess/ExternalTextList/PrimaryLanguage/Text
- 1030 Use: One or more
- 1031 Attributes: See table Table 16.

1032 **Table 16 — Attributes of element Text**

Name	Type	Use	SV	Description
TextId	Id	Required	1.0	Contains the unique ID for referencing a text. The TextId shall be kept unique within the parent element "PrimaryLanguage" or "Language"
Value	String	Required	1.0	Contains the language dependent text.

1033

1034 **A.4.4.26 Language**

- 1035 The Language element contains the text definitions of the specified language.

- 1036 Location:
 1037 //ISO15745Profile/ProfileBody/ApplicationProcess/ExternalTextList/Language

- 1038 Use: One for each language.

- 1039 Attributes: See Table 17.

1040 **Table 17 — Attributes of element Language**

Name	Type	Use	SV	Description
xml:lang	xsd:language	Required	1.0	Specifies the language of the text in subelement "Text/@Value". The two-letter language codes of ISO 639-1:2002 are allowed except for "en", because English texts shall be put into "PrimaryLanguage".

1041

1042 **A.4.4.27 Text**

- 1043 The Text element contains the language dependent text and can be referenced by the TextId attribute of
 1044 other elements.

- 1045 Location:
 1046 //ISO15745Profile/ProfileBody/ApplicationProcess/ExternalTextList/Language/Text

- 1047 Use: One or more

- 1048 Attributes: See Table 16.

1049 **A.4.5 DeviceAccessPoint related elements**

1050 **A.4.5.1 DeviceAccessPointItem**

- 1051 This element describes the characteristics of a DAP.

- 1052 Location:
 1053 //ISO15745Profile/ProfileBody/ApplicationProcess/DeviceAccessPointList/DeviceAccessPointItem

- 1054 Use: One for each DAP.

1055 Each element shall contain attributes as shown in Table 18.

1056 **Table 18 — Attributes of element DeviceAccessPointItem**

Name	Type	Use	SV	Description
ID	Id	Required	1.0	The ID of the element
ModuleIdentNumber	Unsigned32hex0	Required	1.0	The module ident number for identification purposes. The value "0x0" is reserved.
PNIO_Version	VersionString	Required	2.3	The version of the PROFINET IO Specification that this DAP implementation adheres to. Used for version-dependent sanity checks. Allowed values: All versions known today, i.e. "V1.0", "V2.0", "V2.1", "V2.2" and "V2.3", plus future (higher) versions to allow importing newer GSDs in older engineering tools. If the PNIO_Version is higher than the highest version known to the engineering tool, the tool shall do the sanity checks based on its highest known PNIO_Version.
PhysicalSlots	ValueList	Required	1.0	Which slots the DAP supports. The Slotnumber of the DAP itself shall be part of the list. The fieldbus integration slots shall not be part of the list.
MinDeviceInterval	Unsigned16	Required	1.0	The minimum interval time for sending cyclic IO data. The basic clock tick is 31.25 µs. The value of this element contains the multiplier of the basic clock tick. EXAMPLE: MinDeviceInterval = 32 means that the device is able to send / receive data every millisecond.
ImplementationType	NormalizedString	Optional	1.0	Description of the standard implementation in the DAP for example, standard software, controller or ASIC solution.
DNS_CompatibleName	String	Required	1.0	The default name of a device. This attribute may be used by the engineering tool to build a unique name of station. The allowed syntax is described in PNO-2.722 <i>Application Layer protocol</i> "Coding of the field NameOfStationValue" (see Note). The schema checks most of the rules by a regular expression pattern. In addition to this syntax, upper case letters are allowed for better readability. The engineering tool shall convert the name of station to lower case before sending it to the device.
ExtendedAddressAssignmentSupported	Boolean	Optional	1.0	Each IO Device shall implement the Discovery and basic Configuration Protocol (DCP), which may be used for assignment of the IP address. If the DAP supports any additional way of IP address assignment (e.g. DHCP), this attribute has to be set to "true". Default: "false"
AddressAssignment	TokenList	Optional	2.1	Which methods for IP address assignment are supported by the DAP. Each method shall be separated by a semicolon character. The following strings can be used for description: "DCP": Means that the DAP supports IP address assignment via Discovery and basic

Name	Type	Use	SV	Description
				Configuration Protocol (DCP). “DHCP”: Means that the DAP supports the Dynamic Host Configuration Protocol for IP address assignment. “LOCAL”: Means that the DAP supports a device specific method for IP address assignment. The first entry in this attribute shall be used as default by the engineering tool. If this attribute is missing, the default value is “DCP”. If this attribute is present, the attribute “ExtendedAddressAssignmentSupported” shall be ignored. EXAMPLE: “LOCAL;DCP” means that the device supports DCP protocol and also a device specific IP address setting. The device specific method shall be used as default.
AllowedInSlots	ValueList	Optional	1.0	Which slots the DAP can be placed in. If not defined, the DAP cannot be placed into other slots as defined in attribute “FixedInSlots”. If the DAP supports system redundancy device type “R2”, either FixedInSlots shall contain two slots (when system redundancy is mandatory) or FixedInSlots shall contain one slot and AllowedInSlots shall contain two slots: the one from FixedInSlots plus another one.
FixedInSlots	ValueList	Required	1.0	The slot number of the DAP when the device is configured in the engineering tool. For a non redundant DAP only one slot number is allowed in the list. It is recommended to use “0” as slotnumber. If the attribute “AllowedInSlots” is used, the slots defined in “FixedInSlots” shall be a subset of the slots defined by the attribute “AllowedInSlots”.
ObjectUUID_LocalIndex	Unsigned16	Required	1.0	The Instance field of the Object UUID. Note that for devices based on PROFINET IO Version ≥ 2.3 the bits 15-12 of this value should be the number of the interface which also appears in the SubslotNumber of the InterfaceSubmoduleItem as bits 11-8.
RequiredSchemaVersion	VersionString	Optional	2.0	The minimal version of the GSDML Schema needed to create valid configuration data for the DAP. If an engineering tool is not able to interpret this schema version it shall refuse to configure this DAP. For compatibility issues it is recommended to keep the RequiredSchemaVersion as low as possible. The RequiredSchemaVersion shall be no higher than the GSDML version that this GSD is based on. Default: “V1.0”
MultipleWriteSupported	Boolean	Optional	2.0	Defines, if the DAP is able to convey multiple Record Data objects within one APDU of the write service. Default: “false” This attribute shall be present and “true” if the PNIO_Version on the DAP is $\geq$ “V2.3” because IODWriteMultiple is mandatory since PN IO V2.3.
IOXS_Required	Boolean	Optional	2.0	Defines, if the DAP needs IOPS and IOCS information even for submodules which have no

Name	Type	Use	SV	Description
				cyclic IO data. Default: "true"
PhysicalSubslots	ValueList	Optional	2.1	This list describes which physical subslots are available in the DAP. Submodules can be assigned to these subslots by the engineering tool. If the DAP has "UseableSubmodules", this list shall be present and shall contain all subslots listed in "AllowedInSubslots", "UsedInSubslots" and "FixedInSubslots" of the "SubmoduleItemRef" elements. This list shall not include the subslot number of any submodule in the "VirtualSubmoduleList" or "SystemDefinedSubmoduleList" of the DAP.
RemoteApplicationTimeout	Unsigned16	Optional	2.1	This attribute represents the deadline of an RPC call in seconds. Default: "300"
MaxSupportedRecordSize	Unsigned32	Optional	2.2	This attribute describes the maximum size of a PN IO record that the DAP is able to receive and process, in octets. Used for speeding up communication by combining write requests. This is the size of "RecordDataRead" or "RecordDataWrite", i.e. the CL-RPC header, the PNIO header and the IODReadResHeader / IODWriteReqHeader are not included, while the BlockHeader of the RecordData is included. Shall be at least 4068. Default: "4068"
PowerOnToCommReady	Unsigned32	Optional	2.2	Defines the maximum time from power on until the DAP is ready to communicate (including switching functionality), in milliseconds. The value 0 means "undefined". Default: "0"
ParameterizationSpeedupSupported	Boolean	Optional	2.2	Defines that the DAP supports speedup of parameterization by starting up before receiving parameters using the last known parameters. Default: "false"
NameOfStationNotTransferable	Boolean	Optional	2.2	This attribute indicates that the DAP does not support device replacement without re-issuing the NameOfStation (e.g. the device does not have a transferable storage medium which contains the NameOfStation). Default: "false"
SharedDeviceSupported	Boolean	Optional	2.25	Defines, whether the DAP supports Shared Device. The feature shared device allows IO controllers to share the same IO device, by assigning submodules to distinct IO controllers. Only one IO controller at a time is able to use a particular output submodule. An input submodule may be used by multiple IO controllers only, if "SharedInputSupported" is set to true. Default: "false"
SharedInputSupported	Boolean	Optional	2.25	Defines, whether the DAP supports Shared Input. The feature shared input allows two IO controllers to share the same input submodule of a particular IO device. Default: "false"
DeviceAccessSupported	Boolean	Required	2.25	Defines, whether the DAP supports Device

Name	Type	Use	SV	Description
				Access. The feature DeviceAccess allows an IO supervisor to write data records to submodules of an IO device without establishing a full fledged application relation (i.e. with IOCR, ApplicationReady callback...). For further details please refer to PNO-2.722 <i>Application Layer protocol</i> .
WebServer	String	Optional	2.25	If this attribute is present, the device features a web server. The content of this attribute tells how the web server can be reached. It is basically an URL but has no host part and no user / password part. The host part is determined by a device instance and will be inserted later by the engineering tool when contacting a device instance. A user / password part does not make sense here because the GSD is distributed as plain text. But a web server may still require a user / password combination, in which case the engineering tool shall let the user input it. The value shall comply with the following regular expression: https?://(:[0-9]+)?(/[a-zA-Z0-9\-.~:/?#@\$'()*+,;=]*)*/? EXAMPLE: "http://:81/"
AutoConfigurationSupported	Boolean	Optional	2.3	Defines, whether the DAP supports AutoConfiguration. Default: "false"
CIR_Supported	Boolean	Optional	2.3	Defines, whether the DAP supports Configuration in Run using a temporary additional AR according to the Guideline Configure in Run for Distributed Control Systems. Default: "false"
PrmBeginPrmEndSequenceSupported	Boolean	Optional	2.3	Defines, whether the DAP supports the PrmBegin-PrmEnd sequence (re-parameterization sequence). Default: "false"
LLDP_NoD_Supported	Boolean	Optional	2.3	Defines, whether the DAP supports PDInterfaceAdjust and thus MultipleInterfaceMode.NameOfDevice. Default: "false" This attribute shall only be present when the PNIO_Version attribute at the DAP is >= "V2.3".
ResetToFactoryModes	ValueList	Optional	2.3	The DAP supports the DCP-Set ResetToFactory Block with suboption SubOptionResetToFactory if this attribute is present. The content describes the values of Bit 1-15 of the BlockQualifier that the DAP supports. For a device with PNIO_Version >= V2.3, values 2 and 8 are mandatory. See PNO-2.722 <i>Application Layer protocol</i> , "Coding of the field BlockQualifier".

1057

1058 A.4.5.2 ModuleInfo

1059 Location:

1060 //ISO15745Profile/ProfileBody/ApplicationProcess/DeviceAccessPointList/DeviceAccessPointItem/ModuleInfo
1061

1062 Use: Required

1063 Attributes and child elements: See A.4.12.1

1064 A.4.5.3 CertificationInfo1065 This element shall be filled out by the device manufacturer according to the aspired certificate before
1066 handing over the GSD to a PI test lab. After testing, the test lab shall adapt the content of this element
1067 according to the passed or failed tests.1068 Engineering tools may display this information together with the device's properties or may use it as
1069 additional filter criteria for the device catalog.

1070 Location:

1071 //ISO15745Profile/ProfileBody/ApplicationProcess/DeviceAccessPointList/DeviceAccessPointItem/CertificationInfo
1072

1073 Use: Optional

1074 Supported since GSDML version: 2.3

1075 Each element shall contain attributes as shown in Table 19.

1076 **Table 19 — Attributes of element CertificationInfo**

Name	Type	Use	SV	Description
ConformanceClass	String	Required	2.3	The value shall comply with the following regular expression: A B C
ApplicationClass	TokenList	Required	2.3	Allowed tokens: "Isochronous" and "ProcessAutomation"
NetloadClass	String	Optional	2.3	The value shall comply with the following regular expression: I II III (roman numerals 1,2 and 3)

1077

1078 A.4.5.4 ProfileList

1079 Location:

1080 //ISO15745Profile/ProfileBody/ApplicationProcess/DeviceAccessPointList/DeviceAccessPointItem/CertificationInfo/ProfileList
1081

1082 Use: Optional

1083 Supported since GSDML version: 2.3

1084 A.4.5.5 ProfileItem

1085 Location:

1086 //ISO15745Profile/ProfileBody/ApplicationProcess/DeviceAccessPointList/DeviceAccessPointItem/CertificationInfo/ProfileItem
1087

1088 Use: One or more

1089 Supported since GSDML version: 2.3

1090 Each element shall contain attributes as shown in Table 20.

1091 **Table 20 — Attributes of element ProfileItem**

Name	Type	Use	SV	Description
ProfileName	String	Required	2.3	The name of the profile. Use: "PROFI-safe", "PROFI-drive", "PROFI-energy", ... It is expected that future versions of the profile specifications will specify the name to be used here.
Class	String	Optional	2.3	Some profiles provide different classes (levels of support).

1092

1093 **A.4.5.6 SubslotList**

1094 Location:

1095 //ISO15745Profile/ProfileBody/ApplicationProcess/DeviceAccessPointList/DeviceAccessPointItem/SubslotList

1097 Use: Optional

1098 Supported since GSDML version: 2.0

1099 Attributes: None

1100 **A.4.5.7 SubslotItem**

1101 The SubslotItem element describes characteristics of a single subslot of a DAP.

1102 Location:

1103 //ISO15745Profile/ProfileBody/ApplicationProcess/DeviceAccessPointList/DeviceAccessPointItem/SubslotList/SubslotItem

1105 Use: One or more

1106 Supported since GSDML version: 2.0

1107 Each element shall contain attributes as shown in Table 21.

1108 **Table 21 — Attributes of element SubslotItem**

Name	Type	Use	SV	Description
SubslotNumber	Unsigned16	Required	2.0	Contains the number of the subslot. The number shall be unique within the SubslotList. Only the subslot numbers which are actually used by submodules of this DAP should be used. These are the subslot numbers defined in: .../SystemDefinedSubmoduleList/InterfaceSubmoduleItem/@SubslotNumber (or its default value "32768" if the attribute "SubslotNumber" is omitted) .../SystemDefinedSubmoduleList/PortSubmoduleItem/@SubslotNumber

Name	Type	Use	SV	Description
				• ../../VirtualSubmoduleList/VirtualSubmoduleItem/@FixedInSubslots (or its default value "1" if the attribute "FixedInSubslots" is omitted, which is only allowed when there is only one VirtualSubmoduleItem) • ../../@PhysicalSubslots (which is mandatory if there are pluggable submodules and contains the set of all their subslot numbers, i.e. ../../UseableSubmodules/SubmoduleItemRef/@FixedInSubslots, @UsedInSubslots and @AllowedInSubslots) Any other values will be ignored.
TextId	RefId	Required	2.0	Contains the ID of a text as a reference into the "ExternalTextList" and is used to describe the function of the subslot.

1109

1110 **A.4.5.8 IOConfigData**

1111 This element contains information about the quantity of IO data.

1112 Location:

1113 //ISO15745Profile/ProfileBody/ApplicationProcess/DeviceAccessPointList/DeviceAccessPointItem/IOConf

1114 igData

1115 Use: Required

1116 Each element shall contain attributes as shown in Table 22.

1117

Table 22 — Attributes of element IOConfigData

Name	Type	Use	SV	Description
MaxInputLength	Unsigned16	Required	1.0	Contains the maximum length of the data in octets in each Input-CR. This length is defined by the sum of the input data of all used submodules, the corresponding IOPS, and the IOCS of the used output submodules. See note below. Allowed values: 0..1440. Note that for values < 40 the engineering tool shall use 40 instead to reach the minimum ethernet frame size.
MaxOutputLength	Unsigned16	Required	1.0	Contains the maximum length of the data in octets in each Output-CR. This length is defined by the sum of the output data of all used submodules, the corresponding IOPS, and the IOCS of the used input submodules. See note below. Allowed values: 0..1440. Note that for values < 40 the engineering tool shall use 40 instead to reach the minimum ethernet frame size.
MaxDataLength	Unsigned16	Optional	1.0	Contains the maximum length of the input and output data in octets. This is a device-global value, i.e. it is the sum over all Input-CRs and Output-CRs, no matter how they are distributed across the ARs. MaxDataLength shall not be less than the highest value of MaxInputLength or MaxOutputLength. If this keyword is not provided, the maximum length is the sum of MaxInputLength and MaxOutputLength.
MaxApplicationInputLength	Unsigned16	Optional	2.3	Contains the maximum length of the input data in octets from an application point-of-view: The IOxS are not counted and it doesn't matter how the input data are distributed across the CRs and ARs.

Name	Type	Use	SV	Description
MaxApplicationOutputLength	Unsigned16	Optional	2.3	Contains the maximum length of the output data in octets from an application point-of-view: The IOxS are not counted and it doesn't matter how the output data are distributed across the CRs and ARs.
MaxApplicationDataLength	Unsigned16	Optional	2.3	Contains the maximum length of data from an application point-of-view: The IOxS are not counted and it doesn't matter how the data are distributed across the CRs and ARs. If MaxApplicationInputLength and / or MaxApplicationOutputLength are present, MaxApplicationDataLength shall be not less than the highest of these two values. If both MaxApplicationInputLength and MaxApplicationOutputLength are given, MaxApplicationDataLength shall not be more than the sum of these two values.
NOTE According to the PROFINET standard, a submodule without application input and output data is treated as having application input data with a length of 0 octets. I.e. an IOPS in the Input-CR and an IOCS in the Output-CR has to be accounted for. Whether IOxS (IOPS and IOCS) are actually transferred is determined by IOXS_Required being set to "true" in the GSD or by the controller requesting IOxS. Nevertheless, IOxS always have to be included in this calculation.				

- 1118
- 1119 **A.4.5.9 UseableModules**
- 1120 The UseableModules element contains a list of module references referring to modules of the element
- 1121 ModuleList. Only modules within this list are compatible with the DAP.
- 1122 An engineering tool shall not be able to configure other modules for this DAP.
- 1123 Fieldbus integration modules (representing devices of a subordinated fieldbus) are not included in this list;
- 1124 they may be plugged into any fieldbus integration slot of a slot cluster with the same FieldbusType.
- 1125 Location:
- 1126 //ISO15745Profile/ProfileBody/ApplicationProcess/DeviceAccessPointList/DeviceAccessPointItem/Useabl
- 1127 eModules
- 1128 Use: Optional
- 1129 Since up to GSDML V2.0 this element is required, a GSD file without this element may be rejected by an
- 1130 engineering tool without GSDML V2.1 support.
- 1131 Attributes: None
- 1132 **A.4.5.10 ModuleItemRef**
- 1133 This element references a module of the ModuleList being compatible with a DAP. Fieldbus integration
- 1134 modules shall not be referenced by ModuleItemRef.
- 1135 Location:
- 1136 //ISO15745Profile/ProfileBody/ApplicationProcess/DeviceAccessPointList/DeviceAccessPointItem/Useabl
- 1137 eModules/ModuleItemRef
- 1138 Use: One or more
- 1139 Each element shall contain attributes as shown in Table 23.

1140

Table 23 — Attributes of element ModuleItemRef

Name	Type	Use	SV	Description
ModuleItemTarget	RefId	Required	1.0	References a ModuleItem element.
AllowedInSlots	ValueList	Optional	1.0	Defines in which slots the module can be placed. If this attribute is not defined, the module can be placed into all slots.
UsedInSlots	ValueList	Optional	1.0	Defines in which slots the module is placed by default. Those built-in modules can be removed with an engineering tool. If the attribute "AllowedInSlots" is used, the slots defined in "UsedInSlots" shall be a subset of the slots defined by the attribute "AllowedInSlots".
FixedInSlots	ValueList	Optional	1.0	Defines in which slots the module is placed by default. Those built-in modules can not be removed with an engineering tool. If the attribute "AllowedInSlots" is used, the slots defined in "FixedInSlots" shall be a subset of the slots defined by the attribute "AllowedInSlots".

1141

1142 A.4.5.11 ARVendorBlock

1143 This element describes the ARVendorBlock introduced in PN IO Version 2.3. It may be used e.g. for
1144 modular devices that support shared device and which need global parameters for startup of the
1145 communication between the modules.

1146 Location:

1147 //ISO15745Profile/ProfileBody/ApplicationProcess/DeviceAccessPointList/DeviceAccessPointItem/ARVen
1148 dorBlock

1149 Use: Optional

1150 A.4.5.12 Request

1151 This element describes the ARVendorBlock request.

1152 Location:

1153 //ISO15745Profile/ProfileBody/ApplicationProcess/DeviceAccessPointList/DeviceAccessPointItem/ARVen
1154 dorBlock/Request

1155 Use: Required

1156 Attributes and child elements: See A.4.10.

1157 A.4.5.13 VirtualSubmoduleList

1158 This element contains a list of VirtualSubmoduleItems.

1159 Location:

1160 //ISO15745Profile/ProfileBody/ApplicationProcess/DeviceAccessPointList/DeviceAccessPointItem/Virtual
1161 SubmoduleList

1162 Use: Optional. The DAP needs at least one submodule. So either "SystemDefinedSubmoduleList",
1163 "VirtualSubmoduleList" or "UseableSubmodules" shall be present.

1164 Attributes: None

- 1165 Since up to GSDML V2.0 this element is always required, a GSD file without this element may be rejected
1166 by an engineering tool without GSDML V2.1 support.
- 1167 **A.4.5.14 VirtualSubmoduleItem**
- 1168 Location:
1169 //ISO15745Profile/ProfileBody/ApplicationProcess/DeviceAccessPointList/DeviceAccessPointItem/Virtual
1170 SubmoduleList/VirtualSubmoduleItem
- 1171 Use: One or more
- 1172 Attributes and child elements: See A.4.7.2.
- 1173 **A.4.5.15 SystemDefinedSubmoduleList**
- 1174 The SystemDefinedSubmoduleList element contains all types of submodules whose structure is defined
1175 in PNO-2.712 *Application Layer services*, “Physical Device Management ASE”.
- 1176 Location:
1177 //ISO15745Profile/ProfileBody/ApplicationProcess/DeviceAccessPointList/DeviceAccessPointItem/Syste
1178 mDefinedSubmoduleList
- 1179 Use: Required for devices developed according to PROFINET IO specification $\geq$ V2.0. The DAP needs
1180 at least one submodule. So either “SystemDefinedSubmoduleList”, “VirtualSubmoduleList” or
1181 “UseableSubmodules” shall be present.
- 1182 Supported since GSDML version: 2.0
- 1183 Attributes: None
- 1184 **A.4.5.16 InterfaceSubmoduleItem**
- 1185 Represents the network interface of a DAP.
- 1186 Location:
1187 //ISO15745Profile/ProfileBody/ApplicationProcess/DeviceAccessPointList/DeviceAccessPointItem/Syste
1188 mDefinedSubmoduleList/InterfaceSubmoduleItem
- 1189 Supported since GSDML version: 2.0
- 1190 Use: Required
- 1191 Attributes and child elements: See A.4.8.1
- 1192 **A.4.5.17 PortSubmoduleItem** (builtin)
- 1193 The PortSubmoduleItem represents a switch port of a DAP.
- 1194 Location:
1195 //ISO15745Profile/ProfileBody/ApplicationProcess/DeviceAccessPointList/DeviceAccessPointItem/Syste
1196 mDefinedSubmoduleList/PortSubmoduleItem
- 1197 Use: One for every switch port of the DAP
- 1198 Since in GSDML V2.0 the “SystemDefinedSubmoduleList” of a DAP requires at least one
1199 “PortSubmoduleItem” element, a GSD file without this element may be rejected by an engineering tool
1200 without GSDML V2.1 support.
- 1201 Supported since GSDML version: 2.0

1202 Attributes and child elements: See A.4.9.5.

1203 A.4.5.18 Graphics

1204 The Graphics element contains a list of GraphicItemRefs.

1205 Location:

1206 //ISO15745Profile/ProfileBody/ApplicationProcess/DeviceAccessPointList/DeviceAccessPointItem/Graphi
1207 cs

1208 Use: Optional

1209 Attributes: None

1210 A.4.5.19 GraphicItemRef

1211 The GraphicItemRef element references graphical information about a device module or submodule.

1212 Location:

1213 //ISO15745Profile/ProfileBody/ApplicationProcess/DeviceAccessPointList/DeviceAccessPointItem/Graphi
1214 cs/GraphicItemRef

1215 Use: One or more

1216 Each element shall contain attributes as shown in Table 24.

1217 **Table 24 — Attributes of element GraphicItemRef**

Name	Type	Use	SV	Description
Type	Enumeration	Required	1.0	Describes the type of the Graphic Item. One of the following values can be used: "DeviceSymbol" "DeviceIcon" Within one "Graphics" element only one GraphicItemRef of one type shall be used. If a "DeviceSymbol" type is used, the assigned GraphicItem shall be a bitmap file (70*40 pixels (width*height) 16 colors). For a DeviceAccessPointItem the GraphicItem should represent the device. For modules and submodules the GraphicItem should represent the module or submodule. If a "DeviceIcon" type is used, the assigned GraphicItem shall be a file in icon format.
GraphicItemTarget	RefId	Required	1.0	Reference to a "GraphicItem" element.

1218

1219 A.4.5.20 ApplicationRelations

1220 This element shall only be used for old DAPs which were built according to PN IO specification V1.0. For
1221 DAPs built according to PN IO specification >= V2.0 PDEV submodules are mandatory. Use the
1222 ApplicationRelations element of the InterfaceSubmoduleItem defined in A.4.8.7 instead. If an
1223 InterfaceSubmodule is present the element DeviceAccessPointItem/ApplicationRelations is ignored.

1224 This element contains information about the application relations implemented in an IO Device.

1225 The VersionInformation attributes are needed to check, if the structure of the connect PDU
1226 (IODConnectReq) complies with the functionality of an IO Device. The engineering tool has to fill in the
1227 version information into the connect PDU with these attributes.

1228 Location:
 1229 //ISO15745Profile/
 1230 ProfileBody/ApplicationProcess/DeviceAccessPointList/DeviceAccessPointItem/ApplicationRelations

1231 Use: Optional

1232 Each element shall contain attributes as shown in Table 25.

1233 **Table 25 — Attributes of element ApplicationRelations**

Name	Type	Use	SV	Description
AR_BlockVersion	Unsigned16	Required	1.0	Contains the version information about the AR block
IOCR_BlockVersion	Unsigned16	Required	1.0	Contains the version information about the IO communication relationship (CR) block, called CR block.
AlarmCR_BlockVersion	Unsigned16	Required	1.0	Contains the version information about the alarm block
SubmoduleDataBlockVersion	Unsigned16	Required	1.0	Contains the version information about the submodule data block

1234

1235 A.4.5.21 TimingProperties

1236 This element shall only be used for old DAPs which were built according to PN IO specification V1.0.

1237 This element defines the timing behaviour for sending cyclic IO data.

1238 Location:
 1239 //ISO15745Profile/ProfileBody/ApplicationProcess/DeviceAccessPointList/DeviceAccessPointItem/Applic
 1240 ationRelations/TimingProperties

1241 Use: Optional

1242 Each element shall contain attributes as shown in Table 26.

1243 **Table 26 — Attributes of element TimingProperties**

Name	Type	Use	SV	Description
SendClock	ValueList	Optional	1.0	Contains a list of cycle times supported by the DAP for sending cyclic data. Basic clock tick is 31.25 μ s. Each value of this element contains the multiplier of the basic clock tick and can be in the range "1" to "128" (see Example). Default is "32".
ReductionRatio	ValueList	Optional	1.0	The send clock interval can be reduced by a reduction. This attribute describes the supported reduction ratios of a DAP. Each value of this element shall be in the range "1" to "512".
EXAMPLE SendClock="16 32 64" means that the DAP is able to send cyclic data in interval 500 μ s, 1 ms or 2 ms.				

1244

1245 A.4.5.22 SystemRedundancy

1246 If this element is present, the DAP supports system redundancy according to PNO-7.122 *PROFINET IO*
 1247 *System Redundancy*.

- 1248 Location:
 1249 //ISO15745Profile/
 1250 ProfileBody/ApplicationProcess/DeviceAccessPointList/DeviceAccessPointItem/SystemRedundancy
- 1251 Use: Optional
- 1252 Each element shall contain attributes as shown in Table 25.

1253 **Table 27 — Attributes of element SystemRedundancy**

Name	Type	Use	SV	Description
DeviceType	Enumeration	Required	2.3	“S2” is a single NAP with two SR-ARs, “R2” is a redundant NAP (a pair of NAPs) with two SR-ARs per NAP. In case of “R2”, the DAP shall be pluggable or plugged two times (attributes AllowedInSlots, UsedInSlots, FixedInSlots). Allowed values: “S2” or “R2”
MaxSwitchOverTime	Unsigned16	Required	2.3	Time to perform a switchover (= MSOT). Used to calculate the max. takeover time (MTOT). Base: 1 ms Allowed values: 1..60000
PrimaryAR_OnBothNAPsSupported	Boolean	Optional	2.3	An R2-Device is able to handle multiple primary AR in case of shared SR-AR-sets simultaneously on both NAPs. This attribute shall only be present if DeviceType is “R2”.
RT_InputOnBackupAR_Supported	Boolean	Optional	2.3	Whether the NAP is able to to send RT Input data over Backup AR in addition to the Primary AR. Default: “false”
RedundancyAlarmSupported	Boolean	Optional	2.3	Whether the NAP is able to generate the system redundancy alarm. Default: “false”
NumberOfAR_Sets	Unsigned8	Optional	2.3	The maximum number of SR-AR sets. Default: 1 Allowed values: 1..255

- 1254
- 1255 **A.4.5.23 UseableSubmodules**
- 1256 This element describes which submodules of the “SubmoduleList” can be used with this DAP. The
 1257 assigned “VirtualSubmoduleItems” of the DAP shall not be described in this list.
- 1258 Location:
 1259 //ISO15745Profile/ProfileBody/ApplicationProcess/DeviceAccessPointList/DeviceAccessPointItem/Useabl
 1260 eSubmodules
- 1261 Use: Optional. The DAP needs at least one submodule. So either “SystemDefinedSubmoduleList”,
 1262 “VirtualSubmoduleList” or “UseableSubmodules” shall be present.
- 1263 Supported since GSDML version: 2.1
- 1264 Attributes: None

1265 **A.4.5.24 SubmoduleItemRef**

1266 The SubmoduleItemRef element references a submodule which can be used by the assigned DAP.

1267 Location:

1268 //ISO15745Profile/ProfileBody/ApplicationProcess/DeviceAccessPointList/DeviceAccessPointItem/Useabl
1269 eSubmodules/SubmoduleItemRef

1270 Supported since GSDML version: 2.1

1271 Use: One or more

1272 Each element shall contain attributes as shown in Table 28.

1273 **Table 28 — Attributes of element SubmoduleItemRef**

Name	Type	Use	SV	Description
SubmoduleItemTarget	RefId	Required	1.0	References a SubmoduleItem or PortSubmoduleItem element of the SubmoduleList.
AllowedInSubslots	ValueList	Optional	1.0	Defines in which subslots the submodule can be placed. If this attribute is not defined, the submodule can be placed into all subslots listed in "PhysicalSubslots".
UsedInSubslots	ValueList	Optional	1.0	Defines in which subslots the submodule is placed by default. Those built-in submodules can be removed with an engineering tool. If the attribute "AllowedInSubslots" is used, the slots defined in "UsedInSubslots" shall be a subset of the subslots defined by the attribute "AllowedInSubslots".
FixedInSubslots	ValueList	Optional	1.0	Defines in which subslots the submodule is placed by default. Those built-in submodules can not be removed with an engineering tool. If the attribute "AllowedInSubslots" is used, the subslots defined in "FixedInSubslots" shall be a subset of the subslots defined by the attribute "AllowedInSubslots". This attribute has the same effect as using "VirtualSubmoduleItem" elements in the "VirtualSubmoduleList" or "PortSubmoduleItem" elements in the "SystemDefinedSubmoduleList". This attribute should be used instead of a VirtualSubmoduleItem / PortSubmoduleItem if the same type of submodule is used more than one time for one module.
The allowed range(s) for the subslot numbers in "AllowedInSubslots", "UsedInSubslots" and "FixedInSubslots" depend on the type of submodule which is referenced by "SubmoduleItemTarget". For a SubmoduleItem, the allowed value range shall be: 0 to 32767 (0x0000 to 0x7FFF) For a PortSubmoduleItem, the allowed value ranges shall be: 32769 to 33023 (0x8001 to 0x80FF – Interface 1) 33025 to 33279 (0x8101 to 0x81FF – Interface 2) ... 36609 to 36863 (0x8F01 to 0x8FFF – Interface 16) The pattern here is 0x8<i>ipp</i> where <i>i</i> is the interface number (0-15) and <i>pp</i> is the port number within the interface (1-255), while <i>pp</i>=0 represents the interface submodule itself.				

1274

1275 A.4.5.25 SlotList

1276 The SlotList element is used to describe the characteristics of the slot structure of a device.

1277 Location:

1278 //ISO15745Profile/ProfileBody/ApplicationProcess/DeviceAccessPointList/DeviceAccessPointItem/SlotList
1279 t

1280 Use: Optional

1281 Supported since GSDML version: 2.1

1282 Attributes: None

1283 A.4.5.26 SlotItem

1284 The SlotItem element describes characteristics of a single Slot of a device.

1285 Location:

1286 //ISO15745Profile/ProfileBody/ApplicationProcess/DeviceAccessPointList/DeviceAccessPointItem
1287 /SlotList/SlotItem

1288 Use: One or more

1289 Supported since GSDML version: 2.1

1290 Each element shall contain attributes as shown in Table 29.

Table 29 — Attributes of element SlotItem

Name	Type	Use	SV	Description
SlotNumber	Unsigned16	Required	2.1	Contains the number of the slot. The number shall be unique within the SlotList. The number shall only contain values defined in the attribute "PhysicalSlots" of the DAP.
TextId	RefId	Required	2.1	Contains the ID of a text as a reference into the "ExternalTextList" and is used to describe the function of the slot.

1292

1293 A.4.5.27 SlotGroups

1294 The SlotGroups element can be used to define a list of slot groups.

1295 Location:

1296 //ISO15745Profile/ProfileBody/ApplicationProcess/DeviceAccessPointList/DeviceAccessPointItem
1297 /SlotGroups

1298 Use: Optional

1299 Supported since GSDML version: 2.1

1300 Attributes: None

1301 A.4.5.28 SlotGroup

1302 The SlotGroup element allows assigning a slot to a slot group. This slot group can be displayed in the
1303 engineering tool in order to show which slots belong together. One example for a slot group is an
1304 underlying fieldbus line of a PROFINET gateway.

1305 Location:
 1306 //ISO15745Profile/ProfileBody/ApplicationProcess/DeviceAccessPointList/DeviceAccessPointItem
 1307 /SlotGroups/SlotGroup

1308 Use: One or more

1309 Supported since GSDML version: 2.1

1310 Each element shall contain attributes as shown in Table 30.

1311 **Table 30 — Attributes of element SlotGroup**

Name	Type	Use	SV	Description
SlotList	ValueList	Required	2.1	Describes which slots belong to this group. Each slot should only appear in one group. A GSD checker tool shall not reject a GSD file with a slot assigned to more than one group. If SlotGroups are used, all slots defined by the attribute "PhysicalSlots" of the DAP shall be assigned to a slot group.

1312

1313 **A.4.5.29 Name**

1314 This element is used to assign a name to a slot group.

1315 Location:
 1316 //ISO15745Profile/ProfileBody/ApplicationProcess/DeviceAccessPointList/DeviceAccessPointItem
 1317 /SlotGroups/SlotGroup/Name

1318 Use: Required

1319 Each element shall contain attributes as shown in Table 31.

1320 **Table 31 — Attributes of element Name**

Name	Type	Use	SV	Description
TextId	RefId	Required	2.1	Contains the ID of a text as a reference into the "ExternalTextList".

1321

1322 **A.4.5.30 InfoText**

1323 This element contains human readable additional text information about a slot group.

1324 Location:
 1325 //ISO15745Profile/ProfileBody/ApplicationProcess/DeviceAccessPointList/DeviceAccessPointItem
 1326 /SlotGroups/SlotGroup/InfoText

1327 Use: Required

1328 Each element shall contain attributes as shown in Table 32.

1329 **Table 32 — Attributes of element InfoText**

Name	Type	Use	SV	Description
TextId	RefId	Required	1.0	Contains the ID of a text as a reference into the "ExternalTextList".

1330

1331 A.4.5.31 FieldbusIntegrationSlots

1332 This element describes slots to where devices of a subordinated fieldbus may be mapped.

1333 Location:

1334 //ISO15745Profile/ProfileBody/ApplicationProcess/DeviceAccessPointList/DeviceAccessPointItem
1335 /FieldbusIntegrationSlots

1336 Use: Optional

1337 Supported since GSDML version: 2.25

1338 Each element shall contain attributes as shown in Table 33.

1339 **Table 33 — Attributes of element FieldbusIntegrationSlots**

Name	Type	Use	SV	Description
MaxSupported	Unsigned16	Required	2.25	The maximum number of fieldbus integration slots that the DAP supports. Allowed values: 1 to 32767
Range	String	Optional	2.25	A range of slot numbers allowed for fieldbus integration slots, given as a lower and an upper value, both being 16 bit unsigned integer values. The attribute shall comply with the following regular expression: \\d+\\.\\.\\d+ Default: Last slot in PhysicalSlots plus 1 to 32767.

1340

1341 A.4.6 ModuleItem related elements

1342 A.4.6.1 ModuleItem

1343 This element contains subelements to describe the properties of a module.

1344 Location:

1345 //ISO15745Profile/ProfileBody/ApplicationProcess/ModuleList/ModuleItem

1346 Use: One or more

1347 Each element shall contain attributes as shown in Table 34.

1348 **Table 34 — Attributes of element ModuleItem**

Name	Type	Use	SV	Description
ID	Id	Required	1.0	Contains the unique ID to be referenced by a "ModuleItemRef" element.
ModuleIdentNumber	Unsigned32hex 0	Required	1.0	Contains the module ident number for identification purposes. The value "0x0" is reserved.
RequiredSchemaVersion	VersionString	Optional	2.0	Specifies the minimal version of the GSDML Schema needed to create valid configuration data for the module. If an engineering tool is not able to interpret this schema version it shall refuse to configure this module. For compatibility issues it is recommended to keep the content of the "RequiredSchemaVersion" attribute as low as possible. The RequiredSchemaVersion shall be no higher

Name	Type	Use	SV	Description
				than the GSDML version that this GSD is based on. Default: "V1.0"
PhysicalSubslots	ValueList	Optional	2.1	This list describes which physical subslots are available in the module. Submodules can be assigned to these subslots by the engineering tool. If the module has "UseableSubmodules", this list shall be present and shall contain all subslots listed in "AllowedInSubslots", "UsedInSubslots" and "FixedInSubslots" of the "SubmoduleItemRef" elements. This list shall not include the subslot number of any submodule in the "VirtualSubmoduleList" or "SystemDefinedSubmoduleList" of the module.
FieldbusType	Unsigned16	Required on fieldbus integration modules	2.25	This attribute contains the profile specific type of the fieldbus integration profile. A link to a list of these types will be added to the table http://www.profibus.com/IM/Profile_ID_Table.xml in line "Fieldbus Integration", row "Profile Specific Type Specification" after the release of the profile. This value indicates which type of fieldbus this module (= device of a subordinated fieldbus) belongs to.

1349

1350 **A.4.6.2 ModuleInfo**

1351 Location:

1352 //ISO15745Profile/ProfileBody/ApplicationProcess/ModuleList/ModuleItem/ModuleInfo

1353 Use: Required

1354 Attributes and child elements: See A.4.12.1.

1355 **A.4.6.3 SubslotList**

1356 The SubslotList element is used to describe the characteristics of the subslot structure of a module.

1357 Location:

1358 //ISO15745Profile/ProfileBody/ApplicationProcess/ModuleList/ModuleItem/SubslotList

1359 Use: Optional

1360 Supported since GSDML version: 2.0

1361 Attributes: None

1362 **A.4.6.4 SubslotItem**

1363 The SubslotItem element describes characteristics of a single subslot of a module.

1364 Location:

1365 //ISO15745Profile/ProfileBody/ApplicationProcess/ModuleList/ModuleItem/SubslotList/SubslotItem

1366 Use: One or more

1367 Supported since GSDML version: 2.0

1368 Attributes and child elements: See A.4.5.7

1369 A.4.6.5 VirtualSubmoduleList

1370 This element contains all submodules which are implemented as integral part of a module.

1371 Location:

1372 //ISO15745Profile/ProfileBody/ApplicationProcess/ModuleList/ModuleItem/VirtualSubmoduleList

1373 Use: Optional. The module needs at least one submodule. So either “SystemDefinedSubmoduleList”,
1374 “VirtualSubmoduleList” or “UseableSubmodules” shall be present.

1375 Attributes: None

1376 Since up to GSDML V2.0 this element is always required, a GSD file without this element may be rejected
1377 by an engineering tool without GSDML V2.1 support.

1378 A.4.6.6 VirtualSubmoduleItem

1379 Location:

1380 //ISO15745Profile/ProfileBody/ApplicationProcess/ModuleList/ModuleItem/VirtualSubmoduleList/VirtualSu
1381 bmoduleItem

1382 Attributes and child elements: See A.4.7.2.

1383 A.4.6.7 SystemDefinedSubmoduleList

1384 The SystemDefinedSubmoduleList element contains all types of submodules whose structure is defined
1385 in PNO-2.712 *Application Layer services*, “Physical Device Management ASE”.

1386 Location:

1387 //ISO15745Profile/ProfileBody/ApplicationProcess/ModuleList/ModuleItem/SystemDefinedSubmoduleList

1388 Use: Optional. The module needs at least one submodule. So either “SystemDefinedSubmoduleList”,
1389 “VirtualSubmoduleList” or “UseableSubmodules” shall be present.

1390 Supported since GSDML version: 2.0

1391 Attributes: None

1392 A.4.6.8 PortSubmoduleItem (builtin)

1393 The PortSubmoduleItem represents a switch port of a module.

1394 Location:

1395 //ISO15745Profile/ProfileBody/ApplicationProcess/ModuleList/ModuleItem/SystemDefinedSubmoduleList/
1396 PortSubmoduleItem

1397 Use: One for every switch port of the module

1398 Supported since GSDML version: 2.0

1399 Attributes and child elements: See A.4.9.5.

1400 A.4.6.9 UseableSubmodules

1401 This element describes which submodules of the “SubmoduleList” can be used with this module. The
1402 assigned “VirtualSubmoduleItems” of the ModuleItem shall not be described in this list.

1403 Location:

1404 //ISO15745Profile/ProfileBody/ApplicationProcess/ModuleList/ModuleItem/UseableSubmodules

1405 Use: Optional. The module needs at least one submodule. So either “SystemDefinedSubmoduleList”,
1406 “VirtualSubmoduleList” or “UseableSubmodules” shall be present.

1407 Supported since GSDML version: 2.1

1408 Attributes: None

1409 **A.4.6.10 SubmoduleItemRef**

1410 The SubmoduleItemRef element references a submodule which can be used by the assigned module.

1411 Location:

1412 //ISO15745Profile/ProfileBody/ApplicationProcess/ModuleList/ModuleItem/UseableSubmodules/Submodu
1413 leItemRef

1414 Use: One or more

1415 Supported since GSDML version: 2.1

1416 Each element shall contain attributes as shown in Table 28.

1417 **A.4.6.11 Graphics**

1418 The Graphics element contains a list of GraphicItemRef.

1419 Location:

1420 //ISO15745Profile/ProfileBody/ApplicationProcess/ModuleList/ModuleItem/Graphics

1421 Use: Optional

1422 Attributes: None

1423 **A.4.6.12 GraphicItemRef**

1424 The GraphicItemRef element references graphical information about a device module or submodule.

1425 Location:

1426 //ISO15745Profile/ProfileBody/ApplicationProcess/ModuleList/ModuleItem/Graphics/GraphicItemRef

1427 Use: One or more

1428 Each element shall contain attributes as shown in Table 24.

1429 **A.4.7 (Virtual)SubmoduleItem related elements**

1430 **A.4.7.1 SubmoduleItem**

1431 This element defines the characteristics of a submodule which is referenced by a DAP or module.

1432 //ISO15745Profile/ProfileBody/ApplicationProcess/SubmoduleList/SubmoduleItem

1433 Use: Required

1434 Each element shall contain the elements and attributes of the base class SubmoduleItemBase (see
1435 A.4.7.3) plus the attribute shown in Table 37.

1436

Table 35 — Additional attributes of element SubmoduleItem

Name	Type	Use	SV	Description
RequiredSchemaVersion	VersionString	Optional	2.25	The minimal version of the GSDML Schema needed to create valid configuration data for the submodule. If an engineering tool is not able to interpret this schema version it shall refuse to configure this submodule. For compatibility issues it is recommended to keep the RequiredSchemaVersion as low as possible. The RequiredSchemaVersion shall be no higher than the GSDML version that this GSD is based on. Default: "V2.1"

1437

1438 A.4.7.2 VirtualSubmoduleItem

1439 This element defines the characteristics of a virtual (builtin) submodule as part of a DAP or module.

1440 Locations:

1441 //ISO15745Profile/ProfileBody/ApplicationProcess/DeviceAccessPointList/DeviceAccessPointItem/Virtual
1442 SubmoduleList/VirtualSubmoduleItem

1443 //ISO15745Profile/ProfileBody/ApplicationProcess/ModuleList/ModuleItem/VirtualSubmoduleList/VirtualSu
1444 bmoduleItem

1445 Use: Required

1446 Each element shall contain the elements and attributes of the base class SubmoduleItemBase (see
1447 A.4.7.3) plus the attribute shown in Table 36.

1448

Table 36 — Additional attributes of element VirtualSubmoduleItem

Name	Type	Use	SV	Description
FixedInSubslots	ValueList	Required if more than one VirtualSubmoduleItem is defined	2.0	Defines in which Subslots the submodule is placed by default. Those built-in submodules can not be removed with an engineering tool. Only values in the range from 0 to 32767 (0x0 to 0x7FFF) are allowed. The values of this attribute shall be unique. If only one VirtualSubmoduleItem is defined in the VirtualSubmoduleList, the default is "1". Be careful: Subslot 0 was added to the allowed range in GSDML version 2.2. It breaks backward compatibility to engineering tools that don't support GSDML V2.2.

1449

1450 A.4.7.3 SubmoduleItemBase (abstract)

1451 Base class of a submodule of a DAP or module. Used by SubmoduleItem and VirtualSubmoduleItem.

1452 Each derived element shall contain attributes as shown in Table 37.

1453

Table 37 — Attributes of SubmoduleItemBase

Name	Type	Use	SV	Description
ID	Id	Required	1.0	Contains the ID of the element

Name	Type	Use	SV	Description
SubmoduleIdentNumber	Unsigned32hex	Required	1.0	Contains the submodule ident number for identification purposes.
API	Unsigned32	Optional	2.0	This attribute defines the Application Process Identifier (API) of the submodule which can be defined by a device profile. Default: "0"
PROFIsafeSupported	Boolean	Optional	2.0	Describes if the module implements a behaviour according to PROFIsafe Profile V2.5. Default: "false"
Writeable_IM_Records	ValueList	Optional	2.1	Defines, which I&M records as defined in <i>Profile Guidelines Part 1 – Identification & Maintenance Functions</i> can be transferred to the device using the record data index defined in PNO-2.722 <i>Application Layer protocol</i> . In this attribute, the numbers of all writeable I&M records shall be listed. The I&M0 record can't be written, so 0 shall not be contained in this list. EXAMPLE: "1 2" defines that I&M1 and I&M2 can be written to the device with record data index 0xAFF1 and 0xAFF2 Default: No I&M record can be written
Max_iParameterSize	Unsigned32	Optional	2.2	Defines the maximum size of the i-Parameters that this submodule requires, in octets. Default: "0"
SubsysModuleDirIndex	Unsigned16	Optional	2.2	Contains the record index where the object directory of a HART master device can be read out. This attribute shall only be used for HART master modules that were built according to "Profile for HART on PROFIBUS" which are now migrated to PROFINET just by exchanging the head module.
SupportedSubstitutionModes	ValueList	Optional	2.25	The presence of this attribute indicates that the submodule supports to set substitute values for one subslot using the record data index defined in PNO-2.722 <i>Application Layer protocol</i> . The content of the attribute describes which substitution modes are supported. Use the numeric values defined in PNO-2.722 <i>Application Layer protocol</i> , "Coding of the field SubstitutionMode".

1454

1455 **A.4.7.4 IOData**

1456 This element defines characteristics of the IO data of a submodule.

1457 Locations:

1458 //ISO15745Profile/ProfileBody/ApplicationProcess/DeviceAccessPointList/DeviceAccessPointItem/Virtual
1459 SubmoduleList/VirtualSubmoduleItem/IOData1460 //ISO15745Profile/ProfileBody/ApplicationProcess/ModuleList/ModuleItem/VirtualSubmoduleList/VirtualSubmoduleItem/IOData
1461

1462 //ISO15745Profile/ProfileBody/ApplicationProcess/SubmoduleList/SubmoduleItem/IOData

1463 Use: Required

1464 Each element shall contain attributes as shown in Table 38.

1465 **Table 38 — Attributes of element IOData**

Name	Type	Use	SV	Description
IOPS_Length	Unsigned8	Optional	1.0	Contains the length of the IO producer status in octets as part of an IO data object. In this version of GSDML the IOPS_Length is fixed to one octet.
IOCS_Length	Unsigned8	Optional	1.0	Contains the length of the IO consumer status in octets as part of an IO data object. In this version of GSDML the IOCS_Length is fixed to one octet.
F_IO_StructureDescVersion	Unsigned8	Optional Only allowed if the attribute "PROFIsafeSupported" of the corresponding submodule is set to "true".	2.2	Controls the layout of the PROFIsafe IO structure. Currently, it affects the valid range of the F_IO_StructureDescCRC. A value of 1 indicates a 16-bit CRC while a value of 2 indicates a 32-bit CRC. If this attribute is not present, a value of 1 is assumed.
F_IO_StructureDescCRC	Unsigned32 (Unsigned16 in SV 2.0-2.1)	Required if the attribute "PROFIsafeSupported" of the corresponding submodule is set to "true".	2.0	In order to read the PROFIsafe parameter description safely from the GSD file, a CRC code is necessary. The CRC signature has to be calculated according to PROFIsafe Profile V2.5 and certified by a notified assessment body (e.g. TÜV). The value of this parameter will not be transferred to the device but is needed to avoid errors during the parameterization with the configuration tool. The valid range is dependent on the presence and value of attribute F_IO_StructureDescVersion. If missing or 1, the range is limited to 0..65535 (0x0..0xFFFF). If 2, the range is that of Unsigned32. Be careful: The 4-byte CRC was introduced in GSDML version 2.2. It breaks backward compatibility to engineering tools that don't support GSDML V2.2.

1466

1467 **A.4.7.5 Input**

1468 Specifies the input characteristics of a submodule. If input data is available, this element has Dataltem
1469 elements (see A.4.7.6).

1470 Locations:

1471 //ISO15745Profile/ProfileBody/ApplicationProcess/DeviceAccessPointList/DeviceAccessPointItem/Virtual
1472 SubmoduleList/VirtualSubmoduleItem/IOData/Input

1473 //ISO15745Profile/ProfileBody/ApplicationProcess/ModuleList/ModuleItem/VirtualSubmoduleList/VirtualSu
1474 bmoduleItem/IOData/Input

1475 //ISO15745Profile/ProfileBody/ApplicationProcess/SubmoduleList/SubmoduleItem/IOData/Input

1476 Use: Optional

1477 Each element shall contain the attribute as shown in Table 39.

1478 **Table 39 — Attributes of element Input**

Name	Type	Use	SV	Description
Consistency	Enumeration	Optional	1.0	If this attribute is set to "Item consistency", the consistency is only given within a DatalItem. If this attribute is set to "All items consistency" the submodule provides consistency for all data items. Default is "Item consistency".

1479

1480 **A.4.7.6 DatalItem**

1481 The DatalItem element contains information about a single DatalItem.

1482 Locations:

1483 //ISO15745Profile/ProfileBody/ApplicationProcess/DeviceAccessPointList/DeviceAccessPointItem/Virtual
1484 SubmoduleList/VirtualSubmoduleItem/IOData/Input/DatalItem

1485 //ISO15745Profile/ProfileBody/ApplicationProcess/ModuleList/ModuleItem/VirtualSubmoduleList/VirtualSu
1486 bmoduleItem/IOData/Input/DatalItem

1487 //ISO15745Profile/ProfileBody/ApplicationProcess/SubmoduleList/SubmoduleItem/IOData/Input/DatalItem

1488 //ISO15745Profile/ProfileBody/ApplicationProcess/DeviceAccessPointList/DeviceAccessPointItem/Virtual
1489 SubmoduleList/VirtualSubmoduleItem/IOData/Output/DatalItem

1490 //ISO15745Profile/ProfileBody/ApplicationProcess/ModuleList/ModuleItem/VirtualSubmoduleList/VirtualSu
1491 bmoduleItem/IOData/Output/DatalItem

1492 //ISO15745Profile/ProfileBody/ApplicationProcess/SubmoduleList/SubmoduleItem/IOData/Output/Datalte
1493 m

1494 Use: One for every DatalItem.

1495 Each element shall contain attributes as shown in Table 40.

1496

Table 40 — Attributes of element DataItem

Name	Type	Use	SV	Description
DataType	Enumeration	Required	1.0	Defines the data type of the data item. One of the following values is allowed: "Integer8" "Integer16" "Integer32" "Integer64" "Unsigned8" "Unsigned16" "Unsigned32" "Unsigned64" "Float32" "Float64" "Date" "TimeOfDay with date indication" "TimeOfDay without date indication" "TimeDifference with date indication" "TimeDifference without date indication" "NetworkTime" "NetworkTimeDifference" "VisibleString" "OctetString" "Unsigned8+Unsigned8" "Float32+Unsigned8" "F_MessageTrailer4Byte" "F_MessageTrailer5Byte" The DataType "F_MessageTrailer4Byte" and "F_MessageTrailer5Byte" shall only be used if the attribute "PROFIsafeSupported" of the corresponding submodule is set to "true". Be careful: The data types Float32+Status8, F_MessageTrailer4-Byte and F_MessageTrailer5Byte were introduced in GSDML version 2.0. The data types Unsigned8+Unsigned8 and Float32+Unsigned8 were introduced in GSDML version 2.2. Using these data types breaks backward compatibility to engineering tools that don't support the respective GSDML version.
Length	Unsigned16	Optional	1.0	This attribute shall be used if the "DataType" attribute is set to "VisibleString" or "OctetString". This attribute specifies the length of the DataItem element in octets.^a A value of 0 is not allowed. If this attribute is used for data types with fixed length definition, the length shall correspond to the definition in PNO-2.712 <i>Application Layer services</i>, "Data Type ASE".
UseAsBits	Boolean	Optional	1.0	If the attribute is set to "true", the engineering tool should display the data item in a bit representation. Default: "false".
TextId	RefId	Required	1.0	Contains the ID of a text as a reference into the "ExternalTextList".
^a This is necessary because these data types have a variable length.				

1497

1498 A.4.7.7 BitDataItem

1499 The BitDataItem element can be used to describe the bit structure of a DataItem. Each bit is represented
 1500 by a BitDataItem element.

1501 Locations:

1502 //ISO15745Profile/ProfileBody/ApplicationProcess/DeviceAccessPointList/DeviceAccessPointItem/Virtual
 1503 SubmoduleList/VirtualSubmoduleItem/IOData/Input/DataItem/BitDataItem

- 1504 //ISO15745Profile/ProfileBody/ApplicationProcess/ModuleList/ModuleItem/VirtualSubmoduleList/VirtualSubmoduleItem/IOData/Input/DataItem/BitDataItem
- 1506 //ISO15745Profile/ProfileBody/ApplicationProcess/SubmoduleList/SubmoduleItem/IOData/Input/DataItem/BitDataItem
- 1508 //ISO15745Profile/ProfileBody/ApplicationProcess/DeviceAccessPointList/DeviceAccessPointItem/VirtualSubmoduleList/VirtualSubmoduleItem/IOData/Output/DataItem/BitDataItem
- 1511 //ISO15745Profile/ProfileBody/ApplicationProcess/ModuleList/ModuleItem/VirtualSubmoduleList/VirtualSubmoduleItem/IOData/Output/DataItem/BitDataItem
- 1513 //ISO15745Profile/ProfileBody/ApplicationProcess/SubmoduleList/SubmoduleItem/IOData/Output/DataItem/BitDataItem
- 1515 Use: Optional, one for every used bit within the DataItem element

Attribute "UseAsBits" present and "true"?		
n	BitDataItem element(s) present?	
	n	y
	All bits of the DataItem are considered relevant (backward compatibility).	Only those bits of the DataItem described in the BitDataItem elements are considered relevant.

Figure 23 — UseAsBits logic

- 1516
- 1517
- 1518 Supported since GSDML version: 2.0
- 1519 Attributes: See Table 41.

Table 41 — Attributes of element BitDataItem

Name	Type	Use	SV	Description
BitOffset	Unsigned8	Required	2.0	Contains the offset in bits of the BitDataItem element from the beginning of the DataItem element. BitOffset 0 indicates the least significant bit. Valid Range: 0 to BitLength-1 BitLength is defined as the length of the DataItem object in octets, multiplied by 8.
TextId	RefId	Required	2.0	Contains the ID of a text as a reference into the "ExternalTextList".

- 1521
- 1522 **A.4.7.8 Output**
- 1523 The Output element specifies the output characteristics of a submodule. If output data is available, this element contains DataItem elements (see A.4.7.6).
- 1524
- 1525 Locations:
- 1526 //ISO15745Profile/ProfileBody/ApplicationProcess/DeviceAccessPointList/DeviceAccessPointItem/VirtualSubmoduleList/VirtualSubmoduleItem/IOData/Output
- 1527
- 1528 //ISO15745Profile/ProfileBody/ApplicationProcess/ModuleList/ModuleItem/VirtualSubmoduleList/VirtualSubmoduleItem/IOData/Output
- 1529

1530 //ISO15745Profile/ProfileBody/ApplicationProcess/SubmoduleList/SubmoduleItem/IOData/Output

1531 Use: Optional

1532 Attributes: see Table 39.

1533 **A.4.7.9 RecordDataList**

1534 This element contains a list of ParameterRecordDataItems (see A.4.10).

1535 The engineering tool should show the content of the ParameterRecordDataItems in the same order as in
1536 the GSDML file.

1537 Locations:

1538 //ISO15745Profile/ProfileBody/ApplicationProcess/DeviceAccessPointList/DeviceAccessPointItem/Virtual
1539 SubmoduleList/VirtualSubmoduleItem/RecordDataList

1540 //ISO15745Profile/ProfileBody/ApplicationProcess/ModuleList/ModuleItem/VirtualSubmoduleList/VirtualSu
1541 bmoduleItem/RecordDataList

1542 //ISO15745Profile/ProfileBody/ApplicationProcess/SubmoduleList/SubmoduleItem/RecordDataList

1543 Use: Optional

1544 Attributes: None

1545 **A.4.7.10 F_ParameterRecordDataItem**

1546 An F_ParameterRecordDataItem element describes the data structure of a parameter record data object
1547 for PROFIsafe parameters.

1548 Locations:

1549 //ISO15745Profile/ProfileBody/ApplicationProcess/DeviceAccessPointList/DeviceAccessPointItem/Virtual
1550 SubmoduleList/VirtualSubmoduleItem/RecordDataList/F_ParameterRecordDataItem

1551 //ISO15745Profile/ProfileBody/ApplicationProcess/ModuleList/ModuleItem/VirtualSubmoduleList/VirtualSu
1552 bmoduleItem/RecordDataList/F_ParameterRecordDataItem

1553 //ISO15745Profile/ProfileBody/ApplicationProcess/SubmoduleList/SubmoduleItem/RecordDataList/F_Par
1554 ameterRecordDataItem

1555 Use: Required if the attribute "PROFIsafeSupported" of the corresponding submodule is set to "true".

1556 Supported since GSDML version: 2.0

1557 Attributes and child elements: See A.4.11.1.

1558 **A.4.7.11 ModuleInfo**

1559 Locations:

1560 //ISO15745Profile/ProfileBody/ApplicationProcess/DeviceAccessPointList/DeviceAccessPointItem/Virtual
1561 SubmoduleList/VirtualSubmoduleItem/ModuleInfo

1562 //ISO15745Profile/ProfileBody/ApplicationProcess/ModuleList/ModuleItem/VirtualSubmoduleList/VirtualSu
1563 bmoduleItem/ModuleInfo

1564 //ISO15745Profile/ProfileBody/ApplicationProcess/SubmoduleList/SubmoduleItem/ModuleInfo

1565 Use: Required

1566 Attributes and child elements: See A.4.12.1.

1567 **A.4.7.12 Graphics**

1568 The Graphics element contains a list of GraphicItemRef.

1569 Locations:

1570 //ISO15745Profile/ProfileBody/ApplicationProcess/DeviceAccessPointList/DeviceAccessPointItem/Virtual
1571 SubmoduleList/VirtualSubmoduleItem/Graphics

1572 //ISO15745Profile/ProfileBody/ApplicationProcess/ModuleList/ModuleItem/VirtualSubmoduleList/VirtualSu
1573 bmoduleItem/Graphics

1574 //ISO15745Profile/ProfileBody/ApplicationProcess/SubmoduleList/SubmoduleItem/Graphics

1575 Use: Optional

1576 Attributes: None

1577 **A.4.7.13 GraphicItemRef**

1578 The GraphicItemRef element references graphical information about a device module or submodule.

1579 Locations:

1580 //ISO15745Profile/ProfileBody/ApplicationProcess/DeviceAccessPointList/DeviceAccessPointItem/Virtual
1581 SubmoduleList/VirtualSubmoduleItem/Graphics/GraphicItemRef

1582 //ISO15745Profile/ProfileBody/ApplicationProcess/ModuleList/ModuleItem/VirtualSubmoduleList/VirtualSu
1583 bmoduleItem/Graphics/GraphicItemRef

1584 //ISO15745Profile/ProfileBody/ApplicationProcess/SubmoduleList/SubmoduleItem/Graphics/GraphicItem
1585 Ref

1586 Use: One or more

1587 Each element shall contain attributes as shown in Table 24.

1588 **A.4.7.14 IsochroneMode**

1589 This element describes attributes used for isochronous mode.

1590 Locations:

1591 //ISO15745Profile/ProfileBody/ApplicationProcess/DeviceAccessPointList/DeviceAccessPointItem/Virtual
1592 SubmoduleList/VirtualSubmoduleItem/IsochroneMode

1593 //ISO15745Profile/ProfileBody/ApplicationProcess/ModuleList/ModuleItem/VirtualSubmoduleList/VirtualSu
1594 bmoduleItem/IsochroneMode

1595 //ISO15745Profile/ProfileBody/ApplicationProcess/SubmoduleList/SubmoduleItem/IsochroneMode

1596 Use: Optional

1597 Supported since GSDML version: 2.0

1598 Each element shall contain attributes as shown in Table 42.

1599

Table 42 — Attributes of element IsochroneMode

Name	Type	Use	SV	Description
T_DC_Base	Unsigned16	Required	2.0	Time base of T_DC (data exchange cycle time), T_DC_Min and T_DC_Max in units of 31.25 μ s. The smallest possible value shall be declared. Allowed values: 1 to 1024
T_DC_Min	Unsigned16	Required	2.0	Minimum of T_DC (data exchange cycle time) in units of T_DC_Base. Allowed values: 1 to 1024 T_DC_Min shall be less than or equal to T_DC_Max. T_DC_Min multiplied by T_DC_Base shall be less than or equal to 1024.
T_DC_Max	Unsigned16	Required	2.0	Maximum of T_DC (data exchange cycle time) in units of T_DC_Base. Allowed values: 1 to 1024 T_DC_Max shall be greater than or equal to T_DC_Min. T_DC_Max multiplied by T_DC_Base shall be less than or equal to 1024
T_IO_Base	Unsigned32	Required	2.0	Time base of T_IO_Input, T_IO_Output, T_IO_InputMin, T_IO_OutputMin in nanoseconds. T_IO_Input is the point in time when the input values are collected and T_IO_Output is the point in time when the output values are taken over. The smallest possible value shall be declared. Allowed values: 1 to 32000000
T_IO_InputMin	Unsigned32	Required	2.0	The minimum time that is necessary to get and update the input values of an individual input submodule, in units of T_IO_Base. T_IO_InputMin multiplied by T_IO_Base shall be less than or equal to 32000000. $T_IO_InputMin \times T_IO_Base$ shall be less or equal than $T_DC_Max \times T_DC_Base \times 31250$.
T_IO_OutputMin	Unsigned32	Required	2.0	The minimum time that is necessary at the end of the data exchange time to get and update the output values of an individual output submodule, in units of T_IO_Base. T_IO_OutputMin multiplied by T_IO_Base shall be less than or equal to 32000000. $T_IO_OutputMin \times T_IO_Base$ shall be less or equal than $T_DC_Max \times T_DC_Base \times 31250$.
IsochroneModeRequired	Boolean	Optional	2.0	This attribute indicates whether the local isochronous application does require the IO controller to support and use isochronous mode. If the attribute is set to "true", the local isochronous application cannot be operated by an IO controller that does not support isochronous mode. Default: "false"

1600 NOTE: The above attributes follow the engineering view described in PNO-2.712 *Application Layer*
1601 *services*. The T_xx_Base values form a raster for the user settings. In contrast to that, the
1602 IsochronousModeData structure described in PNO-2.722 *Application Layer protocol* does not contain
1603 T_xx_Base values. All the elements there are based on 31.25 μ s (TimeDataCycle) or 1 ns (TimeIOxxx)
1604 respectively.

1605 **A.4.7.15 SlotCluster**

1606 This element describes attributes for a slot cluster used by fieldbus integration.

1607 Locations:

1608 //ISO15745Profile/ProfileBody/ApplicationProcess/DeviceAccessPointList/DeviceAccessPointItem/Virtual
1609 SubmoduleList/VirtualSubmoduleItem/SlotCluster

1610 //ISO15745Profile/ProfileBody/ApplicationProcess/ModuleList/ModuleItem/VirtualSubmoduleList/VirtualSu
1611 bmoduleItem/SlotCluster

1612 //ISO15745Profile/ProfileBody/ApplicationProcess/SubmoduleList/SubmoduleItem/SlotCluster

1613 Use: Required for submodules which are a FAP (Fieldbus Access Point).

1614 Supported since GSDML version: 2.25

1615 Each element shall contain attributes as shown in Table 43.

1616 **Table 43 — Attributes of element SlotCluster**

Name	Type	Use	SV	Description
Count	Unsigned16	Required	2.25	Number of fieldbus integration slots that this FAP requires.
FieldbusType	Unsigned16	Required	2.25	This attribute contains the profile specific type of the fieldbus integration profile. A link to a list of these types will be added to the table http://www.profibus.com/IM/Profile_ID_Table.xml in line "Fieldbus Integration", row "Profile Specific Type Specification" after the release of the profile. This value indicates which type of fieldbus this FAP integrates.

1617

1618 **A.4.7.16 PROFIenergy**

1619 The presence of this element indicates that this submodule supports the PROFIenergy SAP (Service
1620 Access Point).

1621 Locations:

1622 //ISO15745Profile/ProfileBody/ApplicationProcess/DeviceAccessPointList/DeviceAccessPointItem/Virtual
1623 SubmoduleList/VirtualSubmoduleItem/PROFIenergy

1624 //ISO15745Profile/ProfileBody/ApplicationProcess/ModuleList/ModuleItem/VirtualSubmoduleList/VirtualSu
1625 bmoduleItem/PROFIenergy

1626 //ISO15745Profile/ProfileBody/ApplicationProcess/SubmoduleList/SubmoduleItem/PROFIenergy

1627 Use: optional

1628 Supported since GSDML version: 2.25

1629 Attributes and child elements: None

1630 **A.4.8 InterfaceSubmoduleItem related elements**

1631 **A.4.8.1 InterfaceSubmoduleItem**

1632 Represents the network interface of a DAP.

1633 Location:
 1634 //ISO15745Profile/ProfileBody/ApplicationProcess/DeviceAccessPointList/DeviceAccessPointItem/Syste
 1635 mDefinedSubmoduleList/InterfaceSubmoduleItem

1636 Use: Required

1637 Each element shall contain attributes as shown in Table 44.

1638 **Table 44 — Attributes of element InterfaceSubmoduleItem**

Name	Type	Use	SV	Description
ID	Id	Required	2.25	Contains the ID of the element
SubmoduleIdentNumber	Unsigned32hex	Required	2.0	Contains the submodule ident number for identification purposes.
TextId	RefId	Required	2.0	Contains the ID of a text as a reference into the "ExternalTextList".
SubslotNumber	Unsigned16	Optional	2.0	Defines the subslot number of the interface submodule. Only one of the following values is allowed: 32768 (0x8000 – Interface 1) 33024 (0x8100 – Interface 2) ... 36608 (0x8F00 – Interface 16) NOTE: If compatibility to GSDML V1.0 and V2.0 based engineering tools is required it is recommended to set this attribute to 32768. Default: "32768"
SupportedRT_Class	Enumeration	Optional	2.0	Obsolete – it is recommended to use the attribute "SupportedRT_Classes". Defines the Realtime Class of the interface. One of the following values are to use: "Class1" "Class2" "Class3" Default: "Class1" NOTE: "Class2" and "Class3" also include support of RT_CLASS_1. But "Class3" does not include support of RT_CLASS_2. This is because the definition of RT_CLASS_3 was completed before the definition of RT_CLASS_2. See explanation on Realtime Classes below.
SupportedRT_Classes	TokenList1	Optional	2.1	Defines all Realtime Classes supported by the interface. Each class shall be separated by a semicolon. One or more of the following class types can be used: "RT_CLASS_UDP" "RT_CLASS_1" "RT_CLASS_2" "RT_CLASS_3" If this attribute is present, the attribute "SupportedRT_Class" shall be ignored. This attribute shall be present if the attribute PNIO_Version at the DAP is >= "V2.3". See NOTE below regarding SyncBoundaries. See explanation on Realtime Classes below.

Name	Type	Use	SV	Description
IsochroneModeSupported	Boolean	Optional	2.0	Obsolete – it is recommended to use the attribute “IsochroneModeInRT_Classes” Describes if the Interface is able to work in an isochronous environment. Default: “false” If this attribute is set to “true” the attribute “SupportedRT_Class” shall be set to “Class3” or the attribute “SupportedRT_Classes” shall contain “RT_CLASS_3”. If the attribute is not set to “true”, the element “UseableModules” of the corresponding DAP shall not reference to modules, which require isochronous mode. (Attribute “IsochroneModeRequired” from one or more Submodules set to “true”.)
IsochroneModeInRT_Classes	TokenList	Optional	2.2	Describes the RT_Classes in which the interface is able to work in an isochronous environment. Each class shall be separated by a semicolon. One or more of the following class types can be used: “RT_CLASS_2” “RT_CLASS_3” Default: “” The RT classes specified in this list shall also appear in SupportedRT_Classes. If the attribute is empty, the element “UseableModules” of the corresponding DAP shall not reference to modules, which require isochronous mode. (Attribute “IsochroneModeRequired” from one or more Submodules set to “true”.) If this attribute is present, the attribute “IsochroneModeSupported” shall be ignored.
SupportedProtocols	TokenList	Required	2.1	This attribute describes the supported protocols of an interface. Each protocol type shall be separated by a semicolon character in the string. One or both of the following protocol types can be used: “SNMP” “LLDP” The string “SNMP” shall be used if the interface supports the “Simple Network Management Protocol” The string “LLDP” shall be used if the interface supports the “Link Layer Discovery Protocol” NOTE: DHCP and DCP protocol ability is described by the attribute “AddressAssignment” of the corresponding DAP.
SupportedMibs	TokenList	Required	2.1	This attribute describes the supported SNMP MIBs (Management Information Base) of the Interface. Each MIB type shall be separated by a semicolon character in the string. EXAMPLE: “MIB2”: Interface supports the SNMP MIB2 Management Information Base.

Name	Type	Use	SV	Description
NetworkComponent-DiagnosisSupported	Boolean	Optional	2.1	Defines if the ports of this interface are able to send alarms in case of a problem of the network components, e.g. frame dropped, no resources. (ChannelErrorType NETWORK_COMPONENT_FUNCTION_MISMATCH) If this attribute is present and "true" the ports of this interface support PDNCDataCheck. Default: "false"
DCP_HelloSupported	Boolean	Optional	2.2	Defines if the interface supports device initiative. Default: "false"
PTP_BoundarySupported	Boolean	Optional	2.2	If this attribute is present and "true" the ports of this interface support the AdjustPeerToPeerBoundary block in PDPortDataAdjust. Default: "false"
DCP_BoundarySupported	Boolean	Optional	2.2	If this attribute is present and "true" the ports of this interface support the AdjustDCPBoundary block in PDPortDataAdjust. Default: "false"
MulticastBoundarySupported	Boolean	Optional	2.25	If this attribute is present and "true" the ports of this interface support the AdjustMulticastBoundary block in PDPortDataAdjust. Default: "false"
ParameterizationDisallowed	Boolean	Optional	2.25	If this attribute is present and "true", the parameterization on this interface submodule shall not be changed, i.e. writing record data is not allowed on this submodule. Default: "false"
DelayMeasurementSupported	Boolean	Optional	2.25	Defines if the ports of this interface are able to send PTCF-Delay telegrams for delay measurement. Default: "false" NOTE: If SynchronisationMode/@SupportedSync-Protocols contains the protocol type "PTCP", the delay measurement is implicitly supported by the ports.
Writeable_IM_Records	ValueList	Optional	2.25	Defines, which I&M records as defined in Profile Guidelines <i>Part 1 – Identification & Maintenance Functions</i> can be transferred to the interface submodule using the record data index defined in PNO-2.722 <i>Application Layer protocol</i>. In this attribute, the numbers of all writeable I&M records shall be listed. The I&M0 record can't be written, so 0 shall not be contained in this list. EXAMPLE: "1 2" defines that I&M1 and I&M2 can be written to the device with record data index 0xAFF1 and 0xAFF2 Default: No I&M record can be written

NOTE: In RT class 2, synchronization is implemented as broadcasts, which are usually propagated throughout the network. To reduce network load, SyncBoundaries are defined which stop propagation of synchronization broadcasts. There is no separate attribute to indicate this; SyncBoundaries and also CableDelayMismatch alarms are enabled when attribute “SupportedRT_Classes” contains the token “RT_CLASS_2” and attribute “SynchronisationMode/@SupportedRole” exists.

Explanation of Realtime Classes:

The terms RT_CLASS_xxx represent a communication class (a marketing term, e.g. for the user interface of the engineering tool). It is not to be confused with the traffic class or the frame ID range. The following table correlates these terms:

Table 45 — Realtime Classes

Traffic class	Communication class (marketing)	Frame ID range	SupportedRT_Class is	SupportedRT_Classes contains [Priority]
GREEN	RT_CLASS_1	range 7 (legacy)	Class1 or Class2 or Class3	<i>not present</i>
		range 6	<i>don't care</i>	RT_CLASS_1
ORANGE	RT_CLASS_2	range 6	Class2	RT_CLASS_2
RED	RT_CLASS_3	range 3	Class3	RT_CLASS_3
GREEN	RT_CLASS_UDP	range 7	–	RT_CLASS_UDP

The terms “traffic class” and “frame ID range” are defined in PNO-2.722 *Application Layer protocol*.

Allowed realtime class combinations according to PNO-2.742 *Profiles for decentralized periphery*:

- RT_CLASS_UDP is independent of the other classes
- RT_CLASS_1 shall be supported
- When RT_CLASS_2 is supported, RT_CLASS_3 shall also be supported (but RT_CLASS_3 may be supported without RT_CLASS_2)

A.4.8.2 General

This element contains general information about the interface submodule.

Location:

//ISO15745Profile/ProfileBody/ApplicationProcess/ModuleList/ModuleItem/SystemDefinedSubmoduleList/InterfaceSubmoduleItem/General

Use: Optional

Attributes: None

A.4.8.3 DCP_FlashOnceSignalUnit

This element contains a human readable text. It shall describe the location on the device of the signal unit for the DCP flash signal.

Location:

//ISO15745Profile/ProfileBody/ApplicationProcess/ModuleList/ModuleItem/SystemDefinedSubmoduleList/InterfaceSubmoduleItem/General/ DCP_FlashOnceSignalUnit

Use: Optional

Attributes: See Table 46.

1672

Table 46 — Attributes of element DCP_FlashOnceSignalUnit

Name	Type	Use	SV	Description
TextId	RefId	Required	2.1	Contains the ID of a text as a reference into the "ExternalTextList".

1673

1674 A.4.8.4 RecordDataList

1675 This element contains a list of ParameterRecordDataItems (see A.4.10), which define manufacturer
1676 specific record data objects of the interface submodule.

1677 The engineering tool should show the content of the ParameterRecordDataItems in the same order as in
1678 the GSDML file.

1679 Location:
1680 //ISO15745Profile/ProfileBody/ApplicationProcess/ModuleList/ModuleItem/SystemDefinedSubmoduleList/
1681 InterfaceSubmoduleItem/RecordDataList

1682 Use: Optional

1683 Attributes: None

1684 A.4.8.5 RT_Class3Properties

1685 This element describes the characteristics of devices specific for RT class 3. It is ignored if the RT class
1686 of the assigned InterfaceSubmoduleItem is not set to "RT_CLASS_3".

1687 Location:
1688 //ISO15745Profile/ProfileBody/ApplicationProcess/DeviceAccessPointList/DeviceAccessPointItem/Syste
1689 mDefinedSubmoduleList/InterfaceSubmoduleItem/RT_Class3Properties

1690 Use: Required for InterfaceSubmodules of RT_CLASS_3.

1691 Each element shall contain attributes as shown in Table 47.

1692

Table 47 — Attributes of element RT_Class3Properties

Name	Type	Use	SV	Description
StartupMode	TokenList1	Optional	2.3	Default: "Legacy" Allowed values: "Advanced", "Legacy" This attribute describes the supported startup modes for the application relation using RT_CLASS_3. If the attribute is not existent or contains the token "Legacy", the device supports the legacy startup mode for RT_CLASS_3. If the attribute contains the token "Advanced", the device supports the advanced startup mode for RT_CLASS_3.

Name	Type	Use	SV	Description
ForwardingMode	TokenList1	Optional	2.3	Contains the forwarding modes supported by this device. This attribute describes the format of the PDIRData. Each forwarding mode shall be separated by a semicolon character. The following strings can be used for description: “Absolute”: Means that the device supports the absolute forwarding mode. “Relative”: Means that the device supports the relative forwarding mode. This value is applicable for devices with two ports only. If the attribute is present, it shall contain at least one token. In this case the following blocks are generated: • PDIRGlobalData (V1.2) • PDIRFrameData (V1.1) • PDIRBeginEndData (V1.0) If the attribute is missing the following blocks are generated: • PDIRGlobalData (V1.1) • PDIRFrameData (V1.0) • PDIRBeginEndData (V1.0)
MaxBridgeDelay	Unsigned16	Required	2.0	Time delay needed for internal bridging of frames. Base: 1 ns This value assumes an unshortened preamble length (7 octets). See PNO-2.722 <i>Application Layer protocol</i>, “Model parameter for GSDML usage”.
MaxBridgeDelayFFW	Unsigned16	Optional	2.3	Time delay needed for internal bridging of frames using FFW addresses. Base: 1 ns This value assumes an unshortened preamble length (7 octets). See PNO-2.722 <i>Application Layer protocol</i>, “Model parameter for GSDML usage”.
MaxDFP_Feed	Unsigned16	Optional	2.3	The feed time needed to bridge an inbound DFP subframe from the RxPort to the TxPort. Base: 1 ns Allowed values: 0..500 ns
MaxDFP_Frames	Unsigned16	Optional	2.3	The maximum number of ARs simultaneously using DFP supported by the device. Shall not exceed ApplicationRelations/@NumberOfAR. Default: 0
AlignDFP_Subframes	Boolean	Optional	2.3	Indicates whether to align subframes on 16 bit boundaries. Default: “false”
MaxNumberIR_FrameData	Unsigned16	Required	2.0	Maximum number of IRT forwarders supported by the interface. Only those IRFrameData elements in the PDIRData record shall be counted that are not locally received or injected by this node (i.e. that are really forwarded). Sync frames shall not be counted. This parameter indicates the number of forwarding resources for use as absolute forwarder. Allowed values: 0..1024

Name	Type	Use	SV	Description
MaxRangeIR_FrameID	Unsigned16	Optional	2.3	Contains the maximum width of the RedGuard supported by this device. Default: 1024 Allowed values: 0..4096
FragmentationType	Enumeration	Optional	2.3	No Default (neither dynamic, nor static fragmentation is supported) Allowed values: "Dynamic"; "Static" If the attribute is set to "Static", the device supports static fragmentation. If the attribute is set to "Dynamic", the device supports dynamic fragmentation. For both the minimum supported fragment size is 108 byte (including FCS, but excluding preamble, start frame delimiter and IFG).
MaxRedPeriodLength	Unsigned16	Optional	2.3	The maximum length of a RED period supported by this device. Base: 1 µs Default: 3875 µs Allowed values: 500..4000 µs
MinFSO	Unsigned16	Optional	2.3	The minimum frame send offset within the RED period supported by this device. Base: 1 ns Default: 5000 ns Allowed values: 1760..5000 ns

1693

1694 **A.4.8.6 SynchronisationMode**

1695 This element describes attributes used for synchronisation purposes.

1696 Location:

1697 //ISO15745Profile/ProfileBody/ApplicationProcess/DeviceAccessPointList/DeviceAccessPointItem/SystemDefinedSubmoduleList/InterfaceSubmoduleItem/SynchronisationMode

1698

1699 Use: Required for InterfaceSubmodules of RT_CLASS_2 or RT_CLASS_3.

1700 Each element shall contain attributes as shown in Table 48.

1701

Table 48 — Attributes of element SynchronisationMode

Name	Type	Use	SV	Description
SupportedRole	Enumeration	Optional	2.0	Defines the supported synchronisation roles of the interface. One of the following values are to use: “SyncMaster” “SyncMasterRedundant” “SyncSlave” “SyncMaster+SyncSlave” “SyncMasterRedundant+SyncSlave” Default: “SyncSlave” NOTE: “SyncMasterRedundant” includes the capability to act as sync master in a non-redundant setup. See also NOTE in A.4.8.1 regarding SyncBoundaries. Be careful: The values “SyncMasterRedundant” and “SyncMasterRedundant+SyncSlave” were introduced in GSDML version 2.2. Use of these values breaks backward compatibility to engineering tools that don't support GSDML V2.2.
MaxLocalJitter	Unsigned16	Required	2.0	The maximum increase of the jitter between input port and output port. Base: 1 ns
T_PLL_MAX	Unsigned16	Optional	2.0	The maximum value of the jitter regarding to the beginning of a communication cycle. Base: 1 ns Default: “1000” (1 µs)
PeerToPeerJitter	Unsigned16	Optional	2.3	The maximum synchronization jitter between two peers. If a device supports send clocks faster than 250 µs and dynamic frame packing, the PeerToPeerJitter shall be lower than 300 ns. Base: 1 ns Default: T_PLL_MAX
SupportedSyncProtocols	TokenList	Optional	2.2	Defines the supported synchronisation protocols of an interface. Each protocol type shall be separated by a semicolon character in the string. The following protocol types can be used: “PTCP” Default: “”

1702

1703 A.4.8.7 ApplicationRelations

1704 This element contains information about the application relations implemented in an IO Device.

1705 Location:

1706 //ISO15745Profile/ProfileBody/ApplicationProcess/DeviceAccessPointList/DeviceAccessPointItem/SystemDefinedSubmoduleList/InterfaceSubmoduleItem/ApplicationRelations

1708 Use: Optional

1709 Each element shall contain attributes as shown in Table 49.

1710

Table 49 — Attributes of element ApplicationRelations

Name	Type	Use	SV	Description
StartupMode	TokenList1	Optional	2.3	Default: "Legacy" Allowed values: "Advanced", "Legacy" This attribute describes the supported startup modes for the application relation using RT_CLASS_1 and RT_CLASS_2. If the attribute is not existent or contains the token "Legacy", the device supports the legacy startup mode for RT_CLASS_1 and RT_CLASS_2. If the attribute contains the token "Advanced", the device supports the advanced startup mode for RT_CLASS_1 and RT_CLASS_2.
NumberOfAdditionalInputCR	Unsigned16	Optional	2.0	Each PROFINET device has to implement at least one input CR. This attribute defines how many additional Input CRs are supported by the device per AR (see NumberOfAR). Default: "0"
NumberOfAdditionalOutputCR	Unsigned16	Optional	2.0	Each PROFINET device has to implement at least one output CR. This attribute defines how many additional Output CRs are supported by the device per AR (see NumberOfAR). Default: "0"
NumberOfAdditionalMulticastProviderCR	Unsigned16	Optional	2.0	Each PROFINET device has to implement at least one multicast provider CR. This attribute defines how many additional multicast provider CRs are supported by the device per AR (see NumberOfAR). Default: "0"
NumberOfMulticastConsumerCR	Unsigned16	Optional	2.0	This attribute defines how many multicast consumer CRs are supported by the device per AR (see NumberOfAR). Default: "0"
PullModuleAlarmSupported	Boolean	Optional	2.2	Describes if the pull module alarm is supported. Default: "false"
NumberOfAR	Unsigned16	Optional	2.25	The number of application relationships (ARs) that the device supports for e.g. Shared Device and System Redundancy. The Device Access AR and implicit AR do not count here. Default: "1"

1711

1712 A.4.8.8 TimingProperties

1713 This element defines the timing behaviour for sending cyclic IO data.

1714 Location:

1715 //ISO15745Profile/ProfileBody/ApplicationProcess/DeviceAccessPointList/DeviceAccessPointItem/SystemDefinedSubmoduleList/InterfaceSubmoduleItem/ApplicationRelations/TimingProperties

1717 Use: Optional

1718 Each element shall contain attributes as shown in Table 50.

1719 **Table 50 — Attributes of element TimingProperties**

Name	Type	Use	SV	Description
SendClock	ValueList	Optional	2.0	Contains a list of cycle times supported by the DAP for sending cyclic data. Basic clock tick is 31.25 μ s. Each value of this element contains the multiplier of the basic clock tick and can be in the range "1" to "128" (see Example). Default: "32"
ReductionRatio	ValueList	Optional	2.0	The send clock interval can be reduced by a reduction. This attribute describes the supported reduction ratios of a DAP for all SendClocks. Each value of this element shall be a power of 2 in the range "1" to "512". Default: "1 2 4 8 16 32 64 128 256 512" This default applies only when both ReductionRatioPow2 and ReductionRatioNonPow2 are missing.
ReductionRatioPow2	ValueList	Optional	2.2	The send clock interval can be reduced by a reduction. This attribute describes the reduction ratios of a DAP which are supported (in addition to those described in attribute ReductionRatio) for SendClocks which are a power of 2. Each value of this element shall be a power of 2 in the range "1" to "512".
ReductionRatioNonPow2	ValueList	Optional	2.2	The send clock interval can be reduced by a reduction. This attribute describes the reduction ratios of a DAP which are supported (in addition to those described in attribute ReductionRatio) for SendClocks which are not a power of 2. Each value of this element shall be a power of 2 in the range "1" to "512".
PreferredSendClock	Unsigned16	Optional	2.25	The send clock preferred by the device. The value must be one of the send clock factors from the SendClock value list.
EXAMPLE SendClock="16 32 64" means that the DAP is able to send cyclic data in interval 500 μ s, 1 ms or 2 ms.				
NOTE There are further restrictions on SendClock / ReductionRatio pairs specified in PNO-2.722 <i>Application Layer protocol</i> , "Coding of the field SendClockFactor" to be considered.				

1720

1721 A.4.8.9 RT_Class3TimingProperties

1722 This element describes the timing behaviour of devices specific for RT class 3.

1723 Location:

1724 //ISO15745Profile/ProfileBody/ApplicationProcess/DeviceAccessPointList/DeviceAccessPointItem/SystemDefinedSubmoduleList/InterfaceSubmoduleItem/ApplicationRelations/RT_Class3TimingProperties

1726 Use: Optional

1727 Each element shall contain attributes as shown in Table 51.

1728

Table 51 — Attributes of element RT_Class3TimingProperties

Name	Type	Use	SV	Description
SendClock	ValueList	Optional	2.0	Contains a list of cycle times supported by the DAP for sending cyclic data. Basic clock tick is 31.25 μ s. Each value of this element contains the multiplier of the basic clock tick and can be in the range "1" to "128" (see Example). Default: "32".
ReductionRatio	ValueList	Optional	2.0	The send clock interval can be reduced by a reduction. This attribute describes the supported reduction ratios of a DAP for all SendClocks. Each value of this element shall be a power of 2 in the range "1" to "512". Default: "1 2 4 8 16" This default applies only when both ReductionRatioPow2 and ReductionRatioNonPow2 are missing.
ReductionRatioPow2	ValueList	Optional	2.2	The send clock interval can be reduced by a reduction. This attribute describes the reduction ratios of a DAP which are supported (in addition to those described in attribute ReductionRatio) for SendClocks which are a power of 2. Each value of this element shall be a power of 2 in the range "1" to "512".
ReductionRatioNonPow2	ValueList	Optional	2.2	The send clock interval can be reduced by a reduction. This attribute describes the reduction ratios of a DAP which are supported (in addition to those described in attribute ReductionRatio) for SendClocks which are not a power of 2. Each value of this element shall be a power of 2 in the range "1" to "512".
PreferredSendClock	Unsigned16	Optional	2.25	The send clock preferred by the device. The value must be one of the send clock factors from the SendClock value list.
EXAMPLE SendClock="16 32 64" means that the DAP is able to send cyclic data in interval 500 μ s, 1 ms or 2 ms.				
NOTE There are further restrictions on SendClock / ReductionRatio pairs specified in PNO-2.722 <i>Application Layer protocol</i> , "Coding of the field SendClockFactor" to be considered.				

1729

1730 A.4.8.10 MediaRedundancy

1731 If this element is present, the interface submodule supports media redundancy via MRP. In this case, at
1732 least two assigned port submodules shall have their attribute "IsDefaultRingport" set to "true".

1733 If more than two port submodules have the attribute "IsDefaultRingport" set to "true", the engineering tool
1734 shall choose the "first" two port submodules as default ringports. "First" means the port submodule with
1735 the lowest slot number and within the slot the port submodule with the lowest subslot number.

1736 Engineering tools shall allow a ring if either

1737 1. MRP is active on all devices in the ring

1738 or

1739 2. none of the devices have MRP active, and all (might) support a different media redundancy
1740 protocol.

1741 In case 1 the engineering tool configures the media redundancy. The element "MediaRedundancy" is
1742 required in the GSDs of all devices in the ring.

1743 In case 2 the engineering tool can't configure the media redundancy, but shall not prevent a ring. In each
1744 of the GSDs of the devices the element "MediaRedundancy" shall either be present and the attribute

1745 “AdditionalProtocolsSupported” be true, or it shall not be present (MRP is not supported, and it is
1746 unknown whether the device supports other media redundancy protocols).

1747 Location:
1748 //ISO15745Profile/ProfileBody/ApplicationProcess/DeviceAccessPointList/DeviceAccessPointItem/Syste
1749 mDefinedSubmoduleList/InterfaceSubmoduleItem/MediaRedundancy

1750 Use: Optional

1751 Support: GSDML V2.1

1752 Each element shall contain attributes as shown in Table 52.

1753 **Table 52 — Attributes of element MediaRedundancy**

Name	Type	Use	SV	Description
SupportedRole	TokenList1	Optional	2.1	Describes, which roles are supported by the interface. Each role shall be separated by a semicolon character. The following strings can be used for description: “Manager” “Client” The first entry in this attribute shall be used as default by the engineering tool. This default shall also correlate to the behaviour of the device without parameterization, e.g. after factory reset. If this attribute is missing, the default value is “Client”.
MRPD_Supported	Boolean	Optional	2.3	Device supports media redundancy for cyclic realtime data. Default: “false”
MRT_Supported	Boolean	Optional	2.3	Device supports media redundancy timer for cyclic realtime data. Default: “false”
AdditionalProtocolsSupported	Boolean	Optional	2.25	Set this attribute to “true” if the device supports at least one media redundancy protocol other than MRP. In a ring configuration with at least one device not supporting MRP, the engineering tool shall not refuse the ring if all devices in the ring support at least one other media redundancy protocol. Default: “false”

1754

1755 **A.4.9 PortSubmoduleItem related elements**

1756 **A.4.9.1 PortSubmoduleItem** (pluggable)

1757 The PortSubmoduleItem represents a switch port which is referenced by a DAP or module.

1758 //ISO15745Profile/ProfileBody/ApplicationProcess/SubmoduleList/PortSubmoduleItem

1759 Use: One for every type of switch port

1760 Each element shall contain the elements and attributes of the base class PortSubmoduleItemBase (see
1761 A.4.9.6) plus the attribute shown in Table 53 and the elements ModuleInfo and Graphics.

1762 **Table 53 — Additional attributes of element PortSubmoduleItem (pluggable)**

Name	Type	Use	SV	Description
RequiredSchemaVersion	VersionString	Optional	2.25	The minimal version of the GSDML Schema needed to create valid configuration data for the port submodule. If an engineering tool is not able to interpret this schema version it shall refuse to configure this port submodule. For compatibility issues it is recommended to keep the RequiredSchemaVersion as low as possible. The RequiredSchemaVersion shall be no higher than the GSDML version that this GSD is based on. Default: "V2.25"

1763

1764 **A.4.9.2 ModuleInfo**

1765 Locations:

1766 //ISO15745Profile/ProfileBody/ApplicationProcess/SubmoduleList/PortSubmoduleItem/ModuleInfo

1767 Use: Required

1768 Attributes and child elements: See A.4.12.1.

1769 **A.4.9.3 Graphics**

1770 The Graphics element contains a list of GraphicItemRef.

1771 Locations:

1772 //ISO15745Profile/ProfileBody/ApplicationProcess/SubmoduleList/PortSubmoduleItem/Graphics

1773 Use: Optional

1774 Attributes: None

1775 **A.4.9.4 GraphicItemRef**

1776 The GraphicItemRef element references graphical information about a device module or submodule.

1777 Locations:

1778 //ISO15745Profile/ProfileBody/ApplicationProcess/SubmoduleList/PortSubmoduleItem/Graphics/GraphicItemRef

1780 Use: One or more

1781 Each element shall contain attributes as shown in Table 24.

1782 **A.4.9.5 PortSubmoduleItem (builtin)**

1783 This element defines the characteristics of a builtin port submodule representing a switch port of a DAP or
1784 module.

1785 Location:

1786 //ISO15745Profile/ProfileBody/ApplicationProcess/DeviceAccessPointList/DeviceAccessPointItem/SystemDefinedSubmoduleList/PortSubmoduleItem

1788 //ISO15745Profile/ProfileBody/ApplicationProcess/ModuleList/ModuleItem/SystemDefinedSubmoduleList/
1789 PortSubmoduleItem

1790 Use: One for every switch port

1791 Each element shall contain the elements and attributes of the base class PortSubmoduleItemBase (see
 1792 A.4.9.6) plus the attributes shown in Table 54.

1793 **Table 54 — Additional attributes of element PortSubmoduleItem (builtin)**

Name	Type	Use	SV	Description
SubslotNumber	Unsigned16	Required	2.0	Defines the subslot number of the port submodule. The value shall be in one of the following ranges: 32769 to 33023 (0x8001 to 0x80FF – Interface 1) 33025 to 33279 (0x8101 to 0x81FF – Interface 2) ... 36609 to 36863 (0x8F01 to 0x8FFF – Interface 16) The pattern here is 0x8<i>ipp</i> where <i>i</i> is the interface number (0-15) and <i>pp</i> is the port number within the interface (1-255), while <i>pp</i>=0 represents the interface submodule itself. Within the "SystemDefinedSubmoduleList" element the "SubslotNumber" attribute shall be unique.
TextId	RefId	Required	2.0	If a DAP or module has more than one port, a label is needed to identify a certain port by humans. This attribute shall contain this port label even if the DAP or module has only one port. The attribute contains the ID of a text as a reference into the "ExternalTextList".

1794

1795 **A.4.9.6 PortSubmoduleItemBase** (abstract)

1796 The base class of a port submodule. Used by PortSubmoduleItem (pluggable) and PortSubmoduleItem
 1797 (builtin).

1798 Each derived element shall contain attributes as shown in Table 55.

1799 **Table 55 — Attributes of PortSubmoduleItemBase**

Name	Type	Use	SV	Description
ID	Id	Required	2.25	Contains the ID of the element
SubmoduleIdentNumber	Unsigned32hex	Required	2.0	Contains the submodule ident number for identification purposes.
MAUType	Enumeration	Optional	2.0	Obsolete – it is recommended to use the attribute "MAUTypes" This attribute describes the physical characteristics of a port. One of the following values is allowed: "100BASETXFD" "100BASEFXFD" Default: "100BASETXFD"

Name	Type	Use	SV	Description
MAUTypes	ValueList	Optional	2.1	This attribute describes all supported MAU Types of a port. Each MAU Type shall be represented by the value defined in PNO-2.722 <i>Application Layer protocol</i>, "Coding of the field MAUType". If this attribute is present, the attribute "MAUType" shall be ignored. If this attribute is present the port submodule supports the AdjustMAUType block in PDPortDataAdjust. EXAMPLE: "16 24" means that the following MAU Types are supported: "100BASETXFD" "1000BASELXFD"
FiberOpticTypes	ValueList	Optional	2.1	This attribute is mandatory for a fiber optic port. If not present, the engineering tool assumes a copper port. Describes all supported Fiber Optic Types of a port. Each Fiber Optic Type shall be represented by the value defined in PNO-2.722 <i>Application Layer protocol</i>, "Coding of the field FiberOpticType". EXAMPLE: "1 2 3" means that the following fiber optic types can be used: "9 µm single mode fiber" "50 µm multi mode fiber" "62.5 µm multi mode fiber"
MaxPortTxDelay	Unsigned16	Required for RT_CLASS_3 interfaces.	2.0	Port specific time delay needed for sending frames. Base: 1 ns According to IEEE 802.3-2008, section 2, Table 24-3, for 100 Mbit/s full duplex, the maximum value is 14 Tbit = 140 ns. See PNO-2.722 <i>Application Layer protocol</i>, "Model parameter for GSDML usage".
MaxPortRxDelay	Unsigned16	Required for RT_CLASS_3 interfaces.	2.0	Port specific time delay needed for receiving frames. Base: 1 ns According to IEEE 802.3-2008, section 2, Table 24-3, for 100 Mbit/s full duplex, the maximum value is 32 Tbit = 320 ns. See PNO-2.722 <i>Application Layer protocol</i>, "Model parameter for GSDML usage".
PortDeactivationSupported	Boolean	Optional	2.1	Defines if a port can be deactivated by the user of the engineering tool (LinkState.State="Down" in sub block AdjustLinkState of PDPortDataAdjust). Default: "false"
LinkStateDiagnosisCapability	Enumeration	Optional	2.1	Defines if a port submodule is able to send alarms when the link state of the port changes. One of the following value is allowed "Up" "Down" "Up+Down" If this attribute is missing, the port submodule has no link state diagnosis capability.

Name	Type	Use	SV	Description
PowerBudgetControlSupported	Boolean	Optional	2.1	Defines if a port is able to send alarms in case of a fiber optic budget problem. (ChannelErrorType FIBER_OPTIC_MISMATCH) Default: "false" If the attribute FiberOpticTypes is not given, this attribute shall not be present.
SupportsRingportConfig	Boolean	Optional	2.1	Defines, if the port is configurable as ring port. Default: "false"
IsDefaultRingport	Boolean	Optional	2.1	Port has ring port capability. If the attribute "SupportsRingportConfig" is set to false, the ring port capability cannot be switched off by the engineering tool. Default: "false"
ParameterizationDisallowed	Boolean	Optional	2.25	If this attribute is present and "true", the parameterization on this port submodule shall not be changed, i.e. writing record data is not allowed on this submodule. Default: "false"
Writeable_IM_Records	ValueList	Optional	2.25	Defines, which I&M records as defined in Profile Guidelines Part 1 – Identification & Maintenance Functions can be transferred to the port submodule using the record data index defined in PNO-2.722 Application Layer protocol. In this attribute, the numbers of all writeable I&M records shall be listed. The I&M0 record can't be written, so 0 shall not be contained in this list. EXAMPLE: "1 2" defines that I&M1 and I&M2 can be written to the device with record data index 0xAFF1 and 0xAFF2 Default: No I&M record can be written
CheckMAUTypeSupported	Boolean	Optional	2.25	If this attribute is present and "true", the port submodule supports the CheckMAUType and CheckMAUTypeDifference block in PDPortDataCheck. If this attribute is not present, the value of PortDeactivationSupported controls the support of CheckMAUType and CheckMAUTypeDifference.
ShortPreamble100MBitSupported	Boolean	Optional	2.3	If the attribute is missing, the data record PDPortDataAdjust (index 0x802F) shall not contain the subblock AdjustPreambleLength. If the attribute is set to "false", the device supports preambles with length 7 octets only. If the attribute is set to "true", the device supports preambles with length 7 octets and preambles with length 1 octet for 100 MBit.

1800 A.4.9.7 RecordDataList

1801 This element contains a list of ParameterRecordDataItems (see A.4.10), which define manufacturer
1802 specific record data objects of the port submodule.

1803 The engineering tool should show the content of the ParameterRecordDataItems in the same order as in
1804 the GSDML file.

1805 Location:
 1806 //ISO15745Profile/ProfileBody/ApplicationProcess/DeviceAccessPointList/DeviceAccessPointItem/Syste
 1807 mDefinedSubmoduleList/PortSubmoduleItem/RecordDataList

1808 //ISO15745Profile/ProfileBody/ApplicationProcess/ModuleList/ModuleItem/SystemDefinedSubmoduleList/
 1809 PortSubmoduleItem/RecordDataList

1810 Use: Optional

1811 Attributes: None

1812 **A.4.10 RecordDataItem related elements**

1813 The ARVendorBlock request describes a device-global manufacturer specific parameter record which is
 1814 transferred within the IOD_ConnectReq.

1815 ParameterRecordDataItems describe manufacturer specific parameter records which will be transferred
 1816 to the submodules during the start-up procedure of an IO device.

1817 Their exact transfer sequence is:

- 1818 • Within the IO Device, all slots carrying a module are processed in ascending order.
- 1819 • Within each module, all subslots carrying a submodule are processed in ascending order.
- 1820 • Within each submodule, the system defined parameter records will always be transferred first.
 1821 The manufacturer specific parameter records will then be transferred according to their attribute
 1822 @TransferSequence.

1823 **A.4.10.1 RecordDataBase (abstract)**

1824 RecordDataBase is the abstract data type from which both the ARVendorBlock/Request and the
 1825 ParameterRecordDataItem are derived.

1826 Each derived element shall contain attributes as shown in Table 56 plus the sub-elements Const (see
 1827 A.4.10.2), Ref (see A.4.10.3) and MenuList (see A.4.10.4).

1828 **Table 56 — Attributes of RecordDataBase (abstract)**

Name	Type	Use	SV	Description
Length	Unsigned32	Required	1.0	Contains the length of the record data object in octets.

1830 **A.4.10.2 Const**

1831 The Const element is used to initialize the content of a record data object. If the Const definition does not
 1832 describe the complete content of the record data object, the undefined fields shall be set to zero.

1833 If the Const element is missing, the record data object is initialized with octets set to zero.

1834 If more than one Const element is defined no overlapping between the definitions is allowed.

1835 Use: Zero or more

1836 Each element shall contain attributes as shown in Table 57.

1837

Table 57 — Attributes of element Const

Name	Type	Use	SV	Description
ByteOffset	Unsigned32	Optional	1.0	Contains the offset in octets of the referenced data object from the beginning of the record data object. Default: 0 The length of the const definition shall be less than or equal to the Length attribute of the RecordDataItem element.
Data	String	Required	1.0	Contains the data content of the record data object starting at "ByteOffset" (see Example). The string is separated in hexadecimal tokens and shall fulfill the rules of the following regular expression: (0x[0-9a-fA-F][0-9a-fA-F],)*0x[0-9a-fA-F][0-9a-fA-F]
EXAMPLE 0x10,0xff,0x4A				

1838

A.4.10.3 Ref

1840 This element references a data object in a record data block.

1841 Because this element may describe the same object as the "Const" element, overlapping may occur. In
1842 this case, the "Ref" definition shall be used.

1843 If more than one Ref element is defined no overlapping between the definitions is allowed.

1844 Use: Zero or more

1845 Each element shall contain attributes as shown in Table 58.

1846

Table 58 — Attributes of element Ref

Name	Type	Use	SV	Description
ID	IdT	Optional	2.3	ID required for referencing this parameter from a MenuItem.
ValueItemTarget	RefId	Optional	1.0	References a ValueItem Element containing the name and value range of the data object. ValueItems can only have numerical Assignments. Thus referencing a ValueItem that has Assignments is only possible for the "Bit", "BitArea" or one of the "IntegerXX", "UnsignedXX" or "FloatXX" DataTypes.
ByteOffset	Unsigned32	Required	1.0	Contains the offset in octets of the referenced data object from the beginning of the record data object.
BitOffset	Integer	Optional	1.0	Contains the offset in bits of the referenced data object from the beginning of the referenced octet. Valid Range: 0..7 Default: "0" Can only be used in conjunction with data type "Bit" or "BitArea".
BitLength	Integer	Optional	1.0	If the type of this data object is "BitArea", this attribute specifies the length of this field. The attribute is ignored if another data type is used. Default: "1" The BitLength shall be in range from "1" to "15".
DataType	Enumeration	Required	1.0	Specifies the type of the data object. One of the following values are to use: "Bit"

				“BitArea” (see Note) “Integer8” “Integer16” “Integer32” “Integer64” “Unsigned8” “Unsigned16” “Unsigned32” “Unsigned64” “Float32” “Float64” “Date” “TimeOfDay with date indication” “TimeOfDay without date indication” “TimeDifference with date indication” “TimeDifference without date indication” “NetworkTime” “NetworkTimeDifference” “VisibleString” “OctetString” The length of the datatype “VisibleString” or “OctetString” shall be defined by the attribute “Length”.
DefaultValue	String	Required	1.0	Contains the default value for the parameter. For the representation of values see Table 2 and Table 3.
AllowedValues	SignedOrFloatValueList	Optional	1.0	Specifies the possible values for this parameter. If this attribute is not used the following rules shall be applied: If the attribute “ValueItemTarget” is used the value range of the parameter is defined by the attribute “Content” of the “Assignments” elements of the referenced ValueItem. If the attribute “ValueItemTarget” is not used or the referenced ValueItem does not contain “Assignments” elements, the value range of the parameter is defined by the data type defined by the attribute DataType. If the “AllowedValues” attribute and the “ValueItemTarget” attribute are both used, the “AllowedValues” attribute shall only contain values defined as attribute “Content” of the “Assignments” elements of the referenced ValueItem. If the ValueItem does not contain “Assignments” elements, every value within the value range of the data type for “AllowedValues” is valid. AllowedValues can only be specified for numerical ValueItems, i.e. for the “Bit”, “BitArea” or one of the “IntegerXX”, “UnsignedXX” or “FloatXX” DataTypes. For the representation of values see Table 2. Be careful: Support of FloatXX DataTypes was introduced to this attribute in GSDML version 2.2. Entering floating point values here breaks backward compatibility to engineering tools that don't support GSDML V2.2.
Changeable	Boolean	Optional	1.0	If this Attribute is set to “false”, no changes of this parameter are allowed. Default: “true”
Visible	Boolean	Optional	1.0	If this Attribute is set to “false”, this parameter will not be displayed in the engineering tool. The “false” setting does not make sense in combination with “Changeable=“true””. The Schema is not able to assert this mutual exclusion. Default: “true”
TextId	RefId	Required	1.0	Contains the ID of a text as a reference into the “ExternalTextList”.

Length	Unsigned16	Optional	2.1	Defines the length of the used "VisibleString" or "OctetString" data type. This attribute shall be ignored if other data types are used.
NOTE The "BitArea" data type can be used to reduce the size of a record data object because it has a variable bit length. For example if a parameter can only have four different values, a BitArea with the length of 2 bit is sufficient. A "BitArea" parameter can contain codings or separate bits.				

1847

1848 **A.4.10.4 MenuList**

1849 This element describes a multi-level menu structure for the ARVendorBlock/Request or a parameter
 1850 record. If this element is present, the engineering tool shall display the parameters referenced from the
 1851 MenuItems within a hierarchical menu structure instead of a flat list.

1852 Rules:

- 1853 • All parameters should be referenced.
- 1854 • Parameters shall not be referenced twice.
- 1855 • The very first MenuItem in the MenuList represents the root menu. If within a parameter record,
 1856 the Name/@TextId of the root menu shall be the same as the Name/@TextId of the parameter
 1857 record.
- 1858 • All MenuItems except for the first one (the root menu) shall be referenced, exactly once, by a
 1859 MenuRef element of a different MenuItem (i.e. the structure shall be a tree consisting of all the
 1860 MenuItems).
- 1861 • The menu hierarchy shall not exceed a nesting level of three (root menu → menu → submenu →
 1862 sub-submenu).

1863 Use: Optional

1864 Supported since GSDML version: 2.3

1865 The element shall contain the sub-element MenuItem (see A.4.10.5)

1866 **A.4.10.5 MenuItem**

1867 This element describes a menu that references parameters or sub-menus.

1868 Use: One ore more

1869 Supported since GSDML version: 2.3

1870 Each element shall contain attributes as shown in Table 59 plus the sub-elements Name (see A.4.10.6),
 1871 ParameterRef (see A.4.10.7) and MenuRef (see A.4.10.8).

1872 **Table 59 — Attributes of element MenuItem**

Name	Type	Use	SV	Description
ID	Id	Required	2.3	ID required for referencing this parameter from a MenuItem (for a hierarchical menu structure).

1873

1874 **A.4.10.6 Name**

1875 The Name element gives the menu item a human readable name.

1876 Use: Required

1877 Supported since GSDML version: 2.3

1878 Each element shall contain attributes as shown in Table 60.

1879 **Table 60 — Attributes of element Name**

Name	Type	Use	SV	Description
TextId	RefId	Required	2.3	Contains the ID of a text as a reference into the "ExternalTextList".

1880

1881 **A.4.10.7 ParameterRef**

1882 This element references a parameter for inclusion in the menu.

1883 Use: Optional

1884 Supported since GSDML version: 2.3

1885 Each element shall contain attributes as shown in Table 61.

1886 **Table 61 — Attributes of element ParameterRef**

Name	Type	Use	SV	Description
ParameterTarget	RefId	Required	2.3	Contains the ID of a parameter (element Ref) to be displayed in the menu.

1887

1888 **A.4.10.8 MenuRef**

1889 This element references a MenuItem for inclusion in the menu as sub-menu.

1890 Use: Optional

1891 Supported since GSDML version: 2.3

1892 Each element shall contain attributes as shown in Table 62.

1893 **Table 62 — Attributes of element MenuRef**

Name	Type	Use	SV	Description
MenuTarget	RefId	Required	2.3	Contains the ID of a MenuItem to be included in the menu as sub-menu.

1894

1895 **A.4.10.9 ARVendorBlock/Request**

1896 The ARVendorBlock/Request element describes the data structure of the ARVendorBlock within the
1897 IOD_ConnectReq.

1898 Locations:

1899 //ISO15745Profile/ProfileBody/ApplicationProcess/DeviceAccessPointList/DeviceAccessPointItem/ARVen
1900 dorBlock/Request

1901 Use: Optional

1902 Supported since GSDML version: 2.3

1903 The element shall contain the elements and attributes of the base class RecordDataBase (see A.4.10.1).

1904 **A.4.10.10ParameterRecordDataItem**

1905 A ParameterRecordDataItem element describes the data structure of a parameter record data object.

1906 Locations:

1907 //ISO15745Profile/ProfileBody/ApplicationProcess/DeviceAccessPointList/DeviceAccessPointItem/SystemDefinedSubmoduleList/InterfaceSubmoduleItem/RecordDataList/ParameterRecordDataItem

1909 //ISO15745Profile/ProfileBody/ApplicationProcess/DeviceAccessPointList/DeviceAccessPointItem/SystemDefinedSubmoduleList/PortSubmoduleItem/RecordDataList/ParameterRecordDataItem

1911 //ISO15745Profile/ProfileBody/ApplicationProcess/DeviceAccessPointList/DeviceAccessPointItem/VirtualSubmoduleList/VirtualSubmoduleItem/RecordDataList/ParameterRecordDataItem

1913 //ISO15745Profile/ProfileBody/ApplicationProcess/ModuleList/ModuleItem/SystemDefinedSubmoduleList/PortSubmoduleItem/RecordDataList/ParameterRecordDataItem

1915 //ISO15745Profile/ProfileBody/ApplicationProcess/ModuleList/ModuleItem/VirtualSubmoduleList/VirtualSubmoduleItem/RecordDataList/ParameterRecordDataItem

1917 //ISO15745Profile/ProfileBody/ApplicationProcess/SubmoduleList/PortSubmoduleItem/RecordDataList/ParameterRecordDataItem

1919 //ISO15745Profile/ProfileBody/ApplicationProcess/SubmoduleList/SubmoduleItem/RecordDataList/ParameterRecordDataItem

1921 Each element shall contain the elements and attributes of the base class RecordDataBase (see A.4.10.1) plus the attributes as shown in Table 63 plus the sub-element Name (see A.4.10.11).

1923 **Table 63 — Additional attributes of element ParameterRecordDataItem**

Name	Type	Use	SV	Description
Index	Unsigned16	Required	1.0	Contains the unique ID of a record data object that is used to address the object with PROFINET IO Remote Procedure Calls (RPC) calls. The value shall be in range from 0 to 32767 (0x0 to 0x7FFF) if the corresponding submodule is defined in API 0. See PNO-2.712 <i>Application Layer services</i> , "Record Data ASE" for valid range of profile specific Record Data Objects (API >0 of the corresponding submodule)

Name	Type	Use	SV	Description
TransferSequence	Unsigned16	Optional	1.0	This attribute controls the transfer sequence within the manufacturer specific ParameterRecordData objects. The TransferSequence attribute shall be unique within a submodule. The first transferred ParameterRecordData object shall have this attribute set to "1", the following objects shall have this attribute incremented in direct sequence (without gaps). If this attribute is set to "0" (which is the default) then the transfer sequence is undefined.
ChangeableWithBump	Boolean	Optional	2.3	Set this attribute to true if this ParameterRecordDataItem contains parameters which will cause an IO bump when changed in Run. This attribute is useful for devices which support the PrmBegin-PrmEnd sequence and/or CiR. Default: "false"
EXAMPLE 1 A valid configuration: TransferSequence First Object: 1 Second Object: 2 Third Object: 3 EXAMPLE 2 Invalid configurations: 2,3,4: Does not start with "1" 1,2,4: Number "3" is missing 1,2,2: Multiple Use of number "2"				

1924

1925 The following example shows the usage of the various lists to describe a module with one parameter data

1926 record object and two parameters:

1927 EXAMPLE

```

1928 <RecordDataList>
1929   <!--The module has one 20 octet record data object with index "1". The description
1930   of this record data object is "General Parameter" in the English language.-->
1931   <ParameterRecordDataItem Index="1" Length="20">
1932     <Name TextId="General Parameter"/>
1933     <Ref ValueItemTarget="AlarmEnabling" DataType="Bit" ByteOffset="8" BitOffset="0"
1934       DefaultValue="0" AllowedValues="0..1" TextId="ID_TEXT1"/>
1935     <Ref ValueItemTarget="IF Supp" DataType="BitArea" ByteOffset="10" BitOffset="0"
1936       BitLength="2" DefaultValue="0" AllowedValues="0..2" TextId="ID_TEXT2"/>
1937   </ParameterRecordDataItem>
1938 </RecordDataList>

```

```

1939 <ValueList>
1940   <ValueItem ID="AlarmEnabling">
1941     <Help TextId="ID_ENABLE_ALARMS"/>
1942     <Assignments>
1943       <Assign Content="0" TextId="No"/>
1944       <Assign Content="1" TextId="Yes"/>
1945     </Assignments>
1946   </ValueItem>
1947   <ValueItem ID="IF Supp">
1948     <Help TextId="ID_IF_SUPP"/>
1949     <Assignments>
1950       <Assign Content="0" TextId="50 Hz"/>
1951       <Assign Content="1" TextId="60 Hz"/>
1952       <Assign Content="2" TextId="400 Hz"/>
1953     </Assignments>
1954   </ValueItem>
1955 </ValueList>

```

1956 **A.4.10.11Name**

1957 The Name element gives the record data object a human readable name.

1958 NOTE This allows an engineering tool to group the data objects of a record data object so that, for example a
1959 dialog can use this name as dialog title.

1960 Locations:

1961 //ISO15745Profile/ProfileBody/ApplicationProcess/DeviceAccessPointList/DeviceAccessPointItem/Syste
1962 mDefinedSubmoduleList/InterfaceSubmoduleItem/RecordDataList/ParameterRecordDataItem/Name

1963 //ISO15745Profile/ProfileBody/ApplicationProcess/DeviceAccessPointList/DeviceAccessPointItem/Syste
1964 mDefinedSubmoduleList/PortSubmoduleItem/RecordDataList/ParameterRecordDataItem/Name

1965 //ISO15745Profile/ProfileBody/ApplicationProcess/DeviceAccessPointList/DeviceAccessPointItem/Virtual
1966 SubmoduleList/VirtualSubmoduleItem/RecordDataList/ParameterRecordDataItem/Name

1967 //ISO15745Profile/ProfileBody/ApplicationProcess/ModuleList/ModuleItem/SystemDefinedSubmoduleList/
1968 PortSubmoduleItem/RecordDataList/ParameterRecordDataItem/Name

1969 //ISO15745Profile/ProfileBody/ApplicationProcess/ModuleList/ModuleItem/VirtualSubmoduleList/VirtualSu
1970 bmoduleItem/RecordDataList/ParameterRecordDataItem/Name

1971 //ISO15745Profile/ProfileBody/ApplicationProcess/SubmoduleList/SubmoduleItem/RecordDataList/Param
1972 eterRecordDataItem/Name

1973 Use: Required

1974 Each element shall contain attributes as shown in Table 64.

1975 **Table 64 — Attributes of element Name**

Name	Type	Use	SV	Description
TextId	RefId	Required	1.0	Contains the ID of a text as a reference into the "ExternalTextList".

1976

1977 **A.4.11 F_ParameterRecordDataItem related elements**

1978 **A.4.11.1 F_ParameterRecordDataItem**

1979 An F_ParameterRecordDataItem element describes the data structure of a parameter record data object
1980 for PROFIsafe parameters.

1981 Locations:

1982 //ISO15745Profile/ProfileBody/ApplicationProcess/DeviceAccessPointList/DeviceAccessPointItem/Virtual
1983 SubmoduleList/VirtualSubmoduleItem/RecordDataList/F_ParameterRecordDataItem

1984 //ISO15745Profile/ProfileBody/ApplicationProcess/ModuleList/ModuleItem/VirtualSubmoduleList/VirtualSu
1985 bmoduleItem/RecordDataList/F_ParameterRecordDataItem

1986 //ISO15745Profile/ProfileBody/ApplicationProcess/SubmoduleList/SubmoduleItem/RecordDataList/F_Par
1987 ameterRecordDataItem

1988 Use: Required if the attribute "PROFIsafeSupported" of the corresponding submodule is set to "true".

1989 Supported since GSDML version: 2.0

1990 Attributes: The element shall contain attributes as shown in Table 65.

1991 **Table 65 — Attributes of element F_ParameterRecordDataItem**

Name	Type	Use	SV	Description
F_ParamDescCRC	Unsigned16	Required	2.0	In order to read the PROFIsafe parameter description safely from the GSD file, 2 byte of CRC signature are necessary. The CRC signature has to be calculated according to the PROFIsafe specification and certified by a notified assessment body (e.g. TÜV). The value of this parameter will not be transferred to the device but is needed to avoid errors during the parameterization with the configuration tool.
F_SupportedParameters	TokenList	Optional	2.3	Indicates the exact support of optional F parameters in the F parameter record (enhances the information of F_Block_ID/@AllowedValues). Allowed tokens: "F_iPar_CRC" and "F_WD_Time_2" If present, this attribute shall contain "F_iPar_CRC" if the optional F parameter F_iPar_CRC is also present. If this attribute and F_Block_ID/@AllowedValues is present, the latter one shall contain a value or value range that includes the bits that represent the supported optional F parameters.
Index	Unsigned16	Required	2.0	Contains the unique ID of a record data object that is used to address the object with PROFINET IO RPC calls. It shall be in the range from 0 to 32767 (0x0 to 0x7FFF) if the corresponding submodule is defined in API 0. See PNO-2.712 <i>Application Layer services</i> , "Record Data ASE" for valid range of profile specific Record Data Objects (API >0 of the corresponding submodule)
TransferSequence	Unsigned16	Optional	2.0	This attribute controls the transfer sequence within the manufacturer specific ParameterRecordData objects. The TransferSequence attribute shall be unique within a submodule. The first transferred ParameterRecordData object shall have this attribute set to "1", the following objects shall have this attribute incremented in direct sequence (without gaps). If this attribute is set to "0" (which is the default) then the transfer sequence is undefined.
ChangeableWithBump	Boolean	Optional	2.3	Set this attribute to true if at least one F parameter will cause an IO bump when changed in Run. This attribute is useful for devices which support the PrmBegin-PrmEnd sequence and/or CiR. Default: "false"

1992

1993 Within a VirtualSubmoduleItem element the content of the "Index" and "TransferSequence" attribute of the

1994 F_ParameterRecordDataItem and ParameterRecordDataItem shall be unique.

1995 Each of the following F parameters has the following four attributes:

1996 **Table 66 — Attributes of F Parameters (General Information)**

Name	Type	Use	Description
DefaultValue	<a single value>	Optional	Defines the initial value of the parameter. This value shall be part of the values defined by the attribute "AllowedValues".
AllowedValues	<a list of values and / or ranges of values>	Optional	Defines the possible values. Ranges of values are only allowed for numeric data types.

Name	Type	Use	Description
Changeable	Boolean	Optional	Defines if this parameter should be changeable by the user of the engineering tool.
Visible	Boolean	Optional	Defines if this parameter - should be shown to the user of the engineering tool - should be included in the F_ParamDescCRC (= CRC0). IMPORTANT: An F parameter with Visible="false" always has a value of 0², which may be different from its default value.

1997

1998 **A.4.11.2 F_Check_iPar**

1999 The F_Check_iPar element defines, whether individual F device parameters should also be included in
2000 CRC1. See PROFIsafe Profile V2.5 for details.

2001 Locations:

2002 //ISO15745Profile/ProfileBody/ApplicationProcess/DeviceAccessPointList/DeviceAccessPointItem/Virtual
2003 SubmoduleList/VirtualSubmoduleItem/RecordDataList/F_ParameterRecordDataItem/F_Check_iPar

2004 //ISO15745Profile/ProfileBody/ApplicationProcess/ModuleList/ModuleItem/VirtualSubmoduleList/VirtualSu
2005 bmoduleItem/RecordDataList/F_ParameterRecordDataItem/F_Check_iPar

2006 //ISO15745Profile/ProfileBody/ApplicationProcess/SubmoduleList/SubmoduleItem/RecordDataList/F_Par
2007 ameterRecordDataItem/F_Check_iPar

2008 Use: Required

2009 The element shall contain attributes as shown in Table 67.

2010 **Table 67 — Attributes of element F_Check_iPar**

Name	Type	Use	SV	Description
DefaultValue	Enumeration	Optional	2.0	One of the following values are to use: "Check" "NoCheck" Default: "NoCheck"
AllowedValues	String	Optional	2.2	The string is separated in tokens and shall fulfill the rules of the following regular expression: (Check)? ?(NoCheck)? Default: "Check NoCheck"
Changeable	Boolean	Optional	2.0	Default: "false"
Visible	Boolean	Optional	2.0	Default: "false"

2011

2012 **A.4.11.3 F_SIL**

2013 The F_SIL element defines the Safety Integrity Level (SIL) of a submodule. See PROFIsafe Profile V2.5
2014 for details.

² For enumerations, this is equivalent to the first enumeration value.

- 2015 Locations:
 2016 //ISO15745Profile/ProfileBody/ApplicationProcess/DeviceAccessPointList/DeviceAccessPointItem/Virtual
 2017 SubmoduleList/VirtualSubmoduleItem/RecordDataList/F_ParameterRecordDataItem/F_SIL
- 2018 //ISO15745Profile/ProfileBody/ApplicationProcess/ModuleList/ModuleItem/VirtualSubmoduleList/VirtualSubmoduleItem/RecordDataList/F_ParameterRecordDataItem/F_SIL
- 2020 //ISO15745Profile/ProfileBody/ApplicationProcess/SubmoduleList/SubmoduleItem/RecordDataList/F_ParameterRecordDataItem/F_SIL
- 2022 Use: Required
- 2023 The element shall contain attributes as shown in Table 68.

2024 **Table 68 — Attributes of element F_SIL**

Name	Type	Use	SV	Description
DefaultValue	Enumeration	Optional	2.0	One of the following values are to use: "SIL1" "SIL2" "SIL3" "NoSIL" Default: "SIL3"
AllowedValues	String	Optional	2.0	The string is separated in tokens and shall fulfill the rules of the following regular expression: (SIL1)??(SIL2)??(SIL3)??(NoSIL)? Default: "SIL1 SIL2 SIL3 NoSIL"
Changeable	Boolean	Optional	2.0	Default: "true"
Visible	Boolean	Optional	2.0	Fixed: "true"

- 2025
- 2026 **A.4.11.4 F_CRC_Length**
- 2027 The F_CRC_Length element describes the length of the CRC2 signature. See PROFIsafe Profile V2.5 for
 2028 details.
- 2029 Locations:
 2030 //ISO15745Profile/ProfileBody/ApplicationProcess/DeviceAccessPointList/DeviceAccessPointItem/Virtual
 2031 SubmoduleList/VirtualSubmoduleItem/RecordDataList/F_ParameterRecordDataItem/F_CRC_Length
- 2032 //ISO15745Profile/ProfileBody/ApplicationProcess/ModuleList/ModuleItem/VirtualSubmoduleList/VirtualSubmoduleItem/RecordDataList/F_ParameterRecordDataItem/F_CRC_Length
- 2033
- 2034 //ISO15745Profile/ProfileBody/ApplicationProcess/SubmoduleList/SubmoduleItem/RecordDataList/F_ParameterRecordDataItem/F_CRC_Length
- 2035
- 2036 Use: Required
- 2037 The element shall contain attributes as shown in Table 69.

2038 **Table 69 — Attributes of element F_CRC_Length**

Name	Type	Use	SV	Description
DefaultValue	Enumeration	Optional	2.0	One of the following values are to use: "3-Byte-CRC" "4-Byte-CRC" Default: "3-Byte-CRC"

Name	Type	Use	SV	Description
				Note: 2-Byte-CRC is allowed in V1-mode only, which is not applicable with PROFINET.
AllowedValues	String	Optional	2.0	The string is separated in tokens and shall fulfill the rules of the following regular expression: (3-Byte-CRC)? ?(4-Byte-CRC)? Default: "3-Byte-CRC"
Changeable	Boolean	Optional	2.0	Default: "false"
Visible	Boolean	Optional	2.0	Default: "false"

2039

2040 **A.4.11.5 F_Block_ID**

2041 The F_Block_ID element describes the type of the parameter block. Bit 0 indicates the presence of i-
 2042 Parameters (F_iPar_CRC). Bit 1 indicates the presence of the secondary F-Watchdog time
 2043 (F_WD_Time_2). See PROFIsafe Profile V2.5 for details.

2044 Locations:

2045 //ISO15745Profile/ProfileBody/ApplicationProcess/DeviceAccessPointList/DeviceAccessPointItem/Virtual
 2046 SubmoduleList/VirtualSubmoduleItem/RecordDataList/F_ParameterRecordDataItem/F_Block_ID

2047 //ISO15745Profile/ProfileBody/ApplicationProcess/ModuleList/ModuleItem/VirtualSubmoduleList/VirtualSu
 2048 bmoduleItem/RecordDataList/F_ParameterRecordDataItem/F_Block_ID

2049 //ISO15745Profile/ProfileBody/ApplicationProcess/SubmoduleList/SubmoduleItem/RecordDataList/F_Par
 2050 ameterRecordDataItem/F_Block_ID

2051 Use: Required

2052 The element shall contain attributes as shown in Table 70.

2053 **Table 70 — Attributes of element F_Block_ID**

Name	Type	Use	SV	Description
DefaultValue	Integer	Optional	2.0	The value shall be in the range 0 to 7. Default: "0"
AllowedValues	ValueList	Optional	2.2	Only values in the range 0 to 7 are allowed. NOTE: Only one single value or one value range may be entered here because of the PROFIsafe CRC0 algorithm. Default: "0..7"
Changeable	Boolean	Optional	2.2	Default: "false"
Visible	Boolean	Optional	2.2	Default: "true"

2054

2055 **A.4.11.6 F_Par_Version**

2056 The F_Par_Version element describes the version number of the F parameter block. See PROFIsafe
 2057 Profile V2.5 for details.

2058 Locations:

2059 //ISO15745Profile/ProfileBody/ApplicationProcess/DeviceAccessPointList/DeviceAccessPointItem/Virtual
 2060 SubmoduleList/VirtualSubmoduleItem/RecordDataList/F_ParameterRecordDataItem/F_Par_Version

2061 //ISO15745Profile/ProfileBody/ApplicationProcess/ModuleList/ModuleItem/VirtualSubmoduleList/VirtualSu
 2062 bmoduleItem/RecordDataList/F_ParameterRecordDataItem/F_Par_Version

2063 //ISO15745Profile/ProfileBody/ApplicationProcess/SubmoduleList/SubmoduleItem/RecordDataList/F_ParameterRecordDataItem/F_ParameterVersion
 2064

2065 Use: Required

2066 Supported since GSDML version: 2.2

2067 The element shall contain attributes as shown in Table 71.

2068 **Table 71 — Attributes of element F_ParameterVersion**

Name	Type	Use	SV	Description
DefaultValue	Integer	Optional	2.0	Fixed: "1"
AllowedValues	ValueList	Optional	2.2	Fixed: "1"
Changeable	Boolean	Optional	2.2	Fixed: "false"
Visible	Boolean	Optional	2.2	Fixed: "true"

2069

2070 **A.4.11.7 F_Source_Add**

2071 The addresses of the F components of a safety control loop F input, F-CPU and F output shall be
 2072 unambiguous.

2073 Locally, each F device has the configured source-destination relationship of the safe communication link
 2074 with its partner. The F_Source_Add element is used to define the supported range of the F_Source
 2075 address. See PROFIsafe Profile V2.5 for details.

2076 While instantiating the hardware component, the engineering tool should use the DefaultValue of the
 2077 F_Source and F_Dest addresses if they are unique. Otherwise the engineering tool should choose unique
 2078 addresses from the respective AllowedValues ranges. These addresses may later be modified by the
 2079 user.

2080 Locations:

2081 //ISO15745Profile/ProfileBody/ApplicationProcess/DeviceAccessPointList/DeviceAccessPointItem/Virtual
 2082 SubmoduleList/VirtualSubmoduleItem/RecordDataList/F_ParameterRecordDataItem/F_Source_Add

2083 //ISO15745Profile/ProfileBody/ApplicationProcess/ModuleList/ModuleItem/VirtualSubmoduleList/VirtualSubmoduleItem/RecordDataList/F_ParameterRecordDataItem/F_Source_Add

2085 //ISO15745Profile/ProfileBody/ApplicationProcess/SubmoduleList/SubmoduleItem/RecordDataList/F_ParameterRecordDataItem/F_Source_Add

2087 Use: Required

2088 The element shall contain attributes as shown in Table 72.

2089 **Table 72 — Attributes of element F_Source_Add**

Name	Type	Use	SV	Description
DefaultValue	Unsigned16	Optional	2.2	Default: "1"
AllowedValues	ValueList	Optional	2.0	Note that the values 0 and 65535 are not allowed. NOTE: Only one single value or one value range may be entered here because of the PROFIsafe CRC0 algorithm. Default: "1..65534"
Changeable	Boolean	Optional	2.2	Fixed: "true"
Visible	Boolean	Optional	2.2	Fixed: "true"

2090

2091 **A.4.11.8 F_Dest_Add**

2092 The addresses of the F components of a safety control loop F input, F-CPU and F output shall be
 2093 unambiguous.

2094 Locally, each F device has the configured source-destination relationship of the safe communication link

2095 with its partner. The F_Dest_Add element is used to define the supported range of the F_Destination
 2096 address. See PROFIsafe Profile V2.5 for details.

2097 While instantiating the hardware component, the engineering tool should use the DefaultValue of the
 2098 F_Source and F_Dest addresses if they are unique. Otherwise the engineering tool should choose unique
 2099 addresses from the respective AllowedValues ranges. These addresses may later be modified by the
 2100 user.

2101 Locations:
 2102 //ISO15745Profile/ProfileBody/ApplicationProcess/DeviceAccessPointList/DeviceAccessPointItem/Virtual
 2103 SubmoduleList/VirtualSubmoduleItem/RecordDataList/F_ParameterRecordDataItem/F_Dest_Add

2104 //ISO15745Profile/ProfileBody/ApplicationProcess/ModuleList/ModuleItem/VirtualSubmoduleList/VirtualSu
 2105 bmoduleItem/RecordDataList/F_ParameterRecordDataItem/F_Dest_Add

2106 //ISO15745Profile/ProfileBody/ApplicationProcess/SubmoduleList/SubmoduleItem/RecordDataList/F_Par
 2107 ameterRecordDataItem/F_Dest_Add

2108 Use: Required

2109 The element shall contain attributes as shown in Table 73.

2110 **Table 73 — Attributes of element F_Dest_Add**

Name	Type	Use	SV	Description
DefaultValue	Unsigned16	Optional	2.2	Default: "1"
AllowedValues	ValueList	Optional	2.0	Note that the values 0 and 65535 are not allowed. NOTE: Only one single value or one value range may be entered here because of the PROFIsafe CRC0 algorithm. Default: "1..65534"
Changeable	Boolean	Optional	2.2	Fixed: "true"
Visible	Boolean	Optional	2.2	Fixed: "true"

2111

2112 **A.4.11.9 F_WD_Time**

2113 The element F_WD_Time defines the watchdog time in the F unit. See PROFIsafe Profile V2.5 for details.

2114 Locations:
 2115 //ISO15745Profile/ProfileBody/ApplicationProcess/DeviceAccessPointList/DeviceAccessPointItem/Virtual
 2116 SubmoduleList/VirtualSubmoduleItem/RecordDataList/F_ParameterRecordDataItem/F_WD_Time

2117 //ISO15745Profile/ProfileBody/ApplicationProcess/ModuleList/ModuleItem/VirtualSubmoduleList/VirtualSu
 2118 bmoduleItem/RecordDataList/F_ParameterRecordDataItem/F_WD_Time

2119 //ISO15745Profile/ProfileBody/ApplicationProcess/SubmoduleList/SubmoduleItem/RecordDataList/F_Par
 2120 ameterRecordDataItem/F_WD_Time

2121 Use: Required

2122 The element shall contain attributes as shown in Table 74.

2123 **Table 74 — Attributes of element F_WD_Time**

Name	Type	Use	SV	Description
DefaultValue	Unsigned16	Optional	2.0	Note that a value of 0 is not allowed. Base: 1 ms Default: "150"
AllowedValues	ValueList	Optional	2.0	Note that a value of 0 is not allowed. NOTE: Only one single value or one value range may be entered here because of the PROFIsafe CRC0 algorithm. Default: "1..65535"
Changeable	Boolean	Optional	2.2	Fixed: "true"
Visible	Boolean	Optional	2.2	Fixed: "true"

2124

2125 **A.4.11.10F_Par_CRC**

2126 The F_Par_CRC defines the CRC1 for F parameters. See PROFIsafe Profile V2.5 for details.

2127 Locations:

2128 //ISO15745Profile/ProfileBody/ApplicationProcess/DeviceAccessPointList/DeviceAccessPointItem/Virtual
2129 SubmoduleList/VirtualSubmoduleItem/RecordDataList/F_ParameterRecordDataItem/F_Par_CRC

2130 //ISO15745Profile/ProfileBody/ApplicationProcess/ModuleList/ModuleItem/VirtualSubmoduleList/VirtualSubmoduleItem/RecordDataList/F_ParameterRecordDataItem/F_Par_CRC

2132 //ISO15745Profile/ProfileBody/ApplicationProcess/SubmoduleList/SubmoduleItem/RecordDataList/F_ParameterRecordDataItem/F_Par_CRC

2134 Use: Required

2135 The element shall contain attributes as shown in Table 75.

2136 **Table 75 — Attributes of element F_Par_CRC**

Name	Type	Use	SV	Description
DefaultValue	Unsigned16	Optional	2.0	Shall be 0 if F_WD_Time_2 is supported in the default configuration (F_Block_ID/@DefaultValue has bit 1 set). Shall contain the CRC1 of the default configuration (the DefaultValues of the visible F-Parameters) otherwise. Default: "53356"
AllowedValues	ValueList	Optional	2.2	Fixed: "0..65535"
Changeable	Boolean	Optional	2.2	Fixed: "true"
Visible	Boolean	Optional	2.2	Fixed: "true"

2137

2138 **A.4.11.11F_iPar_CRC**

2139 The F_iPar_CRC defines the CRC for i-Parameters. See PROFIsafe Profile V2.5 for details.

- 2140 Locations:
 2141 //ISO15745Profile/ProfileBody/ApplicationProcess/DeviceAccessPointList/DeviceAccessPointItem/Virtual
 2142 SubmoduleList/VirtualSubmoduleItem/RecordDataList/F_ParameterRecordDataItem/F_iPar_CRC
- 2143 //ISO15745Profile/ProfileBody/ApplicationProcess/ModuleList/ModuleItem/VirtualSubmoduleList/VirtualSu
 2144 bmoduleItem/RecordDataList/F_ParameterRecordDataItem/F_iPar_CRC
- 2145 //ISO15745Profile/ProfileBody/ApplicationProcess/SubmoduleList/SubmoduleItem/RecordDataList/F_Par
 2146 ameterRecordDataItem/F_iPar_CRC
- 2147 Use: Required for submodules with i-Parameters, i.e. whose F_Block_ID has bit 0 set to 1.
- 2148 Supported since GSDML version: 2.2
- 2149 The element shall contain attributes as shown in Table 76.

2150 **Table 76 — Attributes of element F_iPar_CRC**

Name	Type	Use	SV	Description
DefaultValue	Unsigned32	Optional	2.1	Default: "0"
AllowedValues	ValueList	Optional	2.2	Fixed: "0..4294967295"
Changeable	Boolean	Optional	2.2	Fixed: "true"
Visible	Boolean	Optional	2.2	Fixed: "true"

- 2151
- 2152 **A.4.12 ModuleInfo related elements**
- 2153 **A.4.12.1 ModuleInfo**
- 2154 This element contains general Information about a module or submodule.
- 2155 Locations:
 2156 //ISO15745Profile/ProfileBody/ApplicationProcess/DeviceAccessPointList/DeviceAccessPointItem/Modul
 2157 eInfo
- 2158 //ISO15745Profile/ProfileBody/ApplicationProcess/DeviceAccessPointList/DeviceAccessPointItem/Virtual
 2159 SubmoduleList/VirtualSubmoduleItem/ModuleInfo
- 2160 //ISO15745Profile/ProfileBody/ApplicationProcess/ModuleList/ModuleItem/ModuleInfo
- 2161 //ISO15745Profile/ProfileBody/ApplicationProcess/ModuleList/ModuleItem/VirtualSubmoduleList/VirtualSu
 2162 bmoduleItem/ModuleInfo
- 2163 //ISO15745Profile/ProfileBody/ApplicationProcess/SubmoduleList/SubmoduleItem/ModuleInfo
- 2164 Each element shall contain attributes as shown in Table 77.

2165 **Table 77 — Attributes of element ModuleInfo**

Name	Type	Use	SV	Description
CategoryRef	RefId	Optional	1.0	Contains the category of a module. The value of CategoryRef shall contain an ID of a "CategoryItem" element.
SubCategory1Ref	RefId	Optional	1.0	Within a category, subcategories can be created. The value of SubCategory1Ref shall contain an ID of a "CategoryItem" element.

- 2166
-
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- Draft in review. Do not claim conformance!**

The ModuleInfo element contains textual information for the engineering tool to build catalog entries and to display them to the user. The information shall not be used for a programmatic comparison with identification parameters (see Table 78) which can be read out from the device. As explained in chapter A.3.9, the combination of VendorID and DeviceID selects the GSD for a PROFINET IO device, the ModuleIdentNumber selects the applicable description of the module inside the GSD, and the SubmoduleIdentNumber selects the applicable description of the submodule inside the GSD. Since a device manufacturer does not need to modify ident numbers as long as the engineering view stays the same (see “Coding of the field SubmoduleIdentNumber” in PNO-2.722 *Application Layer protocol*), there may be multiple versions of a device using the same GSD.

In this case, the GSD author should omit optional ModuleInfo texts like HardwareRelease or SoftwareRelease or should use some wildcards like “*”, “?” or “...” in the texts to signal this ambiguity to the user.

Table 78 — Related identification parameters

GSDML	PNO-2.722 <i>Application Layer protocol</i>		Profile Guidelines <i>Part 1 – Identification & Maintenance Functions</i>
	RPC Endpoint-mapper Response	I&M	
VendorName	VendorID in RPCObjectUUID	VendorID (= VendorIDHigh + VendorIDLow)	MANUFACTURER_ID
OrderNumber	OrderID	OrderID	ORDER_ID
HardwareRelease	HWRevision	IM_Hardware_Revision	HARDWARE_REVISION
SoftwareRelease	SWRevisionPrefix + SWRevision	IM_Software_Revision (= SWRevisionPrefix + IM_SWRevision_Functional_Enhancement + IM_SWRevision_Bug_Fix + IM_SWRevision_Internal_Change)	SOFTWARE_REVISION

A.4.12.2 Name

The Name element contains the language dependent name of a module or submodule.

Use: Required.

Each element shall contain attributes as shown in Table 79.

Table 79 — Attributes of element Name

Name	Type	Use	SV	Description
TextId	RefId	Required	1.0	Contains the ID of a text as a reference into the “ExternalTextList”.

A.4.12.3 InfoText

The InfoText element contains human readable text information about a module or submodule.

Use: Required.

Each element shall contain attributes as shown in Table 80.

Table 80 — Attributes of element InfoText

Name	Type	Use	SV	Description
TextId	RefId	Required	1.0	Contains the ID of a text as a reference into the “ExternalTextList”.

2193 A.4.12.4 VendorName

2194 The VendorName element contains the name of the device vendor. If this element does not exist, the
2195 vendor name of the element "DeviceIdentity/Vendorname" shall be used.

2196 Use: Optional

2197 Each element shall contain attributes as shown in Table 81.

2198 **Table 81 — Attributes of element VendorName**

Name	Type	Use	SV	Description
Value	Token	Required	1.0	Contains the name of the device vendor.

2199

2200 A.4.12.5 OrderNumber

2201 The OrderNumber element contains the order number of a module or submodule.

2202 Use: Optional

2203 Each element shall contain attributes as shown in Table 82.

2204 **Table 82 — Attributes of element OrderNumber**

Name	Type	Use	SV	Description
Value	Token	Required	1.0	Contains the order number of a module or submodule.

2205

2206 A.4.12.6 HardwareRelease

2207 The HardwareRelease element contains the hardware release of a module or submodule.

2208 Use: Optional

2209 Each element shall contain attributes as shown in Table 83.

2210 **Table 83 — Attributes of element HardwareRelease**

Name	Type	Use	SV	Description
Value	Token	Required	1.0	Contains the hardware release of a module or submodule.

2211

2212 A.4.12.7 SoftwareRelease

2213 Contains the software release of a module or submodule.

2214 Use: Optional

2215 Each element shall contain attributes as shown in Table 84.

2216 **Table 84 — Attributes of element SoftwareRelease**

Name	Type	Use	SV	Description
Value	Token	Required	1.0	Contains the software release of a module or submodule.

2217

2218 **A.4.12.8 Family**

2219 The module or submodule shall be assigned to a function class. In addition to the main family, a module
2220 or submodule can be assigned to a vendor specific product family.

2221 Use: Optional

2222 See A.4.3.2.

2223 **A.4.13 (Profile)ChannelDiagItem related elements**2224 **A.4.13.1 ChannelDiagItem**

2225 A ChannelDiagItem contains attributes to assign a text to a ChannelErrorType.

2226 Location:

2227 //ISO15745Profile/ProfileBody/ApplicationProcess/ChannelDiagList/ChannelDiagItem

2228 Use: Zero or more if ProfileChannelDiagItem is used. Otherwise: One or more.

2229 Each element shall contain attributes as shown in Table 85.

2230 **Table 85 — Attributes of element ChannelDiagItem**

Name	Type	Use	SV	Description
ErrorType	Unsigned16	Required	1.0	Contains the error type where the diagnostic text definitions can be assigned. Only manufacturer specific error types shall be used here. Therefore only the range from 15 up to 32767 (0x000F to 0x7FFF) is allowed. Note that the range 16 up to 31 (0x0010 to 0x001F) has a recommended usage and the range 32 up to 255 (0x0020 to 0x00FF) is reserved for common errors across all profiles. See PNO-2.722 <i>Application Layer protocol</i>, "Coding of the field ChannelErrorType".
MaintenanceAlarmState	TokenList1	Optional	2.1	Describes the possible values of the field "ChannelProperties.Maintenance". See PNO-2.722 <i>Application Layer protocol</i>, "Coding of the field ChannelProperties". One or more of the following values are allowed: "D" (Diagnosis) "MR" (MaintenanceRequired) "MD" (MaintenanceDemanded) "QD" (QualifiedDiagnosis) Default: "D" EXAMPLE: "D;MR;MD" means that the ChannelProperties.Maintenance field may contain the values "Diagnosis" or "MaintenanceRequired" or "MaintenanceDemanded".

2231

2232 **A.4.13.2 ProfileChannelDiagItem**

2233 A ProfileChannelDiagItem contains attributes to assign a text to a profile specific ChannelErrorType.

2234 Location:

2235 //ISO15745Profile/ProfileBody/ApplicationProcess/ChannelDiagList/ProfileChannelDiagItem

2236 Use: Zero or more if ChannelDiagItem is used. Otherwise: One or more

2237 Supported since GSDML version: 2.1

2238 Each element shall contain attributes as shown in Table 86.

2239 **Table 86 — Attributes of element ProfileChannelDiagItem**

Name	Type	Use	SV	Description
ErrorType	Unsigned16	Required	2.1	Contains the error type where the diagnostic text definitions can be assigned. Only profile specific error types shall be used here. Therefore only the range from 36864 up to 40959 (0x9000 to 0x9FFF) is allowed.
MaintenanceAlarmState	TokenList1	Optional	2.1	Describes the possible values of the field "ChannelProperties.Maintenance". See PNO-2.722 <i>Application Layer protocol</i> , "Coding of the field ChannelProperties". One or more of the following values are allowed: " D " (Diagnosis) " MR " (MaintenanceRequired) " MD " (MaintenanceDemanded) " QD " (QualifiedDiagnosis) Default: "D" EXAMPLE: "D;MR;MD" means that the ChannelProperties.Maintenance field may contain the values "Diagnosis" or "MaintenanceRequired" or "MaintenanceDemanded".
API	Unsigned32	Required	2.1	This attribute defines the Application Process Identifier (API) of the submodule which can be defined by a device profile.

2240

2241 A.4.13.3 Name

2242 This element contains the language dependent text information for a ChannelDiagItem or
2243 ExtChannelDiagItem.

2244 Locations:

2245 //ISO15745Profile/ProfileBody/ApplicationProcess/ChannelDiagList/ChannelDiagItem/Name

2246 //ISO15745Profile/ProfileBody/ApplicationProcess/ChannelDiagList/ChannelDiagItem/ExtChannelDiagList

2247 /ExtChannelDiagItem/Name

2248 //ISO15745Profile/ProfileBody/ApplicationProcess/ChannelDiagList/ChannelDiagItem/ExtChannelDiagList

2249 /ProfileExtChannelDiagItem/Name

2250 //ISO15745Profile/ProfileBody/ApplicationProcess/ChannelDiagList/ProfileChannelDiagItem/Name

2251 //ISO15745Profile/ProfileBody/ApplicationProcess/ChannelDiagList/ProfileChannelDiagItem/ExtChannelD

2252 iagList/ExtChannelDiagItem/Name

2253 //ISO15745Profile/ProfileBody/ApplicationProcess/ChannelDiagList/ProfileChannelDiagItem/ExtChannelD

2254 iagList/ProfileExtChannelDiagItem/Name

2255 Use: Required

2256 Each element shall contain attributes as shown in Table 87.

2257

Table 87 — Attributes of element Name

Name	Type	Use	SV	Description
TextId	RefId	Required	1.0	Contains the ID of a text as a reference into the "ExternalTextList".

2258

2259 If the element "Name" is used for description of extended channel diagnosis it may contain Format
 2260 Specification Fields to insert the values of the additional information "ExtChannelAddValue".

2261 These Format Specification Fields shall only be interpreted if the "ExtChannelAddValue" element is
 2262 present at the related ExtChannelDiagItem or ProfileExtChannelDiagItem.

2263 As coding for the Format Specification Fields the following expression shall be used:

2264 {index[:formatString]}

2265 The matching braces ("{" and "}") are required.

2266 Table 88 defines the semantics:

2267

Table 88 — Content of a Format Specification Field

Name	Use	Description
index	Required	Refers to the attribute "Id" of the corresponding "DataItem" element.
formatString	Optional	The optional format string shall be introduced by a ":" character. The next character defines how the value shall be formatted. One of the following values shall be used: "d": decimal coding "x": hexadecimal coding "f": floating point coding If floating point coding is used a precision specifier can be added to define the number of decimal places. The precision specifier shall be a number in the range 00 to 99. Default: decimal coding

2268

2269 See A.4.13.9 for an example how this method is used.

2270 **A.4.13.4 Help**

2271 This element contains the language dependent help information for a ChannelDiagItem or
 2272 ExtChannelDiagItem.

2273 Locations:

2274 //ISO15745Profile/ProfileBody/ApplicationProcess/ChannelDiagList/ChannelDiagItem/Help

2275 //ISO15745Profile/ProfileBody/ApplicationProcess/ChannelDiagList/ChannelDiagItem/ExtChannelDiagList

2276 /ExtChannelDiagItem/Help

2277 //ISO15745Profile/ProfileBody/ApplicationProcess/ChannelDiagList/ChannelDiagItem/ExtChannelDiagList

2278 /ProfileExtChannelDiagItem/Help

2279 //ISO15745Profile/ProfileBody/ApplicationProcess/ChannelDiagList/ProfileChannelDiagItem/Help

2280 //ISO15745Profile/ProfileBody/ApplicationProcess/ChannelDiagList/ProfileChannelDiagItem/ExtChannelD

2281 iagList/ExtChannelDiagItem/Help

2282 //ISO15745Profile/ProfileBody/ApplicationProcess/ChannelDiagList/ProfileChannelDiagItem/ExtChannelD

2283 iagList/ProfileExtChannelDiagItem/Help

2284 Use: Optional

2285 Each element shall contain attributes as shown in Table 89.

2286 **Table 89 — Attributes of element Help**

Name	Type	Use	SV	Description
TextId	RefId	Required	1.0	Contains the ID of a text as a reference into the "ExternalTextList".

2287

2288 **A.4.13.5 ExtChannelDiagList**

2289 This element specifies a list of error texts for extended channel diagnostics.

2290 Location:

2291 //ISO15745Profile/ProfileBody/ApplicationProcess/ChannelDiagList/ChannelDiagItem/ExtChannelDiagList

2292 //ISO15745Profile/ProfileBody/ApplicationProcess/ChannelDiagList/ProfileChannelDiagItem/ExtChannelD

2293 iagList

2294 Use: Optional

2295

2296 Supported since GSDML version: 2.0

2297 Attributes: None

2298 **A.4.13.6 ExtChannelDiagItem**

2299 An ExtChannelDiagItem contains attributes to assign a text to an ExtChannelErrorType.

2300 The text from the ExtChannelDiagItem shall be concatenated with the text from the enclosing

2301 ChannelDiagItem.

2302 Location:

2303 //ISO15745Profile/ProfileBody/ApplicationProcess/ChannelDiagList/ChannelDiagItem/ExtChannelDiagList

2304 /ExtChannelDiagItem

2305 Use: Zero or more

2306 Supported since GSDML version: 2.0

2307 Each element shall contain attributes as shown in Table 90.

2308 **Table 90 — Attributes of element ExtChannelDiagItem**

Name	Type	Use	SV	Description
ErrorType	Unsigned16	Required	2.0	Contains the ExtChannelErrorType where diagnostic text definitions can be assigned. Only manufacturer specific error types shall be used here. Therefore only the range from 1 up to 32767 (0x1 to 0x7FFF) is allowed.
MaintenanceAlarmState	TokenList1	Optional	2.1	Describes the possible values of the field "ChannelProperties.Maintenance". See PNO-2.722 <i>Application Layer protocol</i> , "Coding of the field ChannelProperties". One or more of the following values are allowed: "D" (Diagnosis) "MR" (MaintenanceRequired) "MD" (MaintenanceDemanded) "QD" (QualifiedDiagnosis) Default: "D"

Name	Type	Use	SV	Description
				EXAMPLE: “D;MR;MD” means that the ChannelProperties.Maintenance field may contain the values “Diagnosis” or “MaintenanceRequired” or “MaintenanceDemanded”.

- 2309
- 2310 **A.4.13.7 ProfileExtChannelDiagItem**
- 2311 An ProfileExtChannelDiagItem contains attributes to assign a text to an ExtChannelErrorType.
- 2312 The text from the ProfileExtChannelDiagItem shall be concatenated with the text from the enclosing
- 2313 ProfileChannelDiagItem.
- 2314 Location:
- 2315 //ISO15745Profile/ProfileBody/ApplicationProcess/ChannelDiagList/ProfileChannelDiagItem/ExtChannelD
- 2316 iagList/ProfileExtChannelDiagItem
- 2317 Use: Zero or more
- 2318 Supported since GSDML version: 2.1
- 2319 Each element shall contain attributes as shown in Table 91.

2320 **Table 91 — Attributes of element ProfileExtChannelDiagItem**

Name	Type	Use	SV	Description
ErrorType	Unsigned16	Required	2.1	Contains the ExtChannelErrorType where diagnostic text definitions can be assigned. Only profile specific error types shall be used here. Therefore only the range from 36864 up to 40959 (0x9000 to 0x9FFF) is allowed.
MaintenanceAlarmState	TokenList1	Optional	2.1	Describes the possible values of the field “ChannelProperties. Maintenance”. See PNO-2.722 <i>Application Layer protocol</i>, “Coding of the field ChannelProperties”. One or more of the following values are allowed: “D” (Diagnosis) “MR” (MaintenanceRequired) “MD” (MaintenanceDemanded) “QD” (QualifiedDiagnosis) Default: “D” EXAMPLE: “D;MR;MD” means that the ChannelProperties.Maintenance field may contain the values “Diagnosis” or “MaintenanceRequired” or “MaintenanceDemanded”.
API	Unsigned32	Required	2.1	This attribute defines the Application Process Identifier (API) of the submodule which can be defined by a device profile.

- 2321
- 2322 **A.4.13.8 ExtChannelAddValue**
- 2323 An ExtChannelAddValue element holds the description for the additional value of the extended channel
- 2324 diagnosis (ExtChannelAddValue).

2325 Location:
 2326 //ISO15745Profile/ProfileBody/ApplicationProcess/ChannelDiagList/ChannelDiagItem/ExtChannelDiagList
 2327 /ExtChannelDiagItem/ExtChannelAddValue
 2328 //ISO15745Profile/ProfileBody/ApplicationProcess/ChannelDiagList/ProfileChannelDiagItem/ExtChannelD
 2329 iagList/ProfileExtChannelDiagItem/ExtChannelAddValue

2330 Use: Optional

2331 Supported since GSDML version: 2.1

2332 Attributes: None

2333 **A.4.13.9 Dataltem**

2334 A Dataltem element describes a single Dataltem of an ExtChannelAddValue.

2335 The sum of all Dataltem elements shall describe exactly 4 octets.

2336 Location:
 2337 //ISO15745Profile/ProfileBody/ApplicationProcess/ChannelDiagList/ChannelDiagItem/ExtChannelDiagList
 2338 /ExtChannelDiagItem/ExtChannelAddValue/Dataltem
 2339 //ISO15745Profile/ProfileBody/ApplicationProcess/ChannelDiagList/ProfileChannelDiagItem/ExtChannelD
 2340 iagList/ProfileExtChannelDiagItem/ExtChannelAddValue/Dataltem

2341 Use: One or more

2342 Supported since GSDML version: 2.1

2343 Each element shall contain attributes as shown in Table 92.

2344 **Table 92 — Attributes of element Dataltem**

Name	Type	Use	SV	Description
Id	Unsigned8	Required	2.1	Defines the unique Identifier of the Dataltem which can be used in the corresponding error text.
DataType	Enumeration	Required	2.1	Defines the data type of the data item. One of the following values is allowed: "Integer8" "Integer16" "Integer32" "Unsigned8" "Unsigned16" "Unsigned32" "Float32" "TimeOfDay without date indication" "TimeDifference without date indication" "VisibleString" "OctetString"
Length	Unsigned16	Optional	2.2	This attribute shall be used if the "DataType" attribute is set to "VisibleString" or "OctetString". This attribute specifies the length of the Dataltem element in octets. ^a A value of 0 is not allowed. If this attribute is used for data types with fixed length definition, the length shall correspond to the definition in PNO-2.712 <i>Application Layer services</i> , "Data Type ASE".
^a This is necessary because these data types have a variable length.				

2345

2346 Example how to use Dataltems in a Text of a channel error:

2347 A) Description of the Extended Channel Error

```

2348 <ChannelDiagItem ErrorType="35">
2349   <Name TextId="IDT_DIAG_NAME_Chiller"/>
2350   <ExtChannelDiagList>
2351     <ExtChannelDiagItem ErrorType="1">
2352       <Name TextId="IDT_DIAG_NAME_TemperatureHigh"/>
2353       <ExtChannelAddValue>
2354         <DataItem Id="1" DataType="Unsigned16"/>
2355         <DataItem Id="2" DataType="Unsigned16"/>
2356       </ExtChannelAddValue>
2357     </ExtChannelDiagItem>
2358   </ExtChannelDiagList>
2359 </ChannelDiagItem>

```

2360 B) Corresponding Error Text

```

2361 <ExternalTextList>
2362   <PrimaryLanguage>
2363     <Text TextId="IDT_DIAG_NAME_Chiller" Value="Chiller problem:"/>
2364     <Text TextId="IDT_DIAG_NAME_TemperatureHigh" Value="The temperature of the
2365 chiller is {1:d} degrees celsius."/>
2366   </PrimaryLanguage>
2367 </ExternalTextList>

```

2368 C) Resulting Text Output

2369 "Chiller problem: The temperature of the chiller is 67 degrees celsius."

2370 NOTE: Syntax of the format string is defined in Table 88.

2371 A.4.14 (Profile)UnitDiagTypeItem related elements

2372 A.4.14.1 UnitDiagTypeItem

2373 Location:

2374 //ISO15745Profile/ProfileBody/ApplicationProcess/UnitDiagTypeList/UnitDiagTypeItem

2375 Use: Zero or more if ProfileUnitDiagTypeItem is used. Otherwise: One or more

2376 Each element shall contain attributes as shown in Table 93.

2377 **Table 93 — Attributes of element UnitDiagTypeItem**

Name	Type	Use	SV	Description
UserStructureIdentifier	Unsigned16	Required	1.0	Describes the User Structure Identifier of the alarm request block. The UserStructureIdentifier shall be in the range from 0 to 32767 (0x0 to 0x7FFF). Within the UnitDiagTypeList the UserStructureIdentifier shall be unique.

2378

2379 A.4.14.2 ProfileUnitDiagTypeItem

2380 Location:

2381 //ISO15745Profile/ProfileBody/ApplicationProcess/UnitDiagTypeList/ProfileUnitDiagTypeItem

2382 Use: Zero or more if UnitDiagTypeItem is used. Otherwise: One or more

2383 Supported since GSDML version: 2.1

2384 Each element shall contain attributes as shown in Table 94.

2385

Table 94 — Attributes of element ProfileUnitDiagTypeItem

Name	Type	Use	SV	Description
UserStructureIdentifier	Unsigned16	Required	2.1	Describes the profile specific User Structure Identifier of the alarm request block. The UserStructureIdentifier shall be in the range from 36864 up to 40959 (0x9000 to 0x9FFF). Within the UnitDiagTypeList the UserStructureIdentifier for an API shall be unique.
API	Unsigned32	Required	2.1	This attribute defines the Application Process Identifier (API) of the submodule which can be defined by a device profile.

2386

2387 A.4.14.3 Ref

2388 The Ref element contains information about a diagnostic data item within an alarm data object.

2389 If more than one Ref element is defined no overlapping between the definitions is allowed.

2390 Locations:

2391 //ISO15745Profile/ProfileBody/ApplicationProcess/UnitDiagTypeList/UnitDiagTypeItem/Ref

2392 //ISO15745Profile/ProfileBody/ApplicationProcess/UnitDiagTypeList/ProfileUnitDiagTypeItem/Ref

2393 Use: One or more

2394 Attributes: See Table 95.

2395

Table 95 — Attributes of element Ref

Name	Type	Use	SV	Description
ValueItemTarget	RefId	Optional	1.0	References a ValueItem Element containing the name and value range of the data object. ValueItems can only have numerical Assignments. Thus referencing a ValueItem that has Assignments is only possible for the "Bit", "BitArea" or one of the "IntegerXX", "UnsignedXX" or "FloatXX" DataTypes. If ValueItemTarget is specified, and the referenced ValueItem has a matching Assignment/Assign/@Content, the engineering tool shall generate the message by composing the text from Ref/@TextId and the text from ValueItem/Assignment/Assign/@TextId.
ByteOffset	Unsigned32	Required	1.0	Contains the offset in octets of the "additional alarm info" block of an alarm request PDU - the header information is not included.
BitOffset	Integer	Optional	1.0	Contains the offset in bits of the referenced data object from the beginning of the referenced octet. Valid Range: 0..7 Default: "0" Can only be used in conjunction with data type "Bit" or "BitArea".
BitLength	Integer	Optional	1.0	If the type of this data object is "BitArea", this attribute specifies the length of this field. The attribute is ignored if another data type is used. Default: "1" The BitLength shall be in range from "1" to "15".
DataType	Enumeration	Required	1.0	Specifies the type of the data object. One of the following values are to use: "Bit" "BitArea" (see Note)

Name	Type	Use	SV	Description
				"Integer8" "Integer16" "Integer32" "Integer64" "Unsigned8" "Unsigned16" "Unsigned32" "Unsigned64" "Float32" "Float64" "Date" "TimeOfDay with date indication" "TimeOfDay without date indication" "TimeDifference with date indication" "TimeDifference without date indication" "NetworkTime" "NetworkTimeDifference" "VisibleString" "OctetString" The length of the datatype "VisibleString" or "OctetString" shall be defined by the attribute "Length".
DefaultValue	String	Obsolete	1.0	Required for backward compatibility to engineering tools that don't support GSDML V2.2. Use an arbitrary value valid for the DataType. For representation of values see Table 2 and Table 3. The content is checked but ignored by older engineering tools.
TextId	RefId	Required	1.0	Contains the ID of a text as a reference into the "ExternalTextList".
Length	Unsigned16	Optional	2.1	Defines the length of the used "VisibleString" or "OctetString" data type. This attribute shall be ignored if other data types are used.
NOTE The "BitArea" data type can be used to reduce the size of an alarm data object because it has a variable bit length. For example if a diagnostic data item can only have four different values, a BitArea with the length of 2 bit is sufficient. A "BitArea" diagnostic data item can contain codings or separate bits.				

2396

2397 A.4.15 LogBookEntryItem related elements

2398 A.4.15.1 LogBookEntryItem

2399 A LogBookEntryItem defines a manufacturer specific PNIOStatus in the LogBook (see PNO-2.722
 2400 *Application Layer protocol*, "Coding section related to PNIOStatus").

2401 Location:

2402 //ISO15745Profile/ProfileBody/ApplicationProcess/LogBookEntryList/LogBookEntryItem

2403 Use: One or more

2404 Each element shall contain attributes as shown in Table 96.

2405 **Table 96 — Attributes of element LogBookEntryItem**

Name	Type	Use	SV	Description
Status	Unsigned32	Required	2.3	Contains the value of the first three octets of the PNIOStatus, i.e. Errorcode x 65536 + ErrorDecode x 256 + Errorcode1. See Table 97 for the allowed values. Status shall be unique across all LogBookEntryItems.

2406

2407

Table 97 — Vendor specific PNIOStatus

Error Code	Meaning	Error Decode	Meaning	Error Code1	Meaning	Attribute "Status"	Error Code2
0x20-0x3F	Manufacturer	0x82	Manufacturer	0x00-0xFF		2130432-4162303	–
0x81	PNIO	0x81	PNIO	0xFF	User specific	8487423	0x00-0xFF
0xCF	RTA error	0x81	PNIO	0xFD	RTA_ERR_CLS_PROTOCOL	13599229	0xC9-0xFF
0xCF	RTA error	0x81	PNIO	0xFF	User specific	13599231	0x00-0xFF
0xDA	AlarmAck	0x81	PNIO	0xFD	RTA_ERR_CLS_PROTOCOL	14320125	0xC9-0xFF
0xDA	AlarmAck	0x81	PNIO	0xFF	User specific	14320127	0x00-0xFF
0xDB	IODConnectRes	0x81	PNIO	0xFF	User specific	14385663	0x00-0xFF
0xDC	IODReleaseRes	0x81	PNIO	0xFF	User specific	14451199	0x00-0xFF
0xDD	IODControlRes, IOXControlRes	0x81	PNIO	0xFF	User specific	14516735	0x00-0xFF
0xDE	IODReadRes	0x80	PNIORW	0xA3-0xA6	Not specified	14581923-14581926	–
0xDE	IODReadRes	0x80	PNIORW	0xAA-0xAF	User specific	14581930-14581935	–
0xDE	IODReadRes	0x80	PNIORW	0xBB-0xBF	User specific	14581947-14581951	–
0xDE	IODReadRes	0x80	PNIORW	0xC4-0xFF	Not specified / User specific	14581956-14582015	–
0xDF	IODWriteRes	0x80	PNIORW	0xA3-0xA6	Not specified	14647459-14647462	–
0xDF	IODWriteRes	0x80	PNIORW	0xAA-0xAF	User specific	14647466-14647471	–
0xDF	IODWriteRes	0x80	PNIORW	0xBB-0xBF	User specific	14647483-14647487	–
0xDF	IODWriteRes	0x80	PNIORW	0xC4-0xFF	Not specified / User specific	14647492-14647551	–

2408

2409 The treatment of ErrorCode2 depends on the value of the Status attribute. If the row ErrorCode2 in the
 2410 table above contains a value range, the subelement ErrorCode2List shall be present, and all
 2411 ErrorCode2Item/@ErrorCode2 shall be numbers from this value range.

2412 Otherwise, the subelement ErrorCode2Value shall be present.

2413 A.4.15.2 ErrorCode2List

2414 Contains a list of entries for ErrorCode2 values which are decoded in individual messages.

2415 Location:

2416 //ISO15745Profile/ProfileBody/ApplicationProcess/LogBookEntryList/LogBookEntryItem/ErrorCode2List

2417 Use: If Table 97 contains a value range in row ErrorCode2 for the respective Status.

2418 A.4.15.3 ErrorCode2Item

2419 Contains an ErrorCode2 value and a corresponding message.

2420 Location:

2421 //ISO15745Profile/ProfileBody/ApplicationProcess/LogBookEntryList/LogBookEntryItem/ErrorCode2List/E
 2422 rrorCode2Item

2423 Use: One or more.

2424 Each element shall contain attributes as shown in Table 98.

2425 **Table 98 — Attributes of element ErrorCode2Item**

Name	Type	Use	SV	Description
ErrorCode2	Unsigned8	Required	2.3	Contains the value of ErrorDecode2. See Table 97 for the allowed values. Status shall be unique across all ErrorCode2Items.

2426

2427 **A.4.15.4 ErrorCode2Value**

2428 Contains a message for the Status. The value of ErrorCode2 shall be shown as number in addition to this
2429 message.

2430 Location:

2431 //ISO15745Profile/ProfileBody/ApplicationProcess/LogBookEntryList/LogBookEntryItem/ErrorCode2Value

2432 Use: If Table 97 does not contain a value range in row ErrorCode2 for the respective Status.

2433 **A.4.15.5 Name**

2434 This element is used to assign the error text to display with this PNIOSStatus.

2435 Location:

2436 //ISO15745Profile/ProfileBody/ApplicationProcess/LogBookEntryList/LogBookEntryItem/ErrorCode2Value
2437 /Name

2438 //ISO15745Profile/ProfileBody/ApplicationProcess/LogBookEntryList/LogBookEntryItem/ErrorCode2List/E
2439 rrorCode2Item/Name

2440 Use: Required

2441 Each element shall contain attributes as shown in Table 99.

2442 **Table 99 — Attributes of element Name**

Name	Type	Use	SV	Description
TextId	RefId	Required	2.3	Contains the ID of a text as a reference into the "ExternalTextList".

2443

2444 **A.4.15.6 Help**

2445 This element contains the language dependent help information for a PNIOSStatus, describing the likely
2446 cause and a possible remedy.

2447 Location:

2448 //ISO15745Profile/ProfileBody/ApplicationProcess/LogBookEntryList/LogBookEntryItem/ErrorCode2Value
2449 /Help

2450 //ISO15745Profile/ProfileBody/ApplicationProcess/LogBookEntryList/LogBookEntryItem/ErrorCode2List/E
2451 rrorCode2Item/Help

2452 Use: Optional

2453 Each element shall contain attributes as shown in Table 100.

2454 **Table 100 — Attributes of element Help**

Name	Type	Use	SV	Description
TextId	RefId	Required	2.3	Contains the ID of a text as a reference into the "ExternalTextList".

2455

2456

2457 A.5 GSDML schemas

2458 The following schema definitions are part of this specification. They are provided together with the
2459 specification as separate files for integration into engineering tools and for inspecting in an XML Editor.

2460 GSDML-DeviceProfile-v2.3.xsd GSDML Device Profile schema

2461 GSDML-CommNetwork-v2.3.xsd GSDML Communication Network Profile schema

2462 GSDML-Primitives-v2.3.xsd GSDML Primitives schema

2463 The following schema definitions are imported by the GSDML schemas. These schema files are provided
2464 by the World Wide Web Consortium (W3C). They can be downloaded from the W3C web site, usually
2465 using the namespace identifier as an internet address.

2466 xml.xsd The “xml:” namespace

2467 Namespace: <http://www.w3.org/XML/1998/namespace>

2468 Schema location: <http://www.w3.org/2001/xml.xsd>

2469 (= <http://www.w3.org/2009/01/xml.xsd> currently)

2470 xmldsig-core-schema.xsd Schema for XML signatures

2471 Namespace and schema location: <http://www.w3.org/2000/09/xmldsig#>

2472 (= <http://www.w3.org/TR/2002/REC-xmldsig-core-20020212/xmldsig-core-schema.xsd> currently)

2473 Notes on schema validation in engineering tools:

2474 • The xmldsig-core-schema.xsd provided by the W3C contains some embedded DTD part, which
2475 causes errors with several XML parsers. In the version of the schema provided with this specification
2476 this DTD part was cut out to prevent problems.

2477 • When there is no schema cache given, XML parsers use the path and filename in
2478 xsi:schemaLocation to find the appropriate schema file(s). Usually, this requires a temporary copy of
2479 the GSD to a place where the relative path “..xsd” points to a directory containing the GSDML
2480 schema files. If the path in xsi:schemaLocation is wrong, this will lead to an error. It is thus
2481 recommended to use a schema cache, eliminating the need for a copy of the GSD.

2482 • For compatibility (see A.3.2 and A.3.3) the namespace of the GSDML is the same across all GSDML
2483 versions, although this is against XML philosophy. Some XML parsers do not allow different schema
2484 files with the same name space together in the schema cache. In this case it is necessary to
2485 determine the GSDML version first and then fill the schema cache with the schema files of the
2486 appropriate version.

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